

VCU118 EVALUATION BOARD HW-U1-VCU118

(XCVU9P-2FLGA2104)

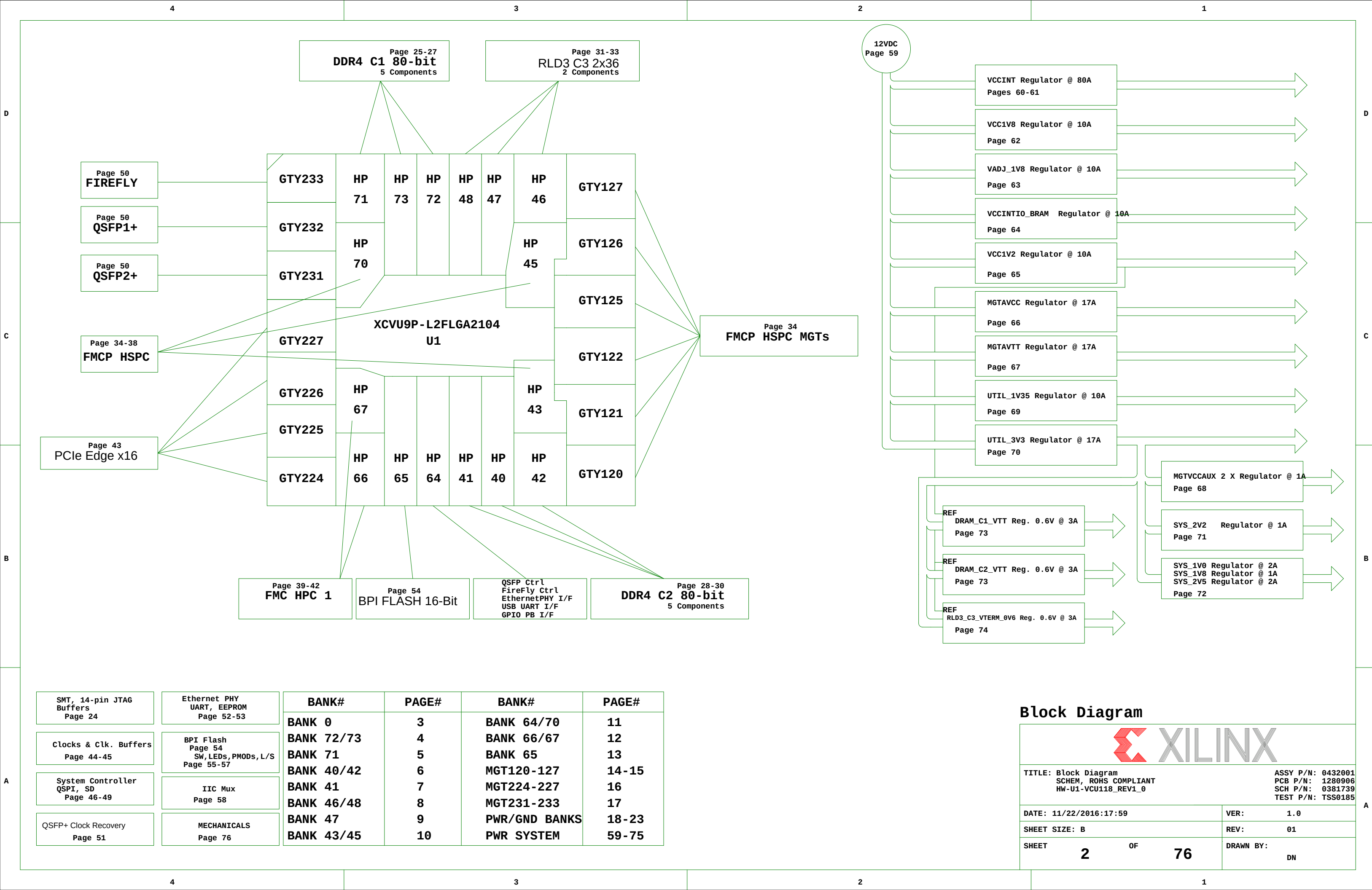
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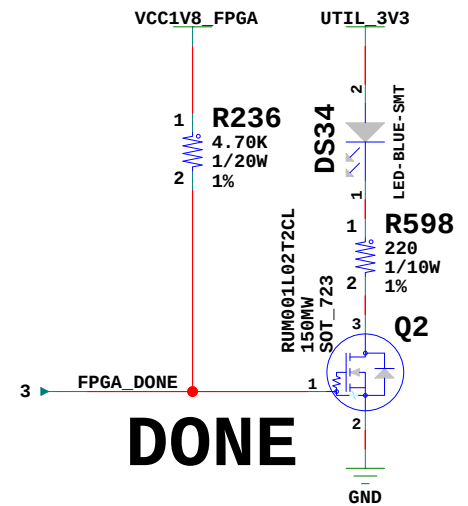
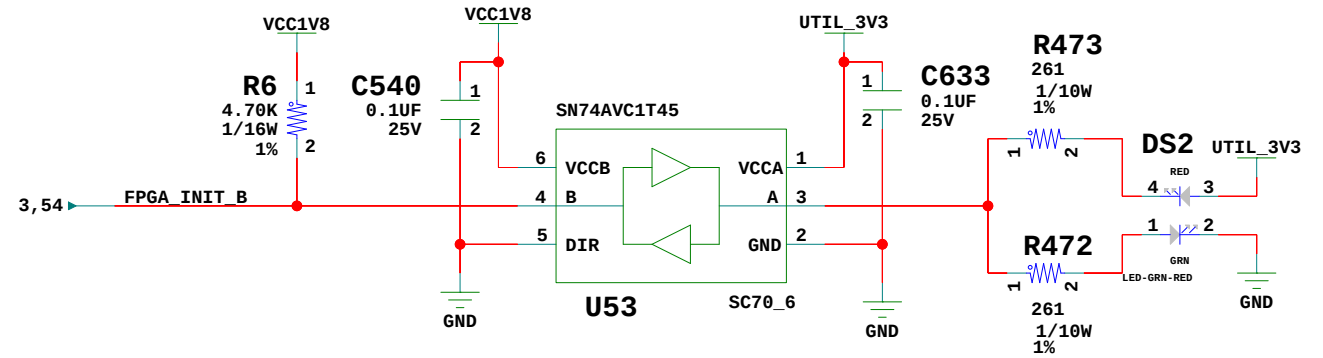
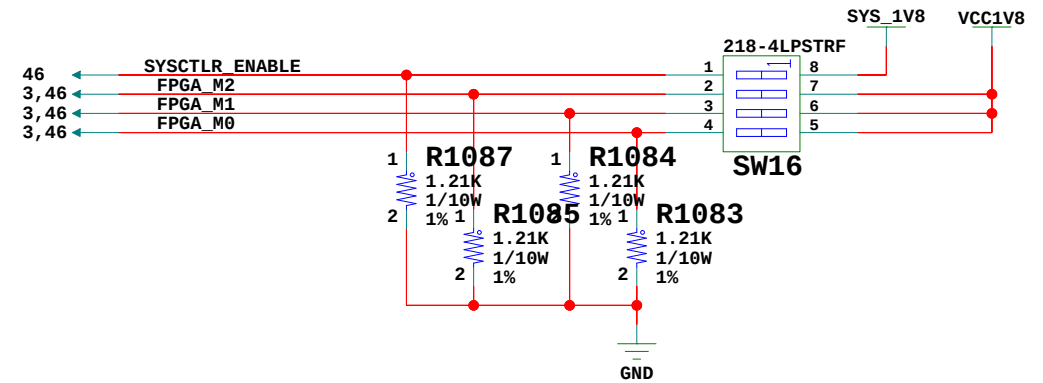
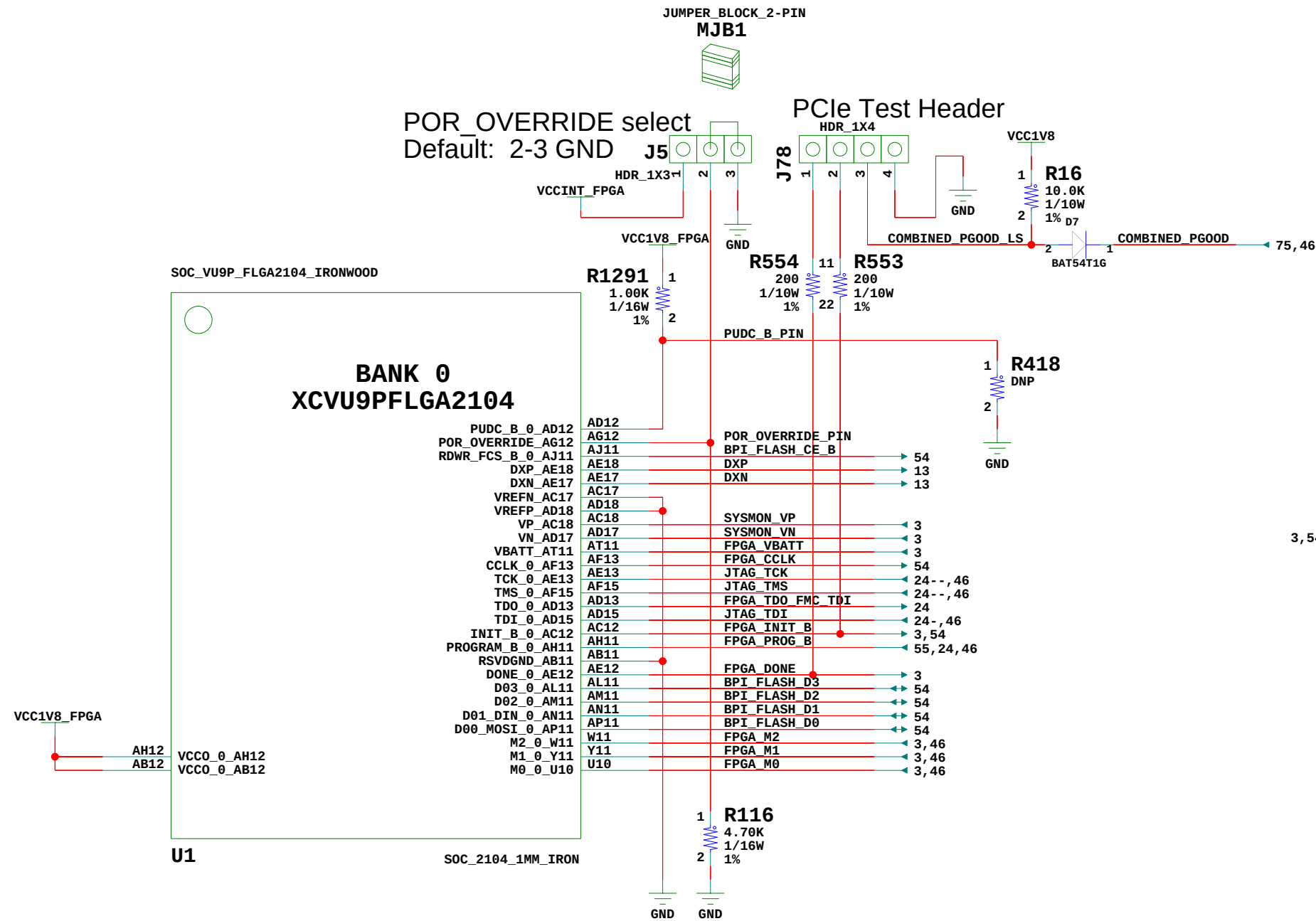
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Title Page			
			
TITLE: Title Page SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0		ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59		VER:	1.0
SHEET SIZE: B		REV:	01
SHEET	1	OF	76
		DRAWN BY:	DN

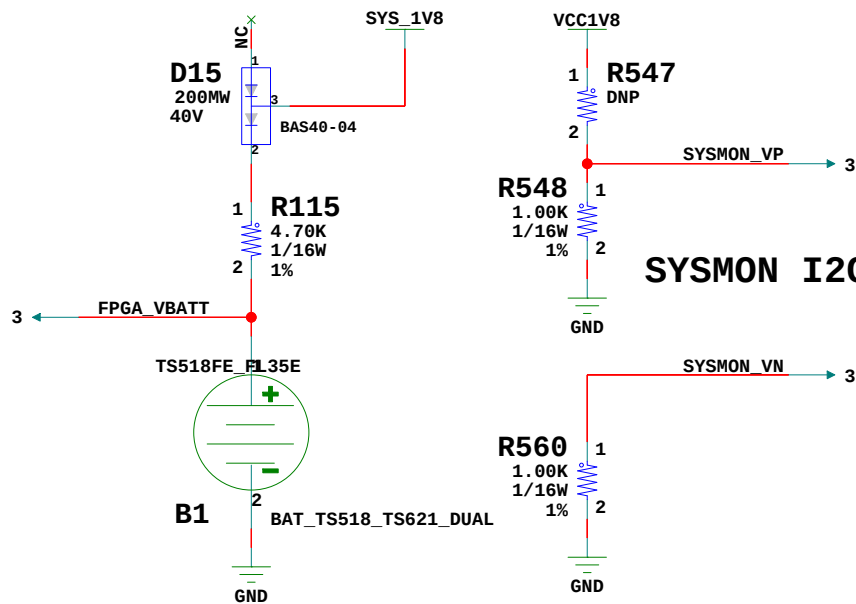


POR_OVERRIDE select
Default: 2-3 GND

PCle Test Header



SYSMON I2C Address jumpers



FPGA Bank 0



TITLE: FPGA Bank 0
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 3 OF 76	DRAWN BY: DN

Bank 73 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 73
XCVU9PFLGA2104

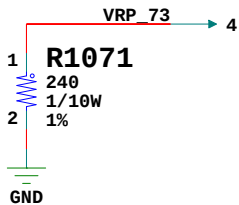
IO_T3U_N12_73_A20	A20	DDR4_C1_TEN	25-, 26-, 27
IO_L24N_T3U_N11_73_B20	B20	DDR4_C1_DQ39	26
IO_L24P_T3U_N10_73_C20	C20	DDR4_C1_DQ38	26
IO_L23N_T3U_N9_73_D19	D19	DDR4_C1_DQ37	26
IO_L23P_T3U_N8_73_D20	D20	DDR4_C1_DQ36	26
IO_L22N_T3U_N7_DBC_AD0N_73_A18	A18	DDR4_C1_DQS4_C	26
IO_L22P_T3U_N6_DBC_AD0P_73_A19	A19	DDR4_C1_DQS4_T	26
IO_L21N_T3L_N5_AD8N_73_C18	C18	DDR4_C1_DQ35	26
IO_L21P_T3L_N4_AD8P_73_C19	C19	DDR4_C1_DQ34	26
IO_L20N_T3L_N3_AD1N_73_C17	C17	DDR4_C1_DQ33	26
IO_L20P_T3L_N2_AD1P_73_D17	D17	DDR4_C1_DQ32	26
IO_L19N_T3L_N1_DBC_AD9N_73_B17	B17	GPIO_DIP_SW1	55
IO_L19P_T3L_N0_DBC_AD9P_73_B18	B18	DDR4_C1_DM4	26
IO_T2U_N12_73_G16	G16	GPIO_DIP_SW2	26
IO_L18N_T2U_N11_AD2N_73_D16	D16	DDR4_C1_DQ31	55
IO_L18P_T2U_N10_AD2P_73_E17	E17	DDR4_C1_DQ30	25
IO_L17N_T2U_N9_AD10N_73_F20	F20	DDR4_C1_DQ29	25
IO_L17P_T2U_N8_AD10P_73_G20	G20	DDR4_C1_DQ28	25
IO_L16N_T2U_N7_QBC_AD3N_73_E16	E16	DDR4_C1_DQS3_C	25
IO_L16P_T2U_N6_QBC_AD3P_73_F16	F16	DDR4_C1_DQS3_T	25
IO_L15N_T2L_N5_AD11N_73_E18	E18	DDR4_C1_DQ27	25
IO_L15P_T2L_N4_AD11P_73_E19	E19	DDR4_C1_DQ26	25
IO_L14N_T2L_N3_GC_73_F18	F18	DDR4_C1_DQ25	25
IO_L14P_T2L_N2_GC_73_F19	F19	DDR4_C1_DQ24	25
IO_L13N_T2L_N1_GC_QBC_73_G17	G17	NC	
IO_L13P_T2L_N0_GC_QBC_73_G18	G18	DDR4_C1_DM3	25
IO_T1U_N12_73_L19	L19	CPU_RESET	55
IO_L12N_T1U_N11_GC_73_H18	H18	DDR4_C1_DQ23	25
IO_L12P_T1U_N10_GC_73_H19	H19	DDR4_C1_DQ22	25
IO_L11N_T1U_N9_GC_73_H17	H17	DDR4_C1_DQ21	25
IO_L11P_T1U_N8_GC_73_J17	J17	DDR4_C1_DQ20	25
IO_L10N_T1U_N7_QBC_AD4N_73_J19	J19	DDR4_C1_DQS2_C	25
IO_L10P_T1U_N6_QBC_AD4P_73_K19	K19	DDR4_C1_DQS2_T	25
IO_L9N_T1L_N5_AD12N_73_K18	K18	DDR4_C1_DQ19	25
IO_L9P_T1L_N4_AD12P_73_L18	L18	DDR4_C1_DQ18	25
IO_L8N_T1L_N3_AD5N_73_K16	K16	DDR4_C1_DQ17	25
IO_L8P_T1L_N2_AD5P_73_L16	L16	DDR4_C1_DQ16	25
IO_L7N_T1L_N1_QBC_AD13N_73_J16	J16	GPIO_DIP_SW3	55
IO_L7P_T1L_N0_QBC_AD13P_73_K17	K17	DDR4_C1_DM2	25
IO_T0U_N12_VRP_73_T18	T18	VRP_73	4
IO_L6N_T0U_N11_AD6N_73_M16	M16	DDR4_C1_DQ15	25
IO_L6P_T0U_N10_AD6P_73_N17	N17	DDR4_C1_DQ14	25
IO_L5N_T0U_N9_AD14N_73_N18	N18	DDR4_C1_DQ13	25
IO_L5P_T0U_N8_AD14P_73_N19	N19	DDR4_C1_DQ12	25
IO_L4N_T0U_N7_DBC_AD7N_73_P16	P16	DDR4_C1_DQS1_C	25
IO_L4P_T0U_N6_DBC_AD7P_73_P17	P17	DDR4_C1_DQS1_T	25
IO_L3N_T0L_N5_AD15N_73_M17	M17	DDR4_C1_DQ11	25
IO_L3P_T0L_N4_AD15P_73_M18	M18	DDR4_C1_DQ10	25
IO_L2N_T0L_N3_73_P19	P19	DDR4_C1_DQ9	25
IO_L2P_T0L_N2_73_R19	R19	DDR4_C1_DQ8	25
IO_L1N_T0L_N1_DBC_73_R17	R17	DDR4_C1_ALERT_B	25
IO_L1P_T0L_N0_DBC_73_R18	R18	DDR4_C1_DM1	25-, 26-, 27
VREF_73_T19	T19	NC	

VCC1V2_FPGA

A17	VCC0_73_A17
D18	VCC0_73_D18
G19	VCC0_73_G19
H16	VCC0_73_H16
L17	VCC0_73_L17
P18	VCC0_73_P18

U1

SOC_2104_1MM_IRON



Bank 72 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 72
XCVU9PFLGA2104

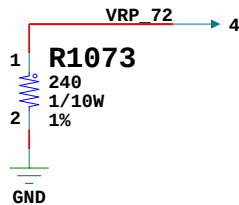
IO_T3U_N12_72_D21	D21	GPIO_DIP_SW4	55
IO_L24N_T3U_N11_72_A21	A21	DDR4_C1_DQ71	27
IO_L24P_T3U_N10_72_B21	B21	DDR4_C1_DQ70	27
IO_L23N_T3U_N9_72_B22	B22	DDR4_C1_DQ69	27
IO_L23P_T3U_N8_72_B23	B23	DDR4_C1_DQ68	27
IO_L22N_T3U_N7_DBC_AD0N_72_C22	C22	DDR4_C1_DQS8_C	27
IO_L22P_T3U_N6_DBC_AD0P_72_D22	D22	DDR4_C1_DQS8_T	27
IO_L21N_T3L_N5_AD8N_72_C23	C23	DDR4_C1_DQ67	27
IO_L21P_T3L_N4_AD8P_72_C24	C24	DDR4_C1_DQ66	27
IO_L20N_T3L_N3_AD1N_72_A23	A23	DDR4_C1_DQ65	27
IO_L20P_T3L_N2_AD1P_72_A24	A24	DDR4_C1_DQ64	27
IO_L19N_T3L_N1_DBC_AD9N_72_D24	D24	NC	
IO_L19P_T3L_N0_DBC_AD9P_72_E24	E24	DDR4_C1_DM8	27
IO_T2U_N12_72_H20	H20	SI5328_INT_ALM_LS	56
IO_L18N_T2U_N11_AD2N_72_F23	F23	DDR4_C1_DQ63	26
IO_L18P_T2U_N10_AD2P_72_F24	F24	DDR4_C1_DQ62	26
IO_L17N_T2U_N9_AD10N_72_E21	E21	DDR4_C1_DQ61	26
IO_L17P_T2U_N8_AD10P_72_F21	F21	DDR4_C1_DQ60	26
IO_L16N_T2U_N7_QBC_AD3N_72_G23	G23	DDR4_C1_DQ57_C	26
IO_L16P_T2U_N6_QBC_AD3P_72_H24	H24	DDR4_C1_DQ57_T	26
IO_L15N_T2L_N5_AD11N_72_E22	E22	DDR4_C1_DQ59	26
IO_L15P_T2L_N4_AD11P_72_E23	E23	DDR4_C1_DQ58	26
IO_L14N_T2L_N3_GC_72_H22	H22	DDR4_C1_DQ57	26
IO_L14P_T2L_N2_GC_72_H23	H23	DDR4_C1_DQ56	26
IO_L13N_T2L_N1_GC_QBC_72_G21	G21	NC	
IO_L13P_T2L_N0_GC_QBC_72_G22	G22	DDR4_C1_DM7	26
IO_T1U_N12_72_J20	J20	NC	
IO_L12N_T1U_N11_GC_72_J22	J22	DDR4_C1_DQ55	26
IO_L12P_T1U_N10_GC_72_K22	K22	DDR4_C1_DQ54	26
IO_L11N_T1U_N9_GC_72_J21	J21	DDR4_C1_DQ53	26
IO_L11P_T1U_N8_GC_72_K21	K21	DDR4_C1_DQ52	26
IO_L10N_T1U_N7_QBC_AD4N_72_L20	L20	DDR4_C1_DQ56_C	26
IO_L10P_T1U_N6_QBC_AD4P_72_M20	M20	DDR4_C1_DQ56_T	26
IO_L9N_T1L_N5_AD12N_72_L21	L21	DDR4_C1_DQ51	26
IO_L9P_T1L_N4_AD12P_72_M21	M21	DDR4_C1_DQ50	26
IO_L8N_T1L_N3_AD5N_72_J24	J24	DDR4_C1_DQ49	26
IO_L8P_T1L_N2_AD5P_72_K24	K24	DDR4_C1_DQ48	26
IO_L7N_T1L_N1_QBC_AD13N_72_K23	K23	NC	
IO_L7P_T1L_N0_QBC_AD13P_72_L23	L23	DDR4_C1_DM6	26
IO_T0U_N12_VRP_72_T21	T21	VRP_72	4
IO_L6N_T0U_N11_AD6N_72_R23	R23	DDR4_C1_DQ47	26
IO_L6P_T0U_N10_AD6P_72_T23	T23	DDR4_C1_DQ46	26
IO_L5N_T0U_N9_AD14N_72_P22	P22	DDR4_C1_DQ45	26
IO_L5P_T0U_N8_AD14P_72_R22	R22	DDR4_C1_DQ44	26
IO_L4N_T0U_N7_DBC_AD7N_72_M22	M22	DDR4_C1_DQ55_C	26
IO_L4P_T0U_N6_DBC_AD7P_72_N22	N22	DDR4_C1_DQ55_T	26
IO_L3N_T0L_N5_AD15N_72_P21	P21	DDR4_C1_DQ43	26
IO_L3P_T0L_N4_AD15P_72_R21	R21	DDR4_C1_DQ42	26
IO_L2N_T0L_N3_72_M23	M23	DDR4_C1_DQ41	26
IO_L2P_T0L_N2_72_N23	N23	DDR4_C1_DQ40	26
IO_L1N_T0L_N1_DBC_72_N20	N20	DDR4_C1_RESET_B	25-, 26-, 27
IO_L1P_T0L_N0_DBC_72_P20	P20	DDR4_C1_DM5	26
VREF_72_T20	T20	NC	

VCC1V2_FPGA

B24	VCC0_72_B24
C21	VCC0_72_C21
F22	VCC0_72_F22
J23	VCC0_72_J23
K20	VCC0_72_K20
N21	VCC0_72_N21
T22	VCC0_72_T22

U1

SOC_2104_1MM_IRON



FPGA Banks 73 72 C1 DDR4 Data

TITLE: FPGA Banks 73 72 C1 DDR4 Data SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 4 OF 76	DRAWN BY: DN

Bank 71 HP

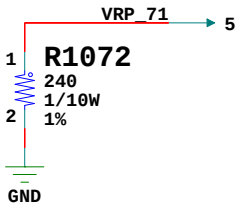
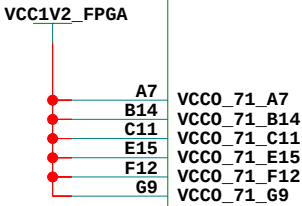
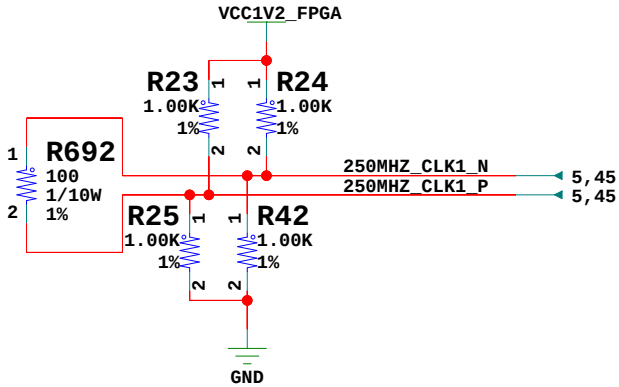
SOC_VU9P_FLGA2104_IRONWOOD

BANK 71
XCVU9PFLGA2104

IO_T3U_N12_71_D14
IO_L24N_T3U_N11_71_B15
IO_L24P_T3U_N10_71_B16
IO_L23N_T3U_N9_71_C14
IO_L23P_T3U_N8_71_C15
IO_L22N_T3U_N7_DBC_AD0N_71_A13
IO_L22P_T3U_N6_DBC_AD0P_71_A14
IO_L21N_T3L_N5_AD8N_71_A15
IO_L21P_T3L_N4_AD8P_71_A16
IO_L20N_T3L_N3_AD1N_71_B12
IO_L20P_T3L_N2_AD1P_71_C12
IO_L19N_T3L_N1_DBC_AD9N_71_B13
IO_L19P_T3L_N0_DBC_AD9P_71_C13
IO_T2U_N12_71_D15
IO_L18N_T2U_N11_AD2N_71_H14
IO_L18P_T2U_N10_AD2P_71_H15
IO_L17N_T2U_N9_AD10N_71_F15
IO_L17P_T2U_N8_AD10P_71_G15
IO_L16N_T2U_N7_QBC_AD3N_71_E14
IO_L16P_T2U_N6_QBC_AD3P_71_F14
IO_L15N_T2L_N5_AD11N_71_G13
IO_L15P_T2L_N4_AD11P_71_H13
IO_L14N_T2L_N3_GC_71_E13
IO_L14P_T2L_N2_GC_71_F13
IO_L13N_T2L_N1_GC_QBC_71_D12
IO_L13P_T2L_N0_GC_QBC_71_E12
IO_T1U_N12_71_A10
IO_L12N_T1U_N11_GC_71_A11
IO_L12P_T1U_N10_GC_71_B11
IO_L11N_T1U_N9_GC_71_B10
IO_L11P_T1U_N8_GC_71_C10
IO_L10N_T1U_N7_QBC_AD4N_71_A8
IO_L10P_T1U_N6_QBC_AD4P_71_A9
IO_L9N_T1L_N5_AD12N_71_B7
IO_L9P_T1L_N4_AD12P_71_B8
IO_L8N_T1L_N3_AD5N_71_C7
IO_L8P_T1L_N2_AD5P_71_D7
IO_L7N_T1L_N1_QBC_AD13N_71_C8
IO_L7P_T1L_N0_QBC_AD13P_71_C9
IO_T0U_N12_VRP_71_D8
IO_L6N_T0U_N11_AD6N_71_D9
IO_L6P_T0U_N10_AD6P_71_E9
IO_L5N_T0U_N9_AD14N_71_G12
IO_L5P_T0U_N8_AD14P_71_H12
IO_L4N_T0U_N7_DBC_AD7N_71_D10
IO_L4P_T0U_N6_DBC_AD7P_71_D11
IO_L3N_T0L_N5_AD15N_71_F9
IO_L3P_T0L_N4_AD15P_71_F10
IO_L2N_T0L_N3_71_E11
IO_L2P_T0L_N2_71_F11
IO_L1N_T0L_N1_DBC_71_G10
IO_L1P_T0L_N0_DBC_71_G11
VREF_71_J15

D14 DDR4 C1 A0 → 25-,26-,27
B15 DDR4 C1 A1 → 25-,26-,27
B16 DDR4 C1 A2 → 25-,26-,27
C14 DDR4 C1 A3 → 25-,26-,27
C15 DDR4 C1 A4 → 25-,26-,27
A13 DDR4 C1 A5 → 25-,26-,27
A14 DDR4 C1 A6 → 25-,26-,27
A15 DDR4 C1 A7 → 25-,26-,27
A16 DDR4 C1 A8 → 25-,26-,27
B12 DDR4 C1 A9 → 25-,26-,27
C12 DDR4 C1 A10 → 25-,26-,27
B13 DDR4 C1 A11 → 25-,26-,27
C13 DDR4 C1 A12 → 25-,26-,27
D15 DDR4 C1 A13 → 25-,26-,27
H14 DDR4 C1 A14_WE_B → 25-,26-,27
H15 DDR4 C1 A15_CAS_B → 25-,26-,27
F15 DDR4 C1 A16_RAS_B → 25-,26-,27
G15 DDR4 C1 BA0 → 25-,26-,27
E14 DDR4 C1 CK_C → 25-,26-,27
F14 DDR4 C1 CK_T → 25-,26-,27
G13 DDR4 C1 BA1 → 25-,26-,27
H13 DDR4 C1 BG0 → 25-,26-,27
E13 DDR4 C1 ACT_B → 25-,26-,27
F13 DDR4 C1 CS_B → 25-,26-,27
D12 250MHZ_CLK1_N → 5,45
E12 250MHZ_CLK1_P → 5,45
A10 DDR4 C1 CKE → 25-,26-,27
A11 DDR4 C1 DQ79 ↔ 27
B11 DDR4 C1 DQ78 ↔ 27
B10 DDR4 C1 DQ77 ↔ 27
C10 DDR4 C1 DQ76 ↔ 27
A8 DDR4 C1 DQS9_C ↔ 27
A9 DDR4 C1 DQS9_T ↔ 27
B7 DDR4 C1 DQ75 ↔ 27
B8 DDR4 C1 DQ74 ↔ 27
C7 DDR4 C1 DQ73 ↔ 27
D7 DDR4 C1 DQ72 ↔ 27
C8 DDR4 C1 ODT → 25-,26-,27
C9 DDR4 C1 DM9 → 27
D8 VRP_71 → 5
D9 DDR4 C1 DQ7 ↔ 25
E9 DDR4 C1 DQ6 ↔ 25
G12 DDR4 C1 DQ5 ↔ 25
H12 DDR4 C1 DQ4 ↔ 25
D10 DDR4 C1 DQS0_C ↔ 25
D11 DDR4 C1 DQS0_T ↔ 25
F9 DDR4 C1 DQ3 ↔ 25
F10 DDR4 C1 DQ2 ↔ 25
E11 DDR4 C1 DQ1 ↔ 25
F11 DDR4 C1 DQ0 ↔ 25
G10 DDR4 C1 PAR → 25
G11 DDR4 C1 DM0 → 25-,26-,27
J15 NC

PCB LAYOUT DIRECTIVE:
TERMINATION RESISTOR
AND BIAS RESISTOR
NETWORK TO BE PLACED
IN "FLY-BY" CONFIGURATION



FPGA Bank 71 C1 DDR4 Addr Ctrl

TITLE: FPGA Bank 71 C1 DDR4 Addr Ctrl SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 5 OF 76	DRAWN BY: DN

Bank 40 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 40
XCVU9PFLGA2104

IO_T3U_N12_40_AR29
IO_L24N_T3U_N11_40_AN31
IO_L24P_T3U_N10_40_AN30
IO_L23N_T3U_N9_40_AR30
IO_L23P_T3U_N8_40_AP30
IO_L22N_T3U_N7_DBC_AD0N_40_AP32
IO_L22P_T3U_N6_DBC_AD0P_40_AP31
IO_L21N_T3L_N5_AD8N_40_AT30
IO_L21P_T3L_N4_AD8P_40_AT29
IO_L20N_T3L_N3_AD1N_40_AT34
IO_L20P_T3L_N2_AD1P_40_AR33
IO_L19N_T3L_N1_DBC_AD9N_40_AT32
IO_L19P_T3L_N0_DBC_AD9P_40_AR32
IO_T2U_N12_40_AV34
IO_L18N_T2U_N11_AD2N_40_AV31
IO_L18P_T2U_N10_AD2P_40_AU31
IO_L17N_T2U_N9_AD10N_40_AU32
IO_L17P_T2U_N8_AD10P_40_AT31
IO_L16N_T2U_N7_QBC_AD3N_40_AV29
IO_L16P_T2U_N6_QBC_AD3P_40_AU29
IO_L15N_T2L_N5_AD11N_40_AU34
IO_L15P_T2L_N4_AD11P_40_AU33
IO_L14N_T2L_N3_GC_40_AW30
IO_L14P_T2L_N2_GC_40_AV30
IO_L13N_T2L_N1_GC_QBC_40_AW33
IO_L13P_T2L_N0_GC_QBC_40_AV33
IO_T1U_N12_40_AY30
IO_L12N_T1U_N11_GC_40_AY33
IO_L12P_T1U_N10_GC_40_AY32
IO_L11N_T1U_N9_GC_40_AW32
IO_L11P_T1U_N8_GC_40_AW31
IO_L10N_T1U_N7_QBC_AD4N_40_BA34
IO_L10P_T1U_N6_QBC_AD4P_40_AY34
IO_L9N_T1L_N5_AD12N_40_BA31
IO_L9P_T1L_N4_AD12P_40_BA30
IO_L8N_T1L_N3_AD5N_40_BB33
IO_L8P_T1L_N2_AD5P_40_BA32
IO_L7N_T1L_N1_QBC_AD13N_40_BB32
IO_L7P_T1L_N0_QBC_AD13P_40_BB31
IO_T0U_N12_VRP_40_BC30
IO_L6N_T0U_N11_AD6N_40_BD31
IO_L6P_T0U_N10_AD6P_40_BC31
IO_L5N_T0U_N9_AD14N_40_BD33
IO_L5P_T0U_N8_AD14P_40_BC33
IO_L4N_T0U_N7_DBC_AD7N_40_BF31
IO_L4P_T0U_N6_DBC_AD7P_40_BF30
IO_L3N_T0L_N5_AD15N_40_BE33
IO_L3P_T0L_N4_AD15P_40_BD32
IO_L2N_T0L_N3_40_BE30
IO_L2P_T0L_N2_40_BD30
IO_L1N_T0L_N1_DBC_40_BF32
IO_L1P_T0L_N0_DBC_40_BE32
VREF_40_AN29

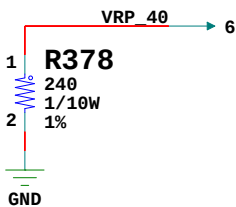
AR29 DDR4_C2_ALERT_B 28-,29-,30
AN31 DDR4_C2_DQ31 28
AN30 DDR4_C2_DQ30 28
AR30 DDR4_C2_DQ29 28
AP30 DDR4_C2_DQ28 28
AP32 DDR4_C2_DQS3_C 28
AP31 DDR4_C2_DQS3_T 28
AT30 DDR4_C2_DQ27 28
AT29 DDR4_C2_DQ26 28
AT34 DDR4_C2_DQ25 28
AR33 DDR4_C2_DQ24 28
AT32 GPIO_LED0 55
AR32 DDR4_C2_DM3 28
AV34 GPIO_LED1 55
AV31 DDR4_C2_DQ23 28
AU31 DDR4_C2_DQ22 28
AU32 DDR4_C2_DQ21 28
AT31 DDR4_C2_DQ20 28
AV29 DDR4_C2_DQS2_C 28
AU29 DDR4_C2_DQS2_T 28
AU34 DDR4_C2_DQ19 28
AU33 DDR4_C2_DQ18 28
AW30 DDR4_C2_DQ17 28
AV30 DDR4_C2_DQ16 28
AW33 FAN_FAIL_LS_B 13
AV33 DDR4_C2_DM2 28
AY30 GPIO_LED2 55
AY33 DDR4_C2_DQ15 28
AY32 DDR4_C2_DQ14 28
AW32 DDR4_C2_DQ13 28
AW31 DDR4_C2_DQ12 28
BA34 DDR4_C2_DQS1_C 28
AY34 DDR4_C2_DQS1_T 28
BA31 DDR4_C2_DQ11 28
BA30 DDR4_C2_DQ10 28
BB33 DDR4_C2_DQ9 28
BA32 DDR4_C2_DQ8 28
BB32 GPIO_LED3 55
BB31 DDR4_C2_DM1 28
BC30 VRP_40 6
BD31 DDR4_C2_DQ7 28
BC31 DDR4_C2_DQ6 28
BD33 DDR4_C2_DQ5 28
BC33 DDR4_C2_DQ4 28
BF31 DDR4_C2_DQS0_C 28
BF30 DDR4_C2_DQS0_T 28
BE33 DDR4_C2_DQ3 28
BD32 DDR4_C2_DQ2 28
BE30 DDR4_C2_DQ1 28
BD30 DDR4_C2_DQ0 28
BF32 GPIO_LED4 55
BE32 DDR4_C2_DM0 28
AN29 NC

VCC1V2_FPGA

AR31
AV32
AW29
BA33
BB30
BE31
VCCO_40_AR31
VCCO_40_AV32
VCCO_40_AW29
VCCO_40_BA33
VCCO_40_BB30
VCCO_40_BE31

U1

SOC_2104_1MM_IRON



Bank 42 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 42
XCVU9PFLGA2104

IO_T3U_N12_42_AU37
IO_L24N_T3U_N11_42_AV39
IO_L24P_T3U_N10_42_AV38
IO_L23N_T3U_N9_42_AU39
IO_L23P_T3U_N8_42_AU38
IO_L22N_T3U_N7_DBC_AD0N_42_AW38
IO_L22P_T3U_N6_DBC_AD0P_42_AW37
IO_L21N_T3L_N5_AD8N_42_AV40
IO_L21P_T3L_N4_AD8P_42_AU40
IO_L20N_T3L_N3_AD1N_42_AW36
IO_L20P_T3L_N2_AD1P_42_AW35
IO_L19N_T3L_N1_DBC_AD9N_42_AV36
IO_L19P_T3L_N0_DBC_AD9P_42_AV35
IO_T2U_N12_42_AY35
IO_L18N_T2U_N11_AD2N_42_AY39
IO_L18P_T2U_N10_AD2P_42_AY38
IO_L17N_T2U_N9_AD10N_42_AY40
IO_L17P_T2U_N8_AD10P_42_AW40
IO_L16N_T2U_N7_QBC_AD3N_42_BA36
IO_L16P_T2U_N6_QBC_AD3P_42_BA35
IO_L15N_T2L_N5_AD11N_42_BA40
IO_L15P_T2L_N4_AD11P_42_BA39
IO_L14N_T2L_N3_GC_42_BB37
IO_L14P_T2L_N2_GC_42_BB36
IO_L13N_T2L_N1_GC_QBC_42_BA37
IO_L13P_T2L_N0_GC_QBC_42_AY37
IO_T1U_N12_42_BC40
IO_L12N_T1U_N11_GC_42_BD38
IO_L12P_T1U_N10_GC_42_BC38
IO_L11N_T1U_N9_GC_42_BB39
IO_L11P_T1U_N8_GC_42_BB38
IO_L10N_T1U_N7_QBC_AD4N_42_BF39
IO_L10P_T1U_N6_QBC_AD4P_42_BE39
IO_L9N_T1L_N5_AD12N_42_BD40
IO_L9P_T1L_N4_AD12P_42_BC39
IO_L8N_T1L_N3_AD5N_42_BE38
IO_L8P_T1L_N2_AD5P_42_BD37
IO_L7N_T1L_N1_QBC_AD13N_42_BF40
IO_L7P_T1L_N0_QBC_AD13P_42_BE40
IO_T0U_N12_VRP_42_BB34
IO_L6N_T0U_N11_AD6N_42_BF37
IO_L6P_T0U_N10_AD6P_42_BF36
IO_L5N_T0U_N9_AD14N_42_BE37
IO_L5P_T0U_N8_AD14P_42_BD36
IO_L4N_T0U_N7_DBC_AD7N_42_BF35
IO_L4P_T0U_N6_DBC_AD7P_42_BE35
IO_L3N_T0L_N5_AD15N_42_BC36
IO_L3P_T0L_N4_AD15P_42_BC35
IO_L2N_T0L_N3_42_BF34
IO_L2P_T0L_N2_42_BE34
IO_L1N_T0L_N1_DBC_42_BD35
IO_L1P_T0L_N0_DBC_42_BC34
VREF_42_AU36

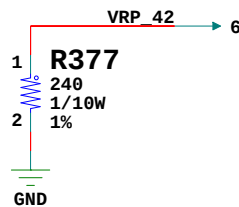
AU37 GPIO_LED5 55
AV39 DDR4_C2_DQ63 29
AV38 DDR4_C2_DQ62 29
AU39 DDR4_C2_DQ61 29
AU38 DDR4_C2_DQ60 29
AW38 DDR4_C2_DQS7_C 29
AW37 DDR4_C2_DQS7_T 29
AV40 DDR4_C2_DQ59 29
AU40 DDR4_C2_DQ58 29
AW36 DDR4_C2_DQ57 29
AW35 DDR4_C2_DQ56 29
AV36 GPIO_LED6 55
AV35 DDR4_C2_DM7 29
AY35 DDR4_C2_TEN 29
AY39 DDR4_C2_DQ55 28-,29-,30
AY38 DDR4_C2_DQ54 29
AY40 DDR4_C2_DQ53 29
AW40 DDR4_C2_DQ52 29
BA36 DDR4_C2_DQ56_C 29
BA35 DDR4_C2_DQ56_T 29
BA40 DDR4_C2_DQ51 29
BA39 DDR4_C2_DQ50 29
BB37 DDR4_C2_DQ49 29
BB36 DDR4_C2_DQ48 29
BA37 GPIO_LED7 55
AY37 DDR4_C2_DM6 29
BC40 FAN_OT_LS_B 13
BD38 DDR4_C2_DQ47 29
BC38 DDR4_C2_DQ46 29
BB39 DDR4_C2_DQ45 29
BB38 DDR4_C2_DQ44 29
BF39 DDR4_C2_DQS5_C 29
BE39 DDR4_C2_DQS5_T 29
BD40 DDR4_C2_DQ43 29
BC39 DDR4_C2_DQ42 29
BE38 DDR4_C2_DQ41 29
BD37 DDR4_C2_DQ40 29
BF40 MAXIM_CABLE_LS_B 56
BE40 DDR4_C2_DM5 29
BB34 VRP_42 6
BF37 DDR4_C2_DQ39 29
BF36 DDR4_C2_DQ38 29
BE37 DDR4_C2_DQ37 29
BD36 DDR4_C2_DQ36 29
BF35 DDR4_C2_DQ54_C 29
BE35 DDR4_C2_DQ54_T 29
BC36 DDR4_C2_DQ35 29
BC35 DDR4_C2_DQ34 29
BF34 DDR4_C2_DQ33 29
BE34 DDR4_C2_DQ32 29
BD35 DDR4_C2_RESET_B 28-,29-,30
BC34 DDR4_C2_DM4 29
AU36 NC

VCC1V2_FPGA

AU35
AW39
AY36
BB40
BC37
BD34
BF38
VCCO_42_AU35
VCCO_42_AW39
VCCO_42_AY36
VCCO_42_BB40
VCCO_42_BC37
VCCO_42_BD34
VCCO_42_BF38

U1

SOC_2104_1MM_IRON



FPGA Banks 40 42 C2 DDR4 Data



TITLE: FPGA Banks 40 42 C2 DDR4 Data SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0		ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59		VER:	1.0
SHEET SIZE: B		REV:	01
SHEET	6 OF 76	DRAWN BY: DN	

Bank 41 HP

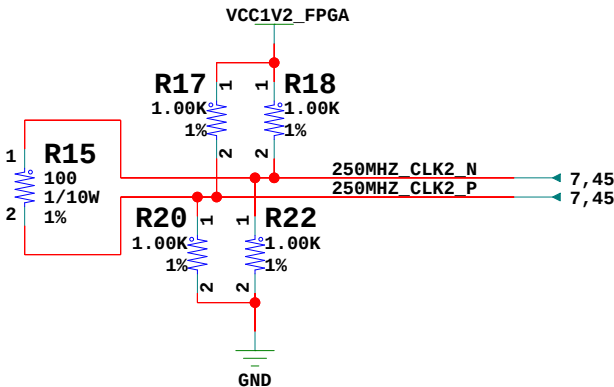
SOC_VU9P_FLGA2104_IRONWOOD

BANK 41
XCVU9PFLGA2104

IO_T3U_N12_41_AN25	
IO_L24N_T3U_N11_41_AM27	
IO_L24P_T3U_N10_41_AL27	
IO_L23N_T3U_N9_41_AP26	
IO_L23P_T3U_N8_41_AP25	
IO_L22N_T3U_N7_DBC_AD0N_41_AN28	
IO_L22P_T3U_N6_DBC_AD0P_41_AM28	
IO_L21N_T3L_N5_AD8N_41_AP28	
IO_L21P_T3L_N4_AD8P_41_AP27	
IO_L20N_T3L_N3_AD1N_41_AN26	
IO_L20P_T3L_N2_AD1P_41_AM26	
IO_L19N_T3L_N1_DBC_AD9N_41_AR28	
IO_L19P_T3L_N0_DBC_AD9P_41_AR27	
IO_T2U_N12_41_AV25	
IO_L18N_T2U_N11_AD2N_41_AT25	
IO_L18P_T2U_N10_AD2P_41_AR25	
IO_L17N_T2U_N9_AD10N_41_AU28	
IO_L17P_T2U_N8_AD10P_41_AU27	
IO_L16N_T2U_N7_QBC_AD3N_41_AT27	
IO_L16P_T2U_N6_QBC_AD3P_41_AT26	
IO_L15N_T2L_N5_AD11N_41_AW28	
IO_L15P_T2L_N4_AD11P_41_AV28	
IO_L14N_T2L_N3_GC_41_AV26	
IO_L14P_T2L_N2_GC_41_AU26	
IO_L13N_T2L_N1_GC_QBC_41_AW27	
IO_L13P_T2L_N0_GC_QBC_41_AW26	
IO_T1U_N12_41_AY29	
IO_L12N_T1U_N11_GC_41_BB27	
IO_L12P_T1U_N10_GC_41_BA27	
IO_L11N_T1U_N9_GC_41_AY28	
IO_L11P_T1U_N8_GC_41_AY27	
IO_L10N_T1U_N7_QBC_AD4N_41_BB26	
IO_L10P_T1U_N6_QBC_AD4P_41_BA26	
IO_L9N_T1L_N5_AD12N_41_BC28	
IO_L9P_T1L_N4_AD12P_41_BB28	
IO_L8N_T1L_N3_AD5N_41_BC26	
IO_L8P_T1L_N2_AD5P_41_BC25	
IO_L7N_T1L_N1_QBC_AD13N_41_BB29	
IO_L7P_T1L_N0_QBC_AD13P_41_BA29	
IO_T0U_N12_VRP_41_BC29	
IO_L6N_T0U_N11_AD6N_41_BD26	
IO_L6P_T0U_N10_AD6P_41_BD25	
IO_L5N_T0U_N9_AD14N_41_BE27	
IO_L5P_T0U_N8_AD14P_41_BD27	
IO_L4N_T0U_N7_DBC_AD7N_41_BF25	
IO_L4P_T0U_N6_DBC_AD7P_41_BE25	
IO_L3N_T0L_N5_AD15N_41_BE28	
IO_L3P_T0L_N4_AD15P_41_BD28	
IO_L2N_T0L_N3_41_BF27	
IO_L2P_T0L_N2_41_BF26	
IO_L1N_T0L_N1_DBC_41_BF29	
IO_L1P_T0L_N0_DBC_41_BE29	
VREF_41_AL26	

AN25	DDR4_C2_ACT_B	28-,29-,30
AM27	DDR4_C2_A0	28-,29-,30
AL27	DDR4_C2_A1	28-,29-,30
AP26	DDR4_C2_A2	28-,29-,30
AP25	DDR4_C2_A3	28-,29-,30
AN28	DDR4_C2_A4	28-,29-,30
AM28	DDR4_C2_A5	28-,29-,30
AP28	DDR4_C2_A6	28-,29-,30
AP27	DDR4_C2_A7	28-,29-,30
AN26	DDR4_C2_A8	28-,29-,30
AM26	DDR4_C2_A9	28-,29-,30
AR28	DDR4_C2_A10	28-,29-,30
AR27	DDR4_C2_A11	28-,29-,30
AV25	DDR4_C2_A12	28-,29-,30
AT25	DDR4_C2_A13	28-,29-,30
AR25	DDR4_C2_BA0	28-,29-,30
AU28	DDR4_C2_BA1	28-,29-,30
AU27	DDR4_C2_B00	28-,29-,30
AT27	DDR4_C2_CK_C	28-,29-,30
AT26	DDR4_C2_CK_T	28-,29-,30
AW28	DDR4_C2_CKE	28-,29-,30
AV28	DDR4_C2_A14_WE_B	28-,29-,30
AV26	DDR4_C2_A16_RAS_B	28-,29-,30
AU26	DDR4_C2_A15_CAS_B	28-,29-,30
AW27	250MHZ_CLK2_N	7,45
AW26	250MHZ_CLK2_P	7,45
AY29	DDR4_C2_CS_B	28-,29-,30
BB27	DDR4_C2_DQ79	30
BA27	DDR4_C2_DQ78	30
AY28	DDR4_C2_DQ77	30
AY27	DDR4_C2_DQ76	30
BB26	DDR4_C2_DQS9_C	30
BA26	DDR4_C2_DQS9_T	30
BC28	DDR4_C2_DQ75	30
BB28	DDR4_C2_DQ74	30
BC26	DDR4_C2_DQ73	30
BC25	DDR4_C2_DQ72	30
BB29	DDR4_C2_ODT	28-,29-,30
BA29	DDR4_C2_DM9	30
BC29	VRP_41	7
BD26	DDR4_C2_DQ71	30
BD25	DDR4_C2_DQ70	30
BE27	DDR4_C2_DQ69	30
BD27	DDR4_C2_DQ68	30
BF25	DDR4_C2_DQS8_C	30
BE25	DDR4_C2_DQS8_T	30
BE28	DDR4_C2_DQ67	30
BD28	DDR4_C2_DQ66	30
BF27	DDR4_C2_DQ65	30
BF26	DDR4_C2_DQ64	30
BF29	DDR4_C2_PAR	28-,29-,30
BE29	DDR4_C2_DM8	30
AL26	NC	30

PCB LAYOUT DIRECTIVE:
TERMINATION RESISTOR
AND BIAS RESISTOR
NETWORK TO BE PLACED
IN "FLY-BY" CONFIGURATION

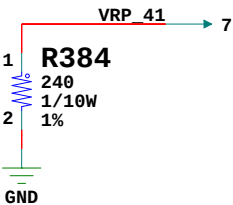


VCC1V2_FPGA

AN27	VCC0_41_AN27
AT28	VCC0_41_AT28
AU25	VCC0_41_AU25
AY26	VCC0_41_AY26
BC27	VCC0_41_BC27
BF28	VCC0_41_BF28

U1

SOC_2104_1MM_IRON



FPGA Bank 41 C2 DDR4 Addr Ctrl

TITLE: FPGA Bank 41 C2 DDR4 Addr Ctrl SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 7 OF 76	DRAWN BY: DN

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

Bank 46 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 46
XCVU9PFLGA2104

IO_T3U_N12_46_D36
IO_L24N_T3U_N11_46_A35
IO_L24P_T3U_N10_46_A34
IO_L23N_T3U_N9_46_A36
IO_L23P_T3U_N8_46_B35
IO_L22N_T3U_N7_DBC_AD0N_46_B37
IO_L22P_T3U_N6_DBC_AD0P_46_B36
IO_L21N_T3L_N5_AD8N_46_C34
IO_L21P_T3L_N4_AD8P_46_D34
IO_L20N_T3L_N3_AD1N_46_C35
IO_L20P_T3L_N2_AD1P_46_D35
IO_L19N_T3L_N1_DBC_AD9N_46_C37
IO_L19P_T3L_N0_DBC_AD9P_46_D37
IO_T2U_N12_46_C38
IO_L18N_T2U_N11_AD2N_46_A38
IO_L18P_T2U_N10_AD2P_46_B38
IO_L17N_T2U_N9_AD10N_46_C40
IO_L17P_T2U_N8_AD10P_46_D40
IO_L16N_T2U_N7_QBC_AD3N_46_A40
IO_L16P_T2U_N6_QBC_AD3P_46_A39
IO_L15N_T2L_N5_AD11N_46_B40
IO_L15P_T2L_N4_AD11P_46_C39
IO_L14N_T2L_N3_GC_46_E38
IO_L14P_T2L_N2_GC_46_E37
IO_L13N_T2L_N1_GC_QBC_46_D39
IO_L13P_T2L_N0_GC_QBC_46_E39
IO_T1U_N12_46_E36
IO_L12N_T1U_N11_GC_46_G37
IO_L12P_T1U_N10_GC_46_G36
IO_L11N_T1U_N9_GC_46_F36
IO_L11P_T1U_N8_GC_46_F35
IO_L10N_T1U_N7_QBC_AD4N_46_G35
IO_L10P_T1U_N6_QBC_AD4P_46_H34
IO_L9N_T1L_N5_AD12N_46_H37
IO_L9P_T1L_N4_AD12P_46_J36
IO_L8N_T1L_N3_AD5N_46_H35
IO_L8P_T1L_N2_AD5P_46_J35
IO_L7N_T1L_N1_QBC_AD13N_46_E34
IO_L7P_T1L_N0_QBC_AD13P_46_F34
IO_T0U_N12_VRP_46_K38
IO_L6N_T0U_N11_AD6N_46_F39
IO_L6P_T0U_N10_AD6P_46_F38
IO_L5N_T0U_N9_AD14N_46_J37
IO_L5P_T0U_N8_AD14P_46_K37
IO_L4N_T0U_N7_DBC_AD7N_46_G38
IO_L4P_T0U_N6_DBC_AD7P_46_H38
IO_L3N_T0L_N5_AD15N_46_F40
IO_L3P_T0L_N4_AD15P_46_G40
IO_L2N_T0L_N3_46_H40
IO_L2P_T0L_N2_46_H39
IO_L1N_T0L_N1_DBC_46_J40
IO_L1P_T0L_N0_DBC_46_J39
VREF_46_J34

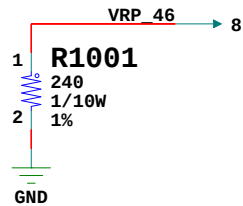
D36 NC
A35 RLD3 C3 72B DM1
A34 RLD3 C3 72B DQ35
A36 RLD3 C3 72B DQ34
B35 RLD3 C3 72B DQ33
B37 RLD3 C3 72B DQ32
B36 RLD3 C3 72B DQ31
C34 RLD3 C3 72B DQ30
D34 RLD3 C3 72B DQ29
C35 RLD3 C3 72B DQ28
D35 RLD3 C3 72B DQ27
C37 RLD3 C3 72B QK3 N
D37 RLD3 C3 72B QK3 P
C38 NC
A38 RLD3 C3 72B QVLD1
B38 RLD3 C3 72B DQ26
C40 RLD3 C3 72B DQ25
D40 RLD3 C3 72B DQ24
A40 RLD3 C3 72B DQ23
A39 RLD3 C3 72B DQ22
B40 RLD3 C3 72B DQ21
C39 RLD3 C3 72B DQ20
E38 RLD3 C3 72B DQ19
E37 RLD3 C3 72B DQ18
D39 RLD3 C3 72B QK2 N
E39 RLD3 C3 72B QK2 P
E36 NC
G37 RLD3 C3 72B QVLD0
G36 RLD3 C3 72B DQ17
F36 RLD3 C3 72B DQ16
F35 RLD3 C3 72B DQ15
G35 RLD3 C3 72B DQ14
H34 RLD3 C3 72B DQ13
H37 RLD3 C3 72B DQ12
J36 RLD3 C3 72B DQ11
H35 RLD3 C3 72B DQ10
J35 RLD3 C3 72B DQ9
E34 RLD3 C3 72B QK1 N
F34 RLD3 C3 72B QK1 P
K38 VRP 46
F39 RLD3 C3 72B DM0
F38 RLD3 C3 72B DQ8
J37 RLD3 C3 72B DQ7
K37 RLD3 C3 72B DQ6
G38 RLD3 C3 72B DQ5
H38 RLD3 C3 72B DQ4
F40 RLD3 C3 72B DQ3
G40 RLD3 C3 72B DQ2
H40 RLD3 C3 72B DQ1
H39 RLD3 C3 72B DQ0
J40 RLD3 C3 72B QK0 N
J39 RLD3 C3 72B QK0 P
J34 NC

VCC1V2_FPGA

D38 VCC0_46_D38
E35 VCC0_46_E35
G39 VCC0_46_G39
A37 VCC0_46_A37
B34 VCC0_46_B34
H36 VCC0_46_H36

U1

SOC_2104_1MM_IRON



Bank 48 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 48
XCVU9PFLGA2104

IO_T3U_N12_48_A25
IO_L24N_T3U_N11_48_B25
IO_L24P_T3U_N10_48_C25
IO_L23N_T3U_N9_48_D26
IO_L23P_T3U_N8_48_D25
IO_L22N_T3U_N7_DBC_AD0N_48_A26
IO_L22P_T3U_N6_DBC_AD0P_48_B26
IO_L21N_T3L_N5_AD8N_48_B27
IO_L21P_T3L_N4_AD8P_48_C27
IO_L20N_T3L_N3_AD1N_48_A28
IO_L20P_T3L_N2_AD1P_48_B28
IO_L19N_T3L_N1_DBC_AD9N_48_C28
IO_L19P_T3L_N0_DBC_AD9P_48_D27
IO_T2U_N12_48_H25
IO_L18N_T2U_N11_AD2N_48_F25
IO_L18P_T2U_N10_AD2P_48_G25
IO_L17N_T2U_N9_AD10N_48_E27
IO_L17P_T2U_N8_AD10P_48_E26
IO_L16N_T2U_N7_QBC_AD3N_48_G28
IO_L16P_T2U_N6_QBC_AD3P_48_H28
IO_L15N_T2L_N5_AD11N_48_E28
IO_L15P_T2L_N4_AD11P_48_F28
IO_L14N_T2L_N3_GC_48_G27
IO_L14P_T2L_N2_GC_48_H27
IO_L13N_T2L_N1_GC_QBC_48_F26
IO_L13P_T2L_N0_GC_QBC_48_G26
IO_T1U_N12_48_J25
IO_L12N_T1U_N11_GC_48_J27
IO_L12P_T1U_N10_GC_48_K27
IO_L11N_T1U_N9_GC_48_J26
IO_L11P_T1U_N8_GC_48_K26
IO_L10N_T1U_N7_QBC_AD4N_48_L25
IO_L10P_T1U_N6_QBC_AD4P_48_L24
IO_L9N_T1L_N5_AD12N_48_K28
IO_L9P_T1L_N4_AD12P_48_L28
IO_L8N_T1L_N3_AD5N_48_L26
IO_L8P_T1L_N2_AD5P_48_M25
IO_L7N_T1L_N1_QBC_AD13N_48_M28
IO_L7P_T1L_N0_QBC_AD13P_48_M27
IO_T0U_N12_VRP_48_M26
IO_L6N_T0U_N11_AD6N_48_N24
IO_L6P_T0U_N10_AD6P_48_P24
IO_L5N_T0U_N9_AD14N_48_N27
IO_L5P_T0U_N8_AD14P_48_P26
IO_L4N_T0U_N7_DBC_AD7N_48_N25
IO_L4P_T0U_N6_DBC_AD7P_48_P25
IO_L3N_T0L_N5_AD15N_48_P27
IO_L3P_T0L_N4_AD15P_48_R27
IO_L2N_T0L_N3_48_R24
IO_L2P_T0L_N2_48_T24
IO_L1N_T0L_N1_DBC_48_R26
IO_L1P_T0L_N0_DBC_48_T26
VREF_48_T25

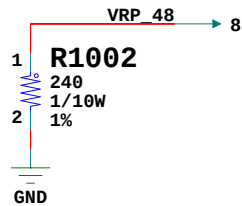
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B25 RLD3 C3 72B DM3
C25 RLD3 C3 72B DQ71
D26 RLD3 C3 72B DQ70
D25 RLD3 C3 72B DQ69
A26 RLD3 C3 72B DQ68
B26 RLD3 C3 72B DQ67
B27 RLD3 C3 72B DQ66
C27 RLD3 C3 72B DQ65
A28 RLD3 C3 72B DQ64
B28 RLD3 C3 72B DQ63
C28 RLD3 C3 72B QK7 N
D27 RLD3 C3 72B QK7 P
H25 NC
F25 RLD3 C3 72B QVLD3
G25 RLD3 C3 72B DQ62
E27 RLD3 C3 72B DQ61
E26 RLD3 C3 72B DQ60
G28 RLD3 C3 72B DQ59
H28 RLD3 C3 72B DQ58
E28 RLD3 C3 72B DQ57
F28 RLD3 C3 72B DQ56
G27 RLD3 C3 72B DQ55
H27 RLD3 C3 72B DQ54
F26 RLD3 C3 72B QK6 N
G26 RLD3 C3 72B QK6 P
J25 NC
J27 RLD3 C3 72B QVLD2
K27 RLD3 C3 72B DQ53
J26 RLD3 C3 72B DQ52
K26 RLD3 C3 72B DQ51
L25 RLD3 C3 72B DQ50
L24 RLD3 C3 72B DQ49
K28 RLD3 C3 72B DQ48
L28 RLD3 C3 72B DQ47
L26 RLD3 C3 72B DQ46
M25 RLD3 C3 72B DQ45
M28 RLD3 C3 72B QK5 N
M27 RLD3 C3 72B QK5 P
M26 VRP 48
N24 RLD3 C3 72B DM2
P24 RLD3 C3 72B DQ44
N27 RLD3 C3 72B DQ43
P26 RLD3 C3 72B DQ42
N25 RLD3 C3 72B DQ41
P25 RLD3 C3 72B DQ40
P27 RLD3 C3 72B DQ39
R27 RLD3 C3 72B DQ38
R24 RLD3 C3 72B DQ37
T24 RLD3 C3 72B DQ36
R26 RLD3 C3 72B QK4 N
T26 RLD3 C3 72B QK4 P
T25 NC

VCC1V2_FPGA

A27 VCC0_48_A27
D28 VCC0_48_D28
E25 VCC0_48_E25
H26 VCC0_48_H26
L27 VCC0_48_L27
M24 VCC0_48_M24
R25 VCC0_48_R25

U1

SOC_2104_1MM_IRON



FPGA Banks 46 48 C3 RLD3 Data

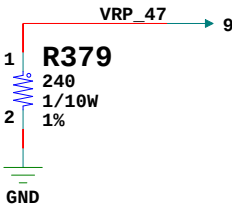
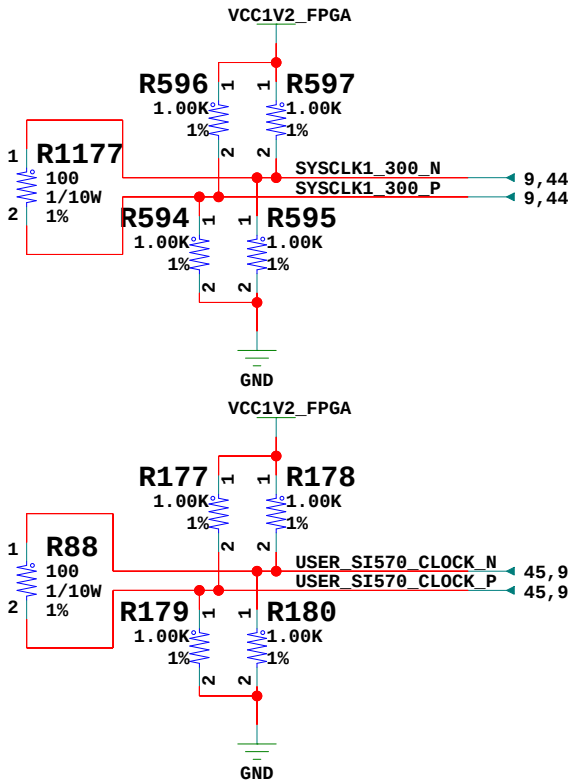
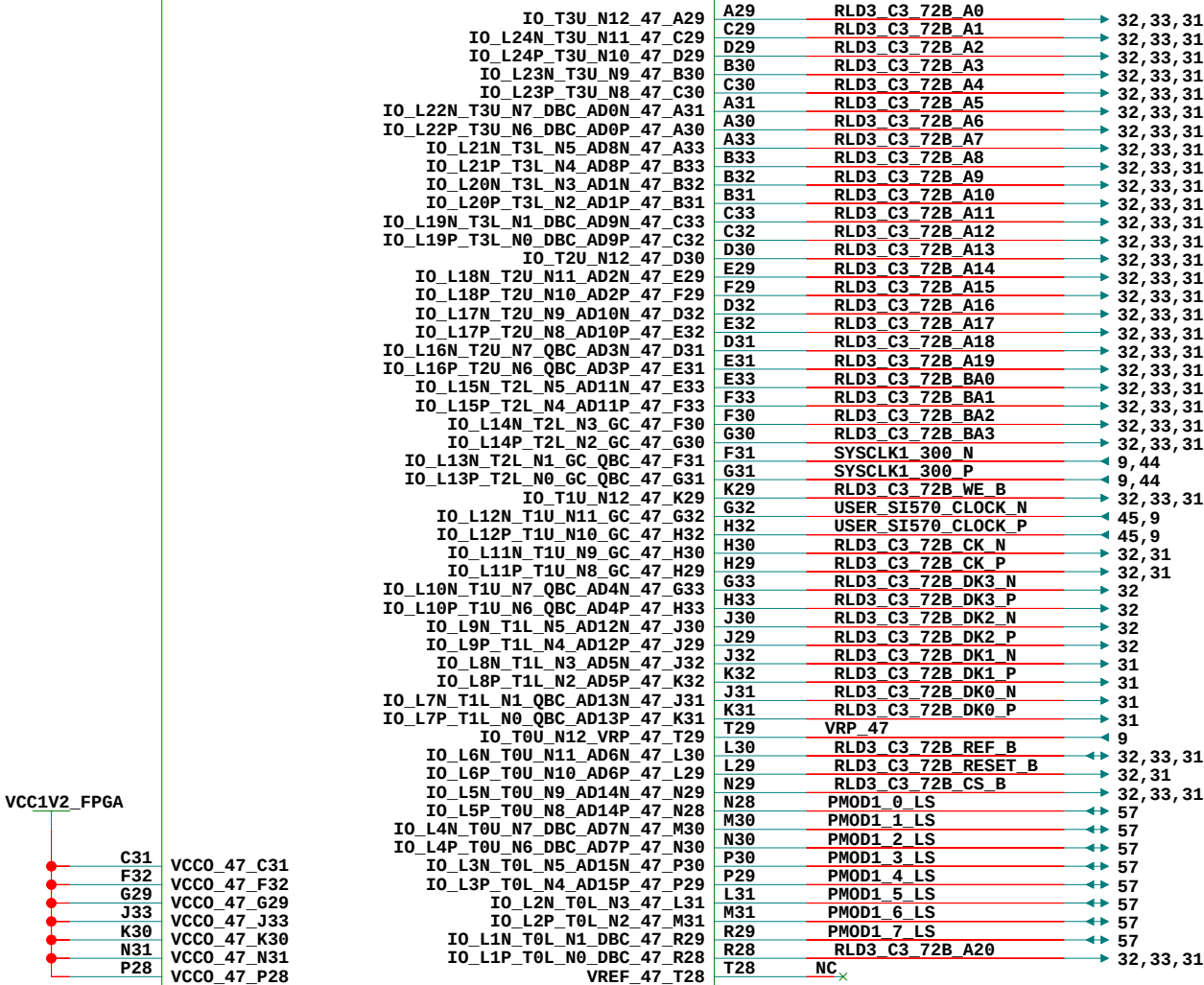


TITLE: FPGA Banks 46 48 C3 RLD3 Data
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 8 OF 76	DRAWN BY: DN

SOC_VU9P_FLGA2104_IRONWOOD

BANK 47
XCVU9PFLGA2104



FPGA Bank 47 C3 RLD3 Addr Ctrl

TITLE: FPGA Bank 47 C3 RLD3 Addr Ctrl		ASSY P/N: 0432001	
SCHEM, ROHS COMPLIANT		PCB P/N: 1280906	
HW-U1-VCU118_REV1_0		SCH P/N: 0381739	
		TEST P/N: TSS0185	
DATE: 11/22/2016:17:59		VER:	1.0
SHEET SIZE: B		REV:	01
SHEET	9	OF	76
		DRAWN BY:	DN

Bank 64 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 64
XCVU9PFLGA2104

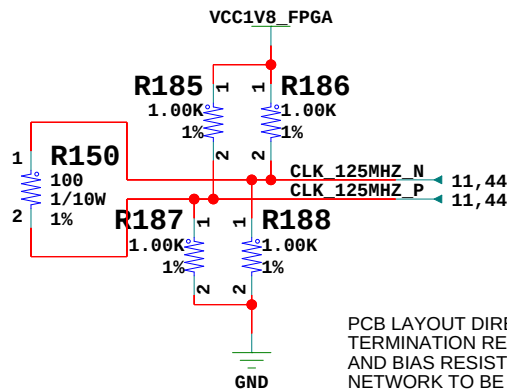
IO_T3U_N12_64_AL25	AL25	IIC_MUX_RESET_B	58
IO_L24N_T3U_N11_64_AM24	AM24	IIC_MAIN_SCL	58,47
IO_L24P_T3U_N10_64_AL24	AL24	IIC_MAIN_SDA	58,47
IO_L23N_T3U_N9_64_AM21	AM21	QSFP1_MODSELL_LS	56
IO_L23P_T3U_N8_64_AL21	AL21	QSFP1_MODPRSL_LS	56
IO_L22N_T3U_N7_DBC_AD0N_64_AM22	AM22	QSFP1_RECCLK_N	56
IO_L22P_T3U_N6_DBC_AD0P_64_AM23	AM23	QSFP1_RECCLK_P	51
IO_L21N_T3L_N5_AD8N_64_AP21	AP21	QSFP1_INTL_LS	56
IO_L21P_T3L_N4_AD8P_64_AN21	AN21	QSFP1_LPMODE_LS	56
IO_L20N_T3L_N3_AD1N_64_AN23	AN23	QSFP2_MODSELL_LS	56
IO_L20P_T3L_N2_AD1P_64_AN24	AN24	QSFP2_MODPRSL_LS	56
IO_L19N_T3L_N1_DBC_AD9N_64_AP22	AP22	QSFP2_RECCLK_N	56
IO_L19P_T3L_N0_DBC_AD9P_64_AP23	AP23	QSFP2_RECCLK_P	51
IO_T2U_N12_64_AT21	AT21	QSFP2_INTL_LS	51
IO_L18N_T2U_N11_AD2N_64_AT24	AT24	QSFP2_LPMODE_LS	56
IO_L18P_T2U_N10_AD2P_64_AR24	AR24	PHY1_PDWN_B_I_INT_B_0	56
IO_L17N_T2U_N9_AD10N_64_AR22	AR22	PHY1_GPIO_0	52
IO_L17P_T2U_N8_AD10P_64_AR23	AR23	PHY1_MDIO	52
IO_L16N_T2U_N7_QBC_AD3N_64_AV24	AV24	PHY1_SGMII_OUT_N	52
IO_L16P_T2U_N6_QBC_AD3P_64_AU24	AU24	PHY1_SGMII_OUT_P	52
IO_L15N_T2L_N5_AD11N_64_AV21	AV21	PHY1_SGMII_IN_N	52
IO_L15P_T2L_N4_AD11P_64_AU21	AU21	PHY1_SGMII_IN_P	52
IO_L14N_T2L_N3_GC_64_AV23	AV23	PHY1_MDC	52
IO_L14P_T2L_N2_GC_64_AU23	AU23	PHY1_CLKOUT	52
IO_L13N_T2L_N1_GC_QBC_64_AU22	AU22	PHY1_SGMII_CLK_N	52
IO_L13P_T2L_N0_GC_QBC_64_AT22	AT22	PHY1_SGMII_CLK_P	52
IO_T1U_N12_64_BA21	BA21	PHY1_RESET_B	52
IO_L12N_T1U_N11_GC_64_AW22	AW22	USER_SI570_CLOCK1_N	52
IO_L12P_T1U_N10_GC_64_AW23	AW23	USER_SI570_CLOCK1_P	11,44
IO_L11N_T1U_N9_GC_64_AY23	AY23	CLK_125MHZ_N	11,44
IO_L11P_T1U_N8_GC_64_AY24	AY24	CLK_125MHZ_P	11,44
IO_L10N_T1U_N7_QBC_AD4N_64_BA22	BA22	QSFP1_RESETL_LS	56
IO_L10P_T1U_N6_QBC_AD4P_64_AY22	AY22	QSFP2_RESETL_LS	56
IO_L9N_T1L_N5_AD12N_64_AY25	AY25	USB_UART_RTS	56
IO_L9P_T1L_N4_AD12P_64_AW25	AW25	USB_UART_TX	53
IO_L8N_T1L_N3_AD5N_64_BB21	BB21	USB_UART_RX	53
IO_L8P_T1L_N2_AD5P_64_BB22	BB22	USB_UART_CTS	53
IO_L7N_T1L_N1_QBC_AD13N_64_BA24	BA24	SYSCTLR_GPIO_7	53
IO_L7P_T1L_N0_QBC_AD13P_64_BA25	BA25	SYSCTLR_GPIO_6	47
IO_T0U_N12_VRP_64_BC24	BC24	PCIE_WAKE_B_LS	47
IO_L6N_T0U_N11_AD6N_64_BD21	BD21	SYSCTLR_GPIO_5	56
IO_L6P_T0U_N10_AD6P_64_BC21	BC21	SI5328_RST_LS_B	47
IO_L5N_T0U_N9_AD14N_64_BB23	BB23	PMBUS_ALERT_FPGA	56
IO_L5P_T0U_N8_AD14P_64_BB24	BB24	GPIO_SW_N	55,47
IO_L4N_T0U_N7_DBC_AD7N_64_BF22	BF22	GPIO_SW_W	55,47
IO_L4P_T0U_N6_DBC_AD7P_64_BE22	BE22	GPIO_SW_S	55,47
IO_L3N_T0L_N5_AD15N_64_BE23	BE23	GPIO_SW_E	55,47
IO_L3P_T0L_N4_AD15P_64_BD23	BD23	GPIO_SW_C	55,47
IO_L2N_T0L_N3_64_BD22	BD22	FIREFLY_MODPRS_LS_B	55,47
IO_L2P_T0L_N2_64_BC23	BC23	FIREFLY_MODSEL_LS_B	56
IO_L1N_T0L_N1_DBC_64_BF24	BF24	FIREFLY_INT_LS_B	56
IO_L1P_T0L_N0_DBC_64_BE24	BE24	FIREFLY_RESET_LS_B	56
VREF_64_AL22	AL22	NC	56

VCC1V8_FPGA

AL23	VCC0_64_AL23
AP24	VCC0_64_AP24
AR21	VCC0_64_AR21
AV22	VCC0_64_AV22
BA23	VCC0_64_BA23
BB20	VCC0_64_BB20
BD24	VCC0_64_BD24
BE21	VCC0_64_BE21

U1

SOC_2104_1MM_IRON



PCB LAYOUT DIRECTIVE:
TERMINATION RESISTOR
AND BIAS RESISTOR
NETWORK TO BE PLACED
IN "FLY-BY" CONFIGURATION

Layout: Place resistor and capacitor for VREF
Underneath the FPGA via array
right next to the via

Bank 70 HP

SOC_VU9P_FLGA2104_IRONWOOD

BANK 70
XCVU9PFLGA2104

IO_T3U_N12_70_J14	J14	NC	
IO_L24N_T3U_N11_70_J12	J12		
IO_L24P_T3U_N10_70_K12	K12		
IO_L23N_T3U_N9_70_K13	K13		
IO_L23P_T3U_N8_70_K14	K14		
IO_L22N_T3U_N7_DBC_AD0N_70_L11	L11		
IO_L22P_T3U_N6_DBC_AD0P_70_M11	M11		
IO_L21N_T3L_N5_AD8N_70_J11	J11		
IO_L21P_T3L_N4_AD8P_70_K11	K11		
IO_L20N_T3L_N3_AD1N_70_L13	L13		
IO_L20P_T3L_N2_AD1P_70_L14	L14		
IO_L19N_T3L_N1_DBC_AD9N_70_M12	M12		
IO_L19P_T3L_N0_DBC_AD9P_70_M13	M13		
IO_T2U_N12_70_N12	N12	NC	
IO_L18N_T2U_N11_AD2N_70_P12	P12		
IO_L18P_T2U_N10_AD2P_70_R12	R12		
IO_L17N_T2U_N9_AD10N_70_L15	L15		
IO_L17P_T2U_N8_AD10P_70_M15	M15		
IO_L16N_T2U_N7_QBC_AD3N_70_P11	P11		
IO_L16P_T2U_N6_QBC_AD3P_70_R11	R11		
IO_L15N_T2L_N5_AD11N_70_N15	N15		
IO_L15P_T2L_N4_AD11P_70_P15	P15		
IO_L14N_T2L_N3_GC_70_P14	P14		
IO_L14P_T2L_N2_GC_70_R14	R14		
IO_L13N_T2L_N1_GC_QBC_70_N13	N13		
IO_L13P_T2L_N0_GC_QBC_70_N14	N14		
IO_T1U_N12_70_R16	R16	NC	
IO_L12N_T1U_N11_GC_70_T13	T13		
IO_L12P_T1U_N10_GC_70_U13	U13		
IO_L11N_T1U_N9_GC_70_R13	R13		
IO_L11P_T1U_N8_GC_70_T14	T14		
IO_L10N_T1U_N7_QBC_AD4N_70_T11	T11		
IO_L10P_T1U_N6_QBC_AD4P_70_U11	U11		
IO_L9N_T1L_N5_AD12N_70_T15	T15		
IO_L9P_T1L_N4_AD12P_70_T16	T16		
IO_L8N_T1L_N3_AD5N_70_U16	U16		
IO_L8P_T1L_N2_AD5P_70_V16	V16		
IO_L7N_T1L_N1_QBC_AD13N_70_U15	U15		
IO_L7P_T1L_N0_QBC_AD13P_70_V15	V15		
IO_T0U_N12_VRP_70_W15	W15		
IO_L6N_T0U_N11_AD6N_70_V14	V14		
IO_L6P_T0U_N10_AD6P_70_W14	W14		
IO_L5N_T0U_N9_AD14N_70_Y12	Y12		
IO_L5P_T0U_N8_AD14P_70_AA12	AA12		
IO_L4N_T0U_N7_DBC_AD7N_70_U12	U12		
IO_L4P_T0U_N6_DBC_AD7P_70_V13	V13		
IO_L3N_T0L_N5_AD15N_70_V12	V12		
IO_L3P_T0L_N4_AD15P_70_W12	W12		
IO_L2N_T0L_N3_70_Y14	Y14		
IO_L2P_T0L_N2_70_AA14	AA14		
IO_L1N_T0L_N1_DBC_70_Y13	Y13		
IO_L1P_T0L_N0_DBC_70_AA13	AA13		
VREF_70_AB13	AB13		

VADJ_1V8_FPGA

J13	VCC0_70_J13
M14	VCC0_70_M14
N11	VCC0_70_N11
R15	VCC0_70_R15
T12	VCC0_70_T12
W13	VCC0_70_W13

U1

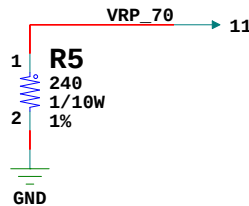
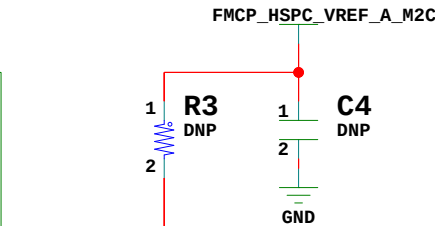
SOC_2104_1MM_IRON

FPGA Banks 64 GPIO 70 FMCP HSPC



TITLE: FPGA Banks 64 GPIO 70 FMCP HSPC SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185
--	--

DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 11 OF 76	DRAWN BY: DN



Layout: Place resistor and capacitor for VREF
Underneath the FPGA via array
right next to the via

Bank 47 HP

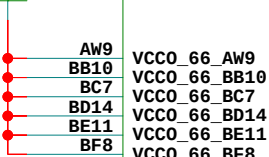
SOC_VU9P_FLGA2104_IRONWOOD

BANK 66
XCVU9PFLGA2104

IO_T3U_N12_66	BB11
IO_L24N_T3U_N11_66	BB12
IO_L24P_T3U_N10_66	BB13
IO_L23N_T3U_N9_66	BB14
IO_L23P_T3U_N8_66	BA14
IO_L22N_T3U_N7_DBC_AD0N_66	BA15
IO_L22P_T3U_N6_DBC_AD0P_66	BA16
IO_L21N_T3L_N5_AD8N_66	BD15
IO_L21P_T3L_N4_AD8P_66	BC15
IO_L20N_T3L_N3_AD1N_66	BC16
IO_L20P_T3L_N2_AD1P_66	BB16
IO_L19N_T3L_N1_DBC_AD9N_66	BC13
IO_L19P_T3L_N0_DBC_AD9P_66	BC14
IO_T2U_N12_66	BA10
IO_L18N_T2U_N11_AD2N_66	AW7
IO_L18P_T2U_N10_AD2P_66	AW8
IO_L17N_T2U_N9_AD10N_66	AY7
IO_L17P_T2U_N8_AD10P_66	AY8
IO_L16N_T2U_N7_QBC_AD3N_66	AV8
IO_L16P_T2U_N6_QBC_AD3P_66	AV9
IO_L15N_T2L_N5_AD11N_66	BB7
IO_L15P_T2L_N4_AD11P_66	BA7
IO_L14N_T2L_N3_GC_66	BB8
IO_L14P_T2L_N2_GC_66	BB9
IO_L13N_T2L_N1_GC_QBC_66	BA9
IO_L13P_T2L_N0_GC_QBC_66	AY9
IO_T1U_N12_66	BF7
IO_L12N_T1U_N11_GC_66	BC8
IO_L12P_T1U_N10_GC_66	BC9
IO_L11N_T1U_N9_GC_66	BD10
IO_L11P_T1U_N8_GC_66	BC10
IO_L10N_T1U_N7_QBC_AD4N_66	BF9
IO_L10P_T1U_N6_QBC_AD4P_66	BF10
IO_L9N_T1L_N5_AD12N_66	BE9
IO_L9P_T1L_N4_AD12P_66	BE10
IO_L8N_T1L_N3_AD5N_66	BE7
IO_L8P_T1L_N2_AD5P_66	BE8
IO_L7N_T1L_N1_QBC_AD13N_66	BD7
IO_L7P_T1L_N0_QBC_AD13P_66	BD8
IO_T0U_N12_VRP_66	BD16
IO_L6N_T0U_N11_AD6N_66	BF15
IO_L6P_T0U_N10_AD6P_66	BE15
IO_L5N_T0U_N9_AD14N_66	BF14
IO_L5P_T0U_N8_AD14P_66	BE14
IO_L4N_T0U_N7_DBC_AD7N_66	BE13
IO_L4P_T0U_N6_DBC_AD7P_66	BD13
IO_L3N_T0L_N5_AD15N_66	BD11
IO_L3P_T0L_N4_AD15P_66	BC11
IO_L2N_T0L_N3_66	BF11
IO_L2P_T0L_N2_66	BF12
IO_L1N_T0L_N1_DBC_66	BE12
IO_L1P_T0L_N0_DBC_66	BD12
VREF_66	BA11

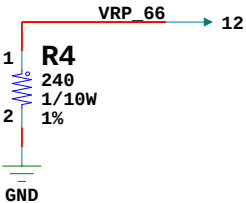
BB11	NC
BB12	FMC HPC1 LA10 N
BB13	FMC HPC1 LA10 P
BB14	FMC HPC1 LA09 N
BA14	FMC HPC1 LA09 P
BA15	FMC HPC1 LA11 N
BA16	FMC HPC1 LA11 P
BD15	FMC HPC1 LA07 N
BC15	FMC HPC1 LA07 P
BC16	FMC HPC1 LA15 N
BB16	FMC HPC1 LA15 P
BC13	FMC HPC1 LA12 N
BC14	FMC HPC1 LA12 P
BA10	NC
AW7	FMC HPC1 LA14 N
AW8	FMC HPC1 LA14 P
AY7	FMC HPC1 LA13 N
AY8	FMC HPC1 LA13 P
AV8	FMC HPC1 LA16 N
AV9	FMC HPC1 LA16 P
BB7	FMC HPC1 PRSNT M2C B LS
BA7	FMC HPC1 PG M2C LS
BB8	NC
BB9	NC
BA9	FMC HPC1 LA00 CC N
AY9	FMC HPC1 LA00 CC P
BF7	NC
BC8	FMC HPC1 CLK0 M2C N
BC9	FMC HPC1 CLK0 M2C P
BD10	NC
BC10	NC
BF9	FMC HPC1 LA01 CC N
BF10	FMC HPC1 LA01 CC P
BE9	NC
BE10	NC
BE7	NC
BE8	NC
BD7	NC
BD8	NC
BD16	VRP_66
BF15	FMC HPC1 LA08 N
BE15	FMC HPC1 LA08 P
BF14	FMC HPC1 LA05 N
BE14	FMC HPC1 LA05 P
BE13	FMC HPC1 LA06 N
BD13	FMC HPC1 LA06 P
BD11	FMC HPC1 LA02 N
BC11	FMC HPC1 LA02 P
BF11	FMC HPC1 LA04 N
BF12	FMC HPC1 LA04 P
BE12	FMC HPC1 LA03 N
BD12	FMC HPC1 LA03 P
BA11	VREF_66

VADJ_1V8_FPGA



U1

SOC_2104_1MM_IRON



Bank 48 HP

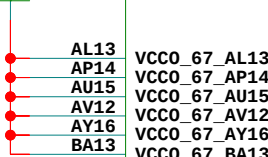
SOC_VU9P_FLGA2104_IRONWOOD

BANK 67
XCVU9PFLGA2104

IO_T3U_N12_67	AM16
IO_L24N_T3U_N11_67	AK13
IO_L24P_T3U_N10_67	AK14
IO_L23N_T3U_N9_67	AM12
IO_L23P_T3U_N8_67	AM13
IO_L22N_T3U_N7_DBC_AD0N_67	AJ12
IO_L22P_T3U_N6_DBC_AD0P_67	AJ13
IO_L21N_T3L_N5_AD8N_67	AL12
IO_L21P_T3L_N4_AD8P_67	AK12
IO_L20N_T3L_N3_AD1N_67	AL15
IO_L20P_T3L_N2_AD1P_67	AK15
IO_L19N_T3L_N1_DBC_AD9N_67	AM14
IO_L19P_T3L_N0_DBC_AD9P_67	AL14
IO_T2U_N12_67	AR15
IO_L18N_T2U_N11_AD2N_67	AP15
IO_L18P_T2U_N10_AD2P_67	AN15
IO_L17N_T2U_N9_AD10N_67	AP16
IO_L17P_T2U_N8_AD10P_67	AN16
IO_L16N_T2U_N7_QBC_AD3N_67	AR12
IO_L16P_T2U_N6_QBC_AD3P_67	AP12
IO_L15N_T2L_N5_AD11N_67	AN13
IO_L15P_T2L_N4_AD11P_67	AN14
IO_L14N_T2L_N3_GC_67	AR13
IO_L14P_T2L_N2_GC_67	AP13
IO_L13N_T2L_N1_GC_QBC_67	AT14
IO_L13P_T2L_N0_GC_QBC_67	AR14
IO_T1U_N12_67	AW16
IO_L12N_T1U_N11_GC_67	AV13
IO_L12P_T1U_N10_GC_67	AV14
IO_L11N_T1U_N9_GC_67	AU13
IO_L11P_T1U_N8_GC_67	AU14
IO_L10N_T1U_N7_QBC_AD4N_67	AY14
IO_L10P_T1U_N6_QBC_AD4P_67	AY15
IO_L9N_T1L_N5_AD12N_67	AW15
IO_L9P_T1L_N4_AD12P_67	AV15
IO_L8N_T1L_N3_AD5N_67	AU16
IO_L8P_T1L_N2_AD5P_67	AU16
IO_L7N_T1L_N1_QBC_AD13N_67	AT15
IO_L7P_T1L_N0_QBC_AD13P_67	AT16
IO_T0U_N12_VRP_67	BA12
IO_L6N_T0U_N11_AD6N_67	AV11
IO_L6P_T0U_N10_AD6P_67	AU11
IO_L5N_T0U_N9_AD14N_67	AY13
IO_L5P_T0U_N8_AD14P_67	AW13
IO_L4N_T0U_N7_DBC_AD7N_67	AW10
IO_L4P_T0U_N6_DBC_AD7P_67	AV10
IO_L3N_T0L_N5_AD15N_67	AY10
IO_L3P_T0L_N4_AD15P_67	AW11
IO_L2N_T0L_N3_67	AY12
IO_L2P_T0L_N2_67	AW12
IO_L1N_T0L_N1_DBC_67	AU12
IO_L1P_T0L_N0_DBC_67	AT12
VREF_67	AL16

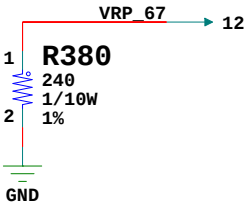
AM16	NC
AK13	FMC HPC1 LA33 N
AK14	FMC HPC1 LA33 P
AM12	FMC HPC1 LA31 N
AM13	FMC HPC1 LA31 P
AJ12	FMC HPC1 LA32 N
AJ13	FMC HPC1 LA32 P
AL12	FMC HPC1 LA30 N
AK12	FMC HPC1 LA30 P
AL15	FMC HPC1 LA26 N
AK15	FMC HPC1 LA26 P
AM14	FMC HPC1 LA27 N
AL14	FMC HPC1 LA27 P
AR15	NC
AP15	FMC HPC1 LA29 N
AN15	FMC HPC1 LA29 P
AP16	FMC HPC1 LA23 N
AN16	FMC HPC1 LA23 P
AR12	FMC HPC1 LA18 CC N
AP12	FMC HPC1 LA18 CC P
AN13	NC
AN14	NC
AR13	FMC HPC1 LA24 N
AP13	FMC HPC1 LA24 P
AT14	FMC HPC1 LA17 CC N
AR14	FMC HPC1 LA17 CC P
AW16	NC
AV13	FMC HPC1 CLK1 M2C N
AV14	FMC HPC1 CLK1 M2C P
AU13	NC
AU14	NC
AY14	PMOD0 0 LS
AY15	PMOD0 1 LS
AW15	PMOD0 2 LS
AV15	PMOD0 3 LS
AV16	PMOD0 4 LS
AU16	PMOD0 5 LS
AT15	PMOD0 6 LS
AT16	PMOD0 7 LS
BA12	VRP_67
AV11	FMC HPC1 LA21 N
AU11	FMC HPC1 LA21 P
AY13	FMC HPC1 LA22 N
AW13	FMC HPC1 LA22 P
AW10	FMC HPC1 LA28 N
AV10	FMC HPC1 LA28 P
AY10	FMC HPC1 LA20 N
AW11	FMC HPC1 LA20 P
AY12	FMC HPC1 LA19 N
AW12	FMC HPC1 LA19 P
AU12	FMC HPC1 LA25 N
AT12	FMC HPC1 LA25 P
AL16	VREF_67

VADJ_1V8_FPGA



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SOC_2104_1MM_IRON



FPGA Banks 66 67 FMC HPC1

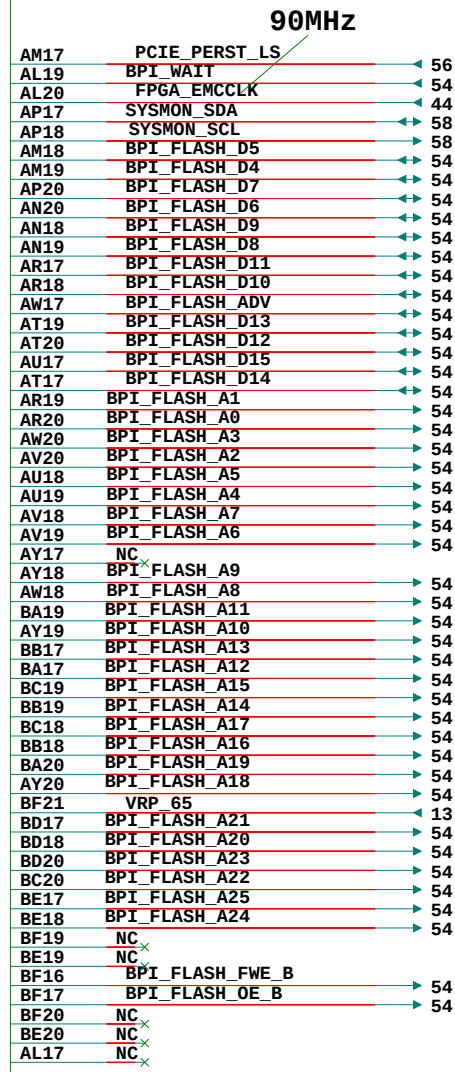
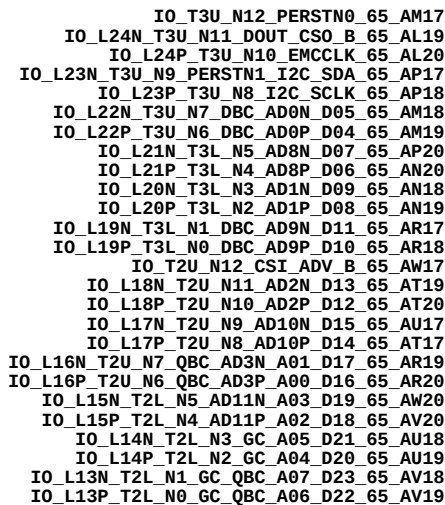


TITLE: FPGA Banks 66 67 FMC HPC1
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 12 OF 76	DRAWN BY: DN

SOC_VU9P_FLGA2104_IRONWOOD



TITLE: FPGA Bank 65 BPI Flash IF	ASSY P/N: 0432001
SCHEM, ROHS COMPLIANT	PCB P/N: 1280906
HW-U1-VCU118_REV1_0	SCH P/N: 0381739
	TEST P/N: TSS0185

DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 13 OF 76	DRAWN BY: DN

SOC_VU9P_FLGA2104_IRONWOOD

BANK 120
XCVU9PFLGA2104

MGTYTXP0_120 BD42
MGTYTXN0_120 BD43
MGTYRXP0_120 BC45
MGTYRXN0_120 BC46
MGTYTXP1_120 BB42
MGTYTXN1_120 BB43
MGTYRXP1_120 BA45
MGTYRXN1_120 BA46
MGTYTXP2_120 AY42
MGTYTXN2_120 AY43
MGTYRXP2_120 AW45
MGTYRXN2_120 AW46
MGTYTXP3_120 AV42
MGTYTXN3_120 AV43
MGTYRXP3_120 AU45
MGTYRXN3_120 AU46
MGTREFCLK0P_120 AN40
MGTREFCLK0N_120 AN41
MGTREFCLK1P_120 AM38
MGTREFCLK1N_120 AM39

BD42 FMCP_HSPC_DP20_C2M_P → 37
BD43 FMCP_HSPC_DP20_C2M_N → 37
BC45 FMCP_HSPC_DP20_M2C_P → 37
BC46 FMCP_HSPC_DP20_M2C_N → 37
BB42 FMCP_HSPC_DP21_C2M_P → 37
BB43 FMCP_HSPC_DP21_C2M_N → 37
BA45 FMCP_HSPC_DP21_M2C_P → 37
BA46 FMCP_HSPC_DP21_M2C_N → 37
AY42 FMCP_HSPC_DP22_C2M_P → 37
AY43 FMCP_HSPC_DP22_C2M_N → 37
AW45 FMCP_HSPC_DP22_M2C_P → 37
AW46 FMCP_HSPC_DP22_M2C_N → 37
AV42 FMCP_HSPC_DP23_C2M_P → 37
AV43 FMCP_HSPC_DP23_C2M_N → 37
AU45 FMCP_HSPC_DP23_M2C_P → 37
AU46 FMCP_HSPC_DP23_M2C_N → 37
AN40 FMCP_HSPC_GBTCLK5_M2C_C_P → 14
AN41 FMCP_HSPC_GBTCLK5_M2C_C_N → 14
AM38 FMCP_HSPC_GBT1_5_P → 14
AM39 FMCP_HSPC_GBT1_5_N → 14

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SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK 121
XCVU9PFLGA2104

MGTYTXP0_121 AT42
MGTYTXN0_121 AT43
MGTYRXP0_121 AR45
MGTYRXN0_121 AR46
MGTYTXP1_121 AP42
MGTYTXN1_121 AP43
MGTYRXP1_121 AN45
MGTYRXN1_121 AN46
MGTYTXP2_121 AM42
MGTYTXN2_121 AM43
MGTYRXP2_121 AL45
MGTYRXN2_121 AL46
MGTYTXP3_121 AL40
MGTYTXN3_121 AL41
MGTYRXP3_121 AJ45
MGTYRXN3_121 AJ46
MGTREFCLK0P_121 AK38
MGTREFCLK0N_121 AK39
MGTREFCLK1P_121 AH38
MGTREFCLK1N_121 AH39
MGTREFCLK1N_121 AH39
MGTRREF_LS_BF43
MGTAVTTRCAL_LS_BF42

AT42 FMCP_HSPC_DP0_C2M_P → 34
AT43 FMCP_HSPC_DP0_C2M_N → 34
AR45 FMCP_HSPC_DP0_M2C_P → 34
AR46 FMCP_HSPC_DP0_M2C_N → 34
AP42 FMCP_HSPC_DP1_C2M_P → 34
AP43 FMCP_HSPC_DP1_C2M_N → 34
AN45 FMCP_HSPC_DP1_M2C_P → 34
AN46 FMCP_HSPC_DP1_M2C_N → 34
AM42 FMCP_HSPC_DP2_C2M_P → 34
AM43 FMCP_HSPC_DP2_C2M_N → 34
AL45 FMCP_HSPC_DP2_M2C_P → 34
AL46 FMCP_HSPC_DP2_M2C_N → 34
AL40 FMCP_HSPC_DP3_C2M_P → 34
AL41 FMCP_HSPC_DP3_C2M_N → 34
AJ45 FMCP_HSPC_DP3_M2C_P → 34
AJ46 FMCP_HSPC_DP3_M2C_N → 34
AK38 FMCP_HSPC_GBT0_0_P → 14
AK39 FMCP_HSPC_GBT0_0_N → 14
AH38 FMCP_HSPC_GBT1_0_P → 14
AH39 FMCP_HSPC_GBT1_0_N → 14
BF43 MGTRREF_121 → 14
BF42

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SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

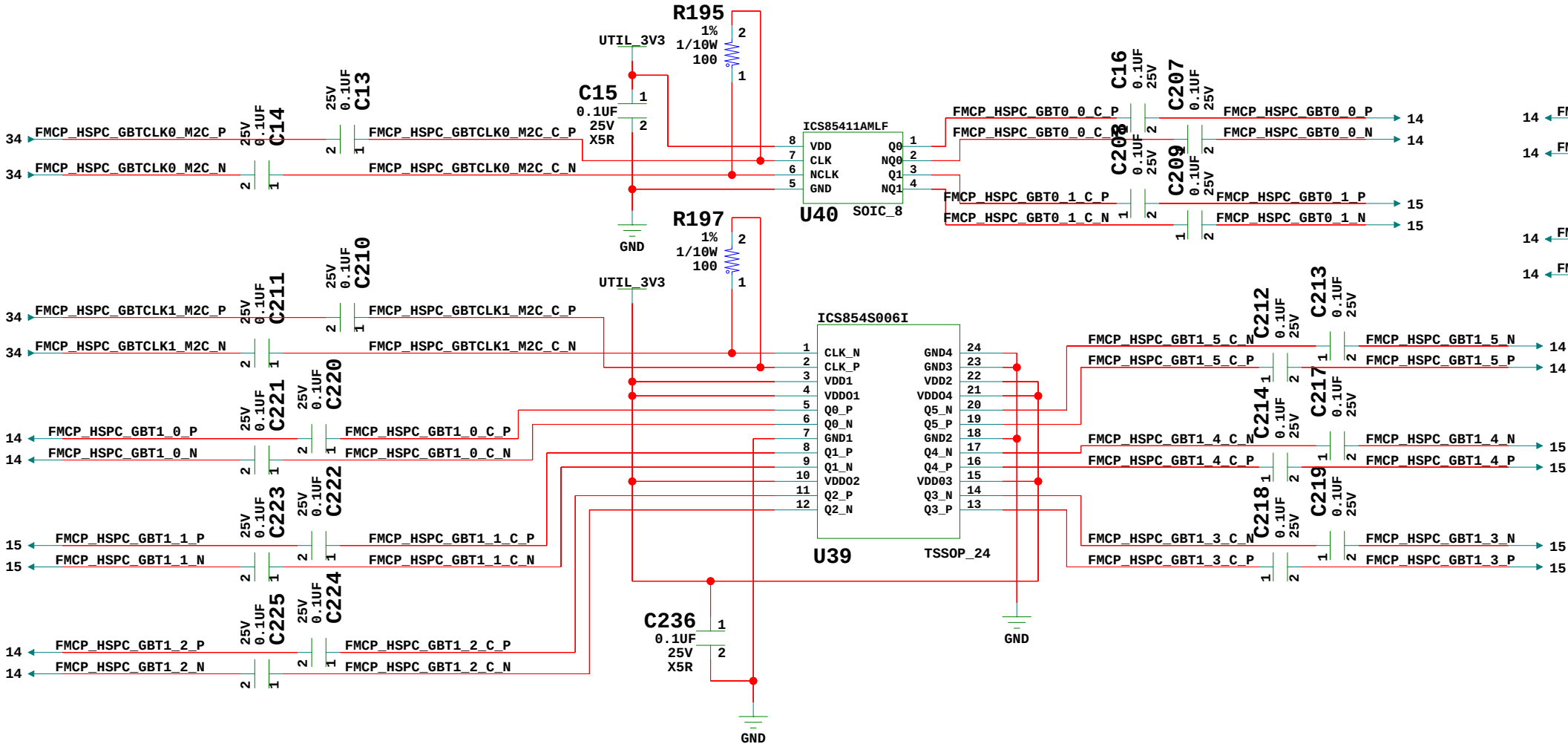
BANK 122
XCVU9PFLGA2104

MGTYTXP0_122 AK42
MGTYTXN0_122 AK43
MGTYRXP0_122 AG45
MGTYRXN0_122 AG46
MGTYTXP1_122 AJ41
MGTYTXN1_122 AJ41
MGTYRXP1_122 AF43
MGTYRXN1_122 AF43
MGTYTXP2_122 AG40
MGTYTXN2_122 AG41
MGTYRXP2_122 AE45
MGTYRXN2_122 AE46
MGTYTXP3_122 AE40
MGTYTXN3_122 AE41
MGTYRXP3_122 AD43
MGTYRXN3_122 AD44
MGTREFCLK0P_122 AF38
MGTREFCLK0N_122 AF39
MGTREFCLK1P_122 AD38
MGTREFCLK1N_122 AD39

AK42 FMCP_HSPC_DP8_C2M_P → 34
AK43 FMCP_HSPC_DP8_C2M_N → 34
AG45 FMCP_HSPC_DP8_M2C_P → 34
AG46 FMCP_HSPC_DP8_M2C_N → 34
AJ41 FMCP_HSPC_DP9_C2M_P → 34
AJ41 FMCP_HSPC_DP9_C2M_N → 34
AF43 FMCP_HSPC_DP9_M2C_P → 34
AF43 FMCP_HSPC_DP9_M2C_N → 34
AF44 FMCP_HSPC_DP9_M2C_P → 34
AF44 FMCP_HSPC_DP9_M2C_N → 34
AG40 FMCP_HSPC_DP10_C2M_P → 37
AG41 FMCP_HSPC_DP10_C2M_N → 37
AE45 FMCP_HSPC_DP10_M2C_P → 37
AE46 FMCP_HSPC_DP10_M2C_N → 37
AE40 FMCP_HSPC_DP11_C2M_P → 37
AE41 FMCP_HSPC_DP11_C2M_N → 37
AD43 FMCP_HSPC_DP11_M2C_P → 37
AD44 FMCP_HSPC_DP11_M2C_N → 37
AF38 FMCP_HSPC_GBTCLK2_M2C_C_P → 14
AF39 FMCP_HSPC_GBTCLK2_M2C_C_N → 14
AD38 FMCP_HSPC_GBT1_2_P → 14
AD39 FMCP_HSPC_GBT1_2_N → 14

U1

SOC_2104_1MM_IRON



FPGA Banks 120 121 122



TITLE: FPGA Banks 120 121 122
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 14 OF 76	DRAWN BY: DN

SOC VU9P FLGA2104 IRONWOOD

BANK 125
XCVU9PFLGA2104

MGITYXP0_125 AC40	AC40	FMCP	HSPC	DP12	C2M	P	37	
MGITYXN0_125 AC41	AC41	FMCP	HSPC	DP12	C2M	N	37	
MGTYRXP0_125 AC45	AC45	FMCP	HSPC	DP12	M2C	P	37	
MGTYRXN0_125 AC46	AC46	FMCP	HSPC	DP12	M2C	N	37	
MGITYXP1_125 AA40	AA40	FMCP	HSPC	DP13	C2M	P	37	
MGITYXN1_125 AA41	AA41	FMCP	HSPC	DP13	C2M	N	37	
MGTYRXP1_125 AB43	AB43	FMCP	HSPC	DP13	M2C	P	37	
MGTYRXN1_125 AB44	AB44	FMCP	HSPC	DP13	M2C	N	37	
MGITYXP2_125 W40	W40	FMCP	HSPC	DP14	C2M	P	37	
MGITYXN2_125 W41	W41	FMCP	HSPC	DP14	C2M	N	37	
MGTYRXP2_125 AA45	AA45	FMCP	HSPC	DP14	M2C	P	37	
MGTYRXN2_125 AA46	AA46	FMCP	HSPC	DP14	M2C	N	37	
MGITYXP3_125 U40	U40	FMCP	HSPC	DP15	C2M	P	37	
MGITYXN3_125 U41	U41	FMCP	HSPC	DP15	C2M	N	37	
MGTYRXP3_125 Y43	Y43	FMCP	HSPC	DP15	M2C	P	37	
MGTYRXN3_125 Y44	Y44	FMCP	HSPC	DP15	M2C	N	37	
MGTREFCLK0P_125 AB38	AB38	FMCP	HSPC	GBTCLK3	M2C	C	37	
MGTREFCLK0N_125 AB39	AB39	FMCP	HSPC	GBTCLK3	M2C	C	N	15
MGTREFCLK1P_125 Y38	Y38	FMCP	HSPC	GBT1	3	P	14	
MGTREFCLK1N_125 Y39	Y39	FMCP	HSPC	GBT1	3	N	14	

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SOC 2104 1MM IRON

SOC VU9P FLGA2104 IRONWOOD

BANK 126
XCVU9PFLGA2104

MGTYTXP0_126_T42	T42	FMCP	HSPC	DP4	C2M	P	34
MGTYTXN0_126_T43	T43	FMCP	HSPC	DP4	C2M	N	34
MGTYRXP0_126_W45	W45	FMCP	HSPC	DP4	M2C	P	34
MGTYRXN0_126_W46	W46	FMCP	HSPC	DP4	M2C	N	34
MGTYTXP1_126_P42	P42	FMCP	HSPC	DP5	C2M	P	34
MGTYTXN1_126_P43	P43	FMCP	HSPC	DP5	C2M	N	34
MGTYRXP1_126_U45	U45	FMCP	HSPC	DP5	M2C	P	34
MGTYRXN1_126_U46	U46	FMCP	HSPC	DP5	M2C	N	34
MGTYTXP2_126_M42	M42	FMCP	HSPC	DP6	C2M	P	34
MGTYTXN2_126_M43	M43	FMCP	HSPC	DP6	C2M	N	34
MGTYRXP2_126_R45	R45	FMCP	HSPC	DP6	M2C	P	34
MGTYRXN2_126_R46	R46	FMCP	HSPC	DP6	M2C	N	34
MGTYTXP3_126_K42	K42	FMCP	HSPC	DP7	C2M	P	34
MGTYTXN3_126_K43	K43	FMCP	HSPC	DP7	C2M	N	34
MGTYRXP3_126_N45	N45	FMCP	HSPC	DP7	M2C	P	34
MGTYRXN3_126_N46	N46	FMCP	HSPC	DP7	M2C	N	34
MGTREFCLKP0_126_V38	V38	FMCP	HSPC	GBT0	1	P	14
MGTREFCLKN0_126_V39	V39	FMCP	HSPC	GBT0	1	N	14
MGTREFCLKP1_126_T38	T38	FMCP	HSPC	GBT1	1	P	14
MGTREFCLKN1_126_T39	T39	FMCP	HSPC	GBT1	1	N	14
MGTRREF_LN_L41	L41	MGTRREF	126				
MGTAVITRCAL_LN_L40	L40						

U1

SOC 2104 1MM IRON

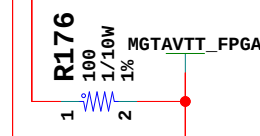
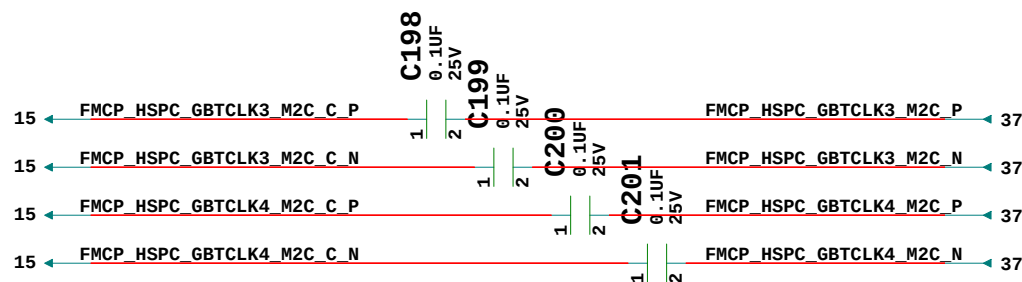
SOC VU9P FLGA2104 IRONWOOD

BANK 127
XCVU9PFLGA2104

MGTYTXP0_127_H42	H42	FMCP	HSPC	DP16	C2M	P		
MGTYTXN0_127_H43	H43	FMCP	HSPC	DP16	C2M	N		37
MGTYRXP0_127_L45	L45	FMCP	HSPC	DP16	M2C	P		37
MGTYRXN0_127_L46	L46	FMCP	HSPC	DP16	M2C	N		37
MGTYTXP1_127_F42	F42	FMCP	HSPC	DP17	C2M	P		37
MGTYTXN1_127_F43	F43	FMCP	HSPC	DP17	C2M	N		37
MGTYRXP1_127_J45	J45	FMCP	HSPC	DP17	M2C	P		37
MGTYRXN1_127_J46	J46	FMCP	HSPC	DP17	M2C	N		37
MGTYTXP2_127_D42	D42	FMCP	HSPC	DP18	C2M	P		37
MGTYTXN2_127_D43	D43	FMCP	HSPC	DP18	C2M	N		37
MGTYRXP2_127_G45	G45	FMCP	HSPC	DP18	M2C	P		37
MGTYRXN2_127_G46	G46	FMCP	HSPC	DP18	M2C	N		37
MGTYTXP3_127_B42	B42	FMCP	HSPC	DP19	C2M	P		37
MGTYTXN3_127_B43	B43	FMCP	HSPC	DP19	C2M	N		37
MGTYRXP3_127_E45	E45	FMCP	HSPC	DP19	M2C	P		37
MGTYRXN3_127_E46	E46	FMCP	HSPC	DP19	M2C	N		37
MGTRFCLK0P_127_R40	R40	FMCP	HSPC	GBTCLK4	M2C	C		37
MGTRFCLK0N_127_R41	R41	FMCP	HSPC	GBTCLK4	M2C	C	N	15
MGTRFCLK1P_127_N40	N40	FMCP	HSPC	GBT1	4	P		15
MGTRFCLK1N_127_N41	N41	FMCP	HSPC	GBT1	4	N		14

U1

SOC 2104 1MM IRON



FPGA Banks 125 126 127



**TITLE: FPGA Banks 125 126 127
SCHEM, ROHS COMPLIANT
HW-U1-VCU118 REV1 0**

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59

VER:	1.0
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SHEET SIZE: B

REV:	01
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SHEET **15** OF **76**

DRAWN BY:

SOC_VU9P_FLGA2104_IRONWOOD

BANK 224
XCVU9PFLGA2104

MGTYTXP0_224_BE5	BE5	PCIE TX15 P	43
MGTYTXN0_224_BE4	BE4	PCIE TX15 N	43
MGTYRXP0_224_BB2	BB2	PCIE RX15 P	43
MGTYRXN0_224_BB1	BB1	PCIE RX15 N	43
MGTYTXP1_224_BC5	BC5	PCIE TX14 P	43
MGTYTXN1_224_BC4	BC4	PCIE TX14 N	43
MGTYRXP1_224_AY2	AY2	PCIE RX14 P	43
MGTYRXN1_224_AY1	AY1	PCIE RX14 N	43
MGTYTXP2_224_BA5	BA5	PCIE TX13 P	43
MGTYTXN2_224_BA4	BA4	PCIE TX13 N	43
MGTYRXP2_224_AV2	AV2	PCIE RX13 P	43
MGTYRXN2_224_AV1	AV1	PCIE RX13 N	43
MGTYTXP3_224_AW5	AW5	PCIE TX12 P	43
MGTYTXN3_224_AW4	AW4	PCIE TX12 N	43
MGTYRXP3_224_AT2	AT2	PCIE RX12 P	43
MGTYRXN3_224_AT1	AT1	PCIE RX12 N	43
MGTREFCLK0P_224_AR9	AR9	NC	43
MGTREFCLK0N_224_AR8	AR8	NC	43
MGTREFCLK1P_224_AN9	AN9	NC	43
MGTREFCLK1N_224_AN8	AN8	NC	43

U1

SOC_2104_1MM_IRON

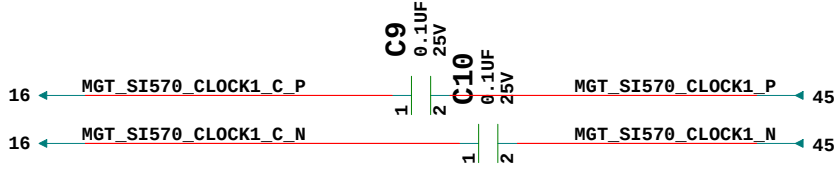
SOC_VU9P_FLGA2104_IRONWOOD

BANK 225
XCVU9PFLGA2104

MGTYTXP0_225_AU5	AU5	PCIE TX11 P	43
MGTYTXN0_225_AU4	AU4	PCIE TX11 N	43
MGTYRXP0_225_AP2	AP2	PCIE RX11 P	43
MGTYRXN0_225_AP1	AP1	PCIE RX11 N	43
MGTYTXP1_225_AT7	AT7	PCIE TX10 P	43
MGTYTXN1_225_AT6	AT6	PCIE TX10 N	43
MGTYRXP1_225_AM2	AM2	PCIE RX10 P	43
MGTYRXN1_225_AM1	AM1	PCIE RX10 N	43
MGTYTXP2_225_AR5	AR5	PCIE TX9 P	43
MGTYTXN2_225_AR4	AR4	PCIE TX9 N	43
MGTYRXP2_225_AK2	AK2	PCIE RX9 P	43
MGTYRXN2_225_AK1	AK1	PCIE RX9 N	43
MGTYTXP3_225_AP7	AP7	PCIE TX8 P	43
MGTYTXN3_225_AP6	AP6	PCIE TX8 N	43
MGTYRXP3_225_AJ4	AJ4	PCIE RX8 P	43
MGTYRXN3_225_AJ3	AJ3	PCIE RX8 N	43
MGTREFCLK0P_225_AL9	AL9	PCIE CLK1 P	16
MGTREFCLK0N_225_AL8	AL8	PCIE CLK1 N	16
MGTREFCLK1P_225_AJ9	AJ9	MGT SI570 CLOCK1 C P	16
MGTREFCLK1N_225_AJ8	AJ8	MGT SI570 CLOCK1 C N	16

U1

SOC_2104_1MM_IRON



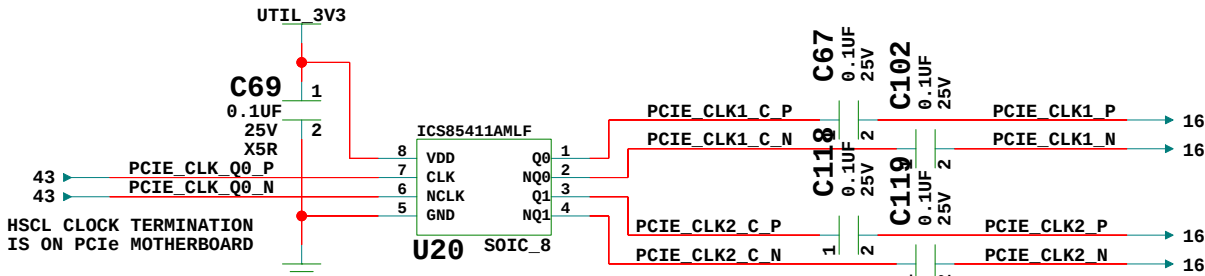
SOC_VU9P_FLGA2104_IRONWOOD

BANK 227
XCVU9PFLGA2104

MGTYTXP0_227_AF7	AF7	PCIE TX3 P	43
MGTYTXN0_227_AF6	AF6	PCIE TX3 N	43
MGTYRXP0_227_AD2	AD2	PCIE RX3 P	43
MGTYRXN0_227_AD1	AD1	PCIE RX3 N	43
MGTYTXP1_227_AD7	AD7	PCIE TX2 P	43
MGTYTXN1_227_AD6	AD6	PCIE TX2 N	43
MGTYRXP1_227_AC4	AC4	PCIE RX2 P	43
MGTYRXN1_227_AC3	AC3	PCIE RX2 N	43
MGTYTXP2_227_AB7	AB7	PCIE TX1 P	43
MGTYTXN2_227_AB6	AB6	PCIE TX1 N	43
MGTYRXP2_227_AB2	AB2	PCIE RX1 P	43
MGTYRXN2_227_AB1	AB1	PCIE RX1 N	43
MGTYTXP3_227_Y7	Y7	PCIE TX0 P	43
MGTYTXN3_227_Y6	Y6	PCIE TX0 N	43
MGTYRXP3_227_AA4	AA4	PCIE RX0 P	43
MGTYRXN3_227_AA3	AA3	PCIE RX0 N	43
MGTREFCLK0P_227_AC9	AC9	PCIE CLK2 P	16
MGTREFCLK0N_227_AC8	AC8	PCIE CLK2 N	16
MGTREFCLK1P_227_AA9	AA9	NC	43
MGTREFCLK1N_227_AA8	AA8	NC	43

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SOC_2104_1MM_IRON



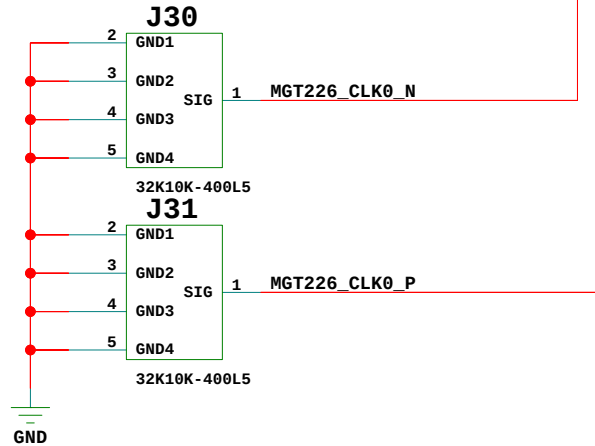
SOC_VU9P_FLGA2104_IRONWOOD

BANK 226
XCVU9PFLGA2104

MGTYTXP0_226_AN5	AN5	PCIE TX7 P	43
MGTYTXN0_226_AN4	AN4	PCIE TX7 N	43
MGTYRXP0_226_AH2	AH2	PCIE RX7 P	43
MGTYRXN0_226_AH1	AH1	PCIE RX7 N	43
MGTYTXP1_226_AM7	AM7	PCIE TX6 P	43
MGTYTXN1_226_AM6	AM6	PCIE TX6 N	43
MGTYRXP1_226_AG4	AG4	PCIE RX6 P	43
MGTYRXN1_226_AG3	AG3	PCIE RX6 N	43
MGTYTXP2_226_AK7	AK7	PCIE TX5 P	43
MGTYTXN2_226_AK6	AK6	PCIE TX5 N	43
MGTYRXP2_226_AF2	AF2	PCIE RX5 P	43
MGTYRXN2_226_AF1	AF1	PCIE RX5 N	43
MGTYTXP3_226_AH7	AH7	PCIE TX4 P	43
MGTYTXN3_226_AH6	AH6	PCIE TX4 N	43
MGTYRXP3_226_AE4	AE4	PCIE RX4 P	43
MGTYRXN3_226_AE3	AE3	PCIE RX4 N	43
MGTREFCLK0P_226_AG9	AG9	NC	43
MGTREFCLK0N_226_AG8	AG8	NC	43
MGTREFCLK1P_226_AE9	AE9	NC	43
MGTREFCLK1N_226_AE8	AE8	NC	43
MGTREFCLK1N_226_AE8	BD2	MGTREF_RS_BD2	43
MGTREFCLK1N_226_AE8	BD3	MGTREF_RS_BD3	43
MGTREFCLK1N_226_AE8	BD3	MGTREF_RS_BD3	43

U1

SOC_2104_1MM_IRON



FPGA Banks 224 225 226 227

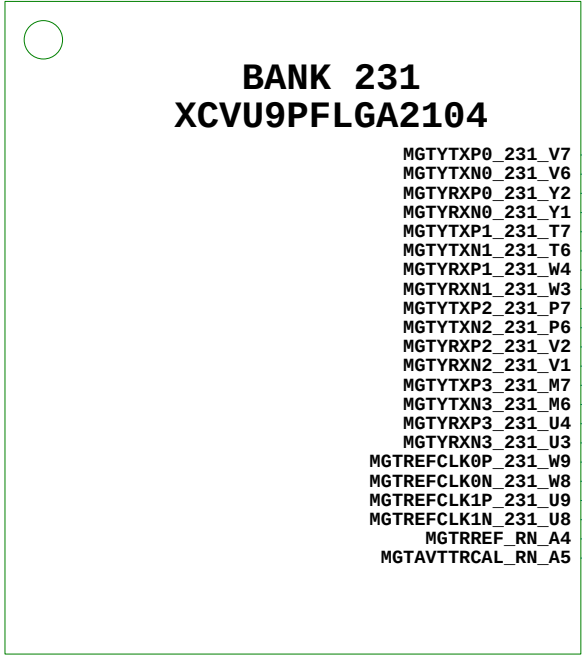


TITLE: FPGA Banks 224 225 226 227
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 16 OF 76	DRAWN BY: DN

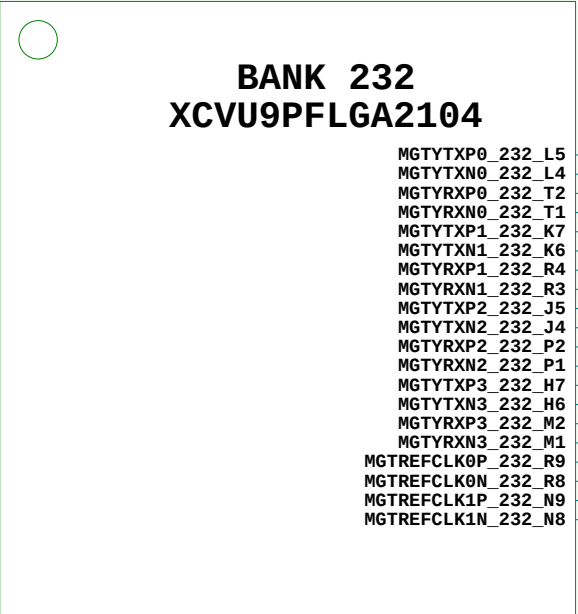
SOC_VU9P_FLGA2104_IRONWOOD



U1

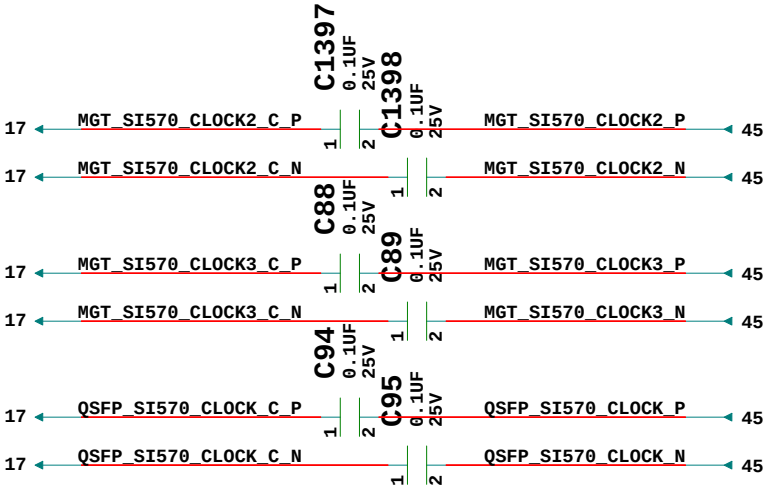
SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

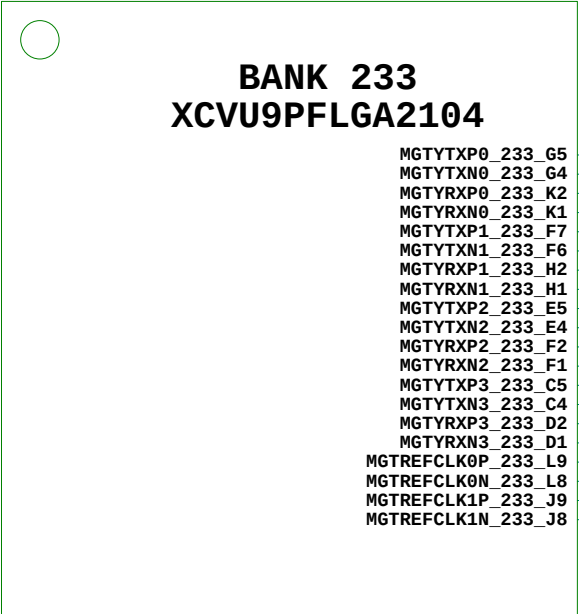


U1

SOC_2104_1MM_IRON

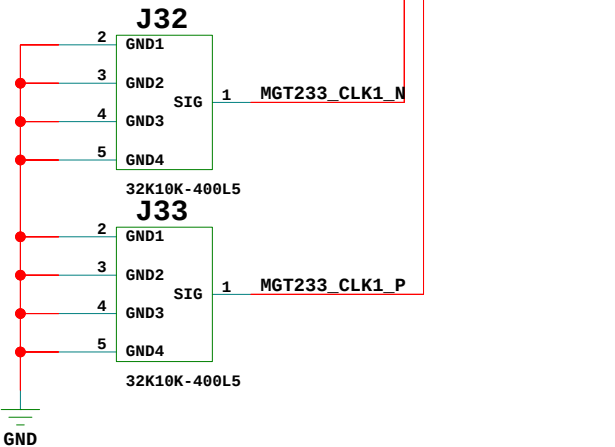


SOC_VU9P_FLGA2104_IRONWOOD



U1

SOC_2104_1MM_IRON



FPGA Banks 231 232 233



TITLE: FPGA Banks 231 232 233
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

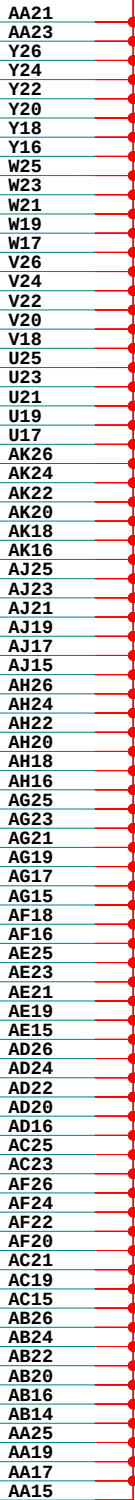
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 17 OF 76	DRAWN BY: DN

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCINT
XCVU9PFLGA2104

VCCINT_AA21
VCCINT_AA23
VCCINT_Y26
VCCINT_Y24
VCCINT_Y22
VCCINT_Y20
VCCINT_Y18
VCCINT_Y16
VCCINT_W25
VCCINT_W23
VCCINT_W21
VCCINT_W19
VCCINT_W17
VCCINT_V26
VCCINT_V24
VCCINT_V22
VCCINT_V20
VCCINT_V18
VCCINT_U25
VCCINT_U23
VCCINT_U21
VCCINT_U19
VCCINT_U17
VCCINT_AK26
VCCINT_AK24
VCCINT_AK22
VCCINT_AK20
VCCINT_AK18
VCCINT_AK16
VCCINT_AJ25
VCCINT_AJ23
VCCINT_AJ21
VCCINT_AJ19
VCCINT_AJ17
VCCINT_AJ15
VCCINT_AH26
VCCINT_AH24
VCCINT_AH22
VCCINT_AH20
VCCINT_AH18
VCCINT_AH16
VCCINT_AG25
VCCINT_AG23
VCCINT_AG21
VCCINT_AG19
VCCINT_AG17
VCCINT_AG15
VCCINT_AF18
VCCINT_AF16
VCCINT_AE25
VCCINT_AE23
VCCINT_AE21
VCCINT_AE19
VCCINT_AE15
VCCINT_AD26
VCCINT_AD24
VCCINT_AD22
VCCINT_AD20
VCCINT_AD16
VCCINT_AC25
VCCINT_AC23
VCCINT_AF26
VCCINT_AF24
VCCINT_AF22
VCCINT_AF20
VCCINT_AC21
VCCINT_AC19
VCCINT_AC15
VCCINT_AB26
VCCINT_AB24
VCCINT_AB22
VCCINT_AB20
VCCINT_AB16
VCCINT_AB14
VCCINT_AA25
VCCINT_AA19
VCCINT_AA17
VCCINT_AA15

VCCINT_FPGA



U1

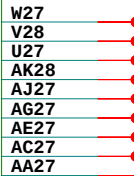
SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCINT_IO
XCVU9PFLGA2104

VCCINT_IO_W27
VCCINT_IO_V28
VCCINT_IO_U27
VCCINT_IO_AK28
VCCINT_IO_AJ27
VCCINT_IO_AG27
VCCINT_IO_AE27
VCCINT_IO_AC27
VCCINT_IO_AA27

VCCINTIO_BRAM_FPGA



U1

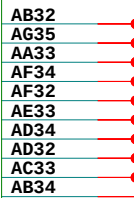
SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCAUX
XCVU9PFLGA2104

VCCAUX_AB32
VCCAUX_AG35
VCCAUX_AA33
VCCAUX_AF34
VCCAUX_AF32
VCCAUX_AE33
VCCAUX_AD34
VCCAUX_AD32
VCCAUX_AC33
VCCAUX_AB34

VCC1V8_FPGA



U1

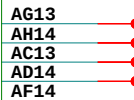
SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCBRAM
XCVU9PFLGA2104

VCCBRAM_AG13
VCCBRAM_AH14
VCCBRAM_AC13
VCCBRAM_AD14
VCCBRAM_AF14

VCCINTIO_BRAM_FPGA



U1

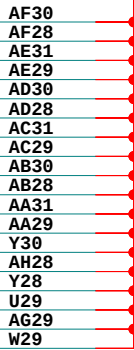
SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCAUX_IO
XCVU9PFLGA2104

VCCAUX_IO_AF30
VCCAUX_IO_AF28
VCCAUX_IO_AE31
VCCAUX_IO_AE29
VCCAUX_IO_AD30
VCCAUX_IO_AD28
VCCAUX_IO_AC31
VCCAUX_IO_AC29
VCCAUX_IO_AB30
VCCAUX_IO_AB28
VCCAUX_IO_AA31
VCCAUX_IO_AA29
VCCAUX_IO_Y30
VCCAUX_IO_AH28
VCCAUX_IO_Y28
VCCAUX_IO_U29
VCCAUX_IO_AG29
VCCAUX_IO_W29

VCC1V8_FPGA



U1

SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK VCCADC
XCVU9PFLGA2104

VCCADC_AB18

AB18

VCC1V8_FPGA

FERRITE-600

L13

FPGA_SYSMON_VCC

U1

SOC_2104_1MM_IRON

SOC_VU9P_FLGA2104_IRONWOOD

BANK GNDADC
XCVU9PFLGA2104

GNDADC_AB17

AB17

FERRITE-600

L87

SYSMON_AGND

GND

U1

SOC_2104_1MM_IRON

FPGA Power 1



TITLE: FPGA Power 1
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59

VER: 1.0

SHEET SIZE: B

REV: 01

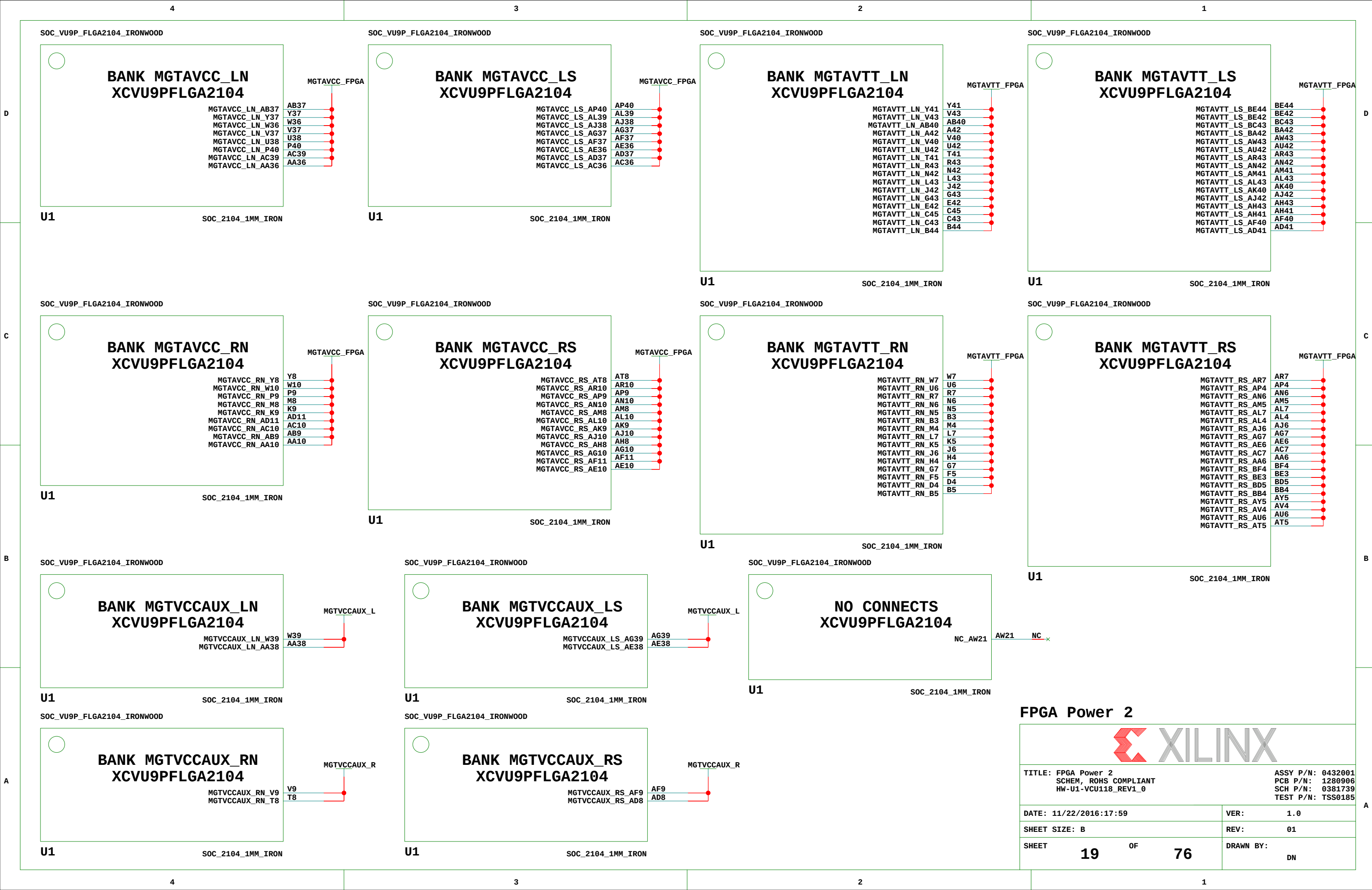
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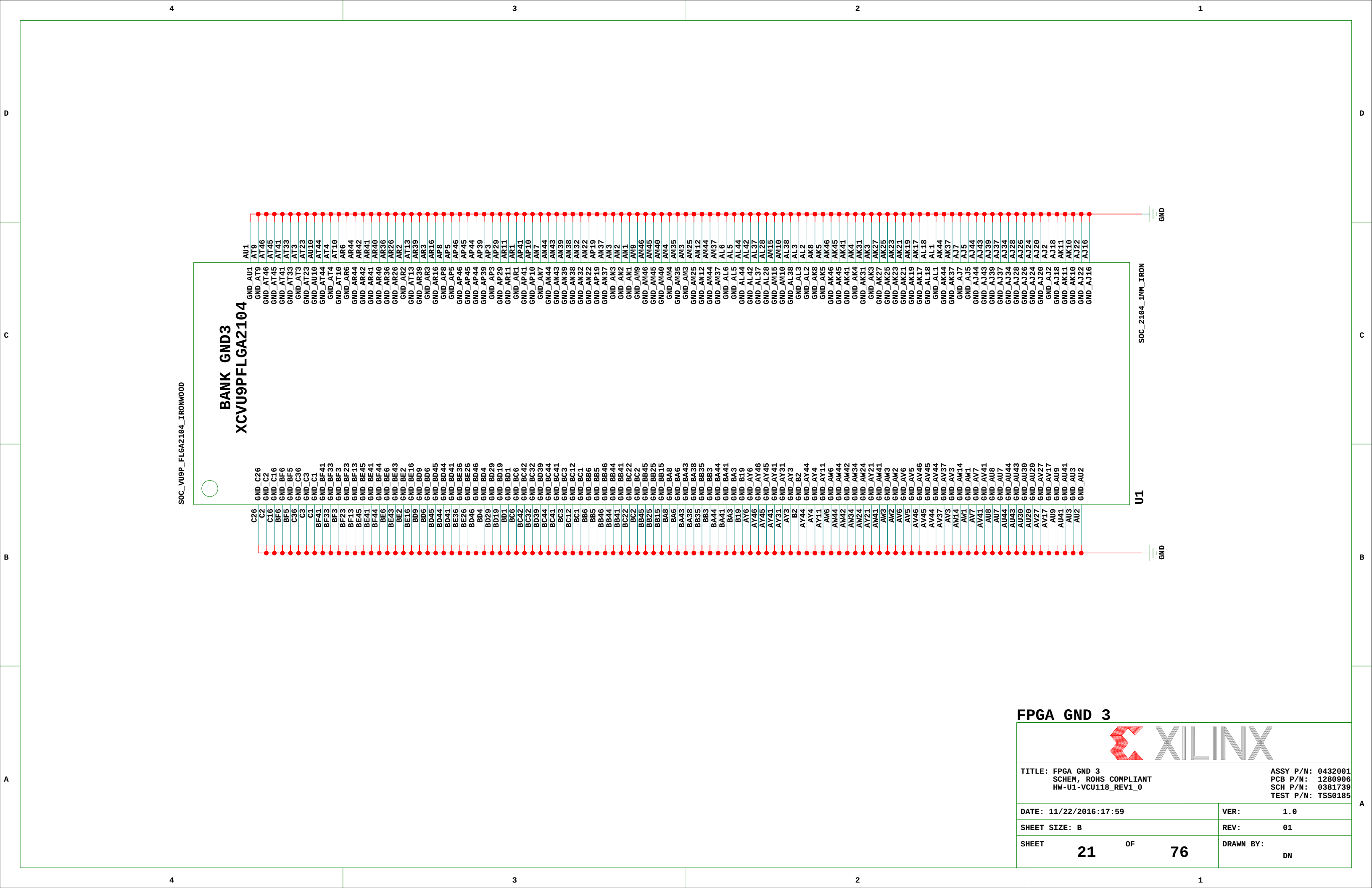
OF

76

DRAWN BY:

DN

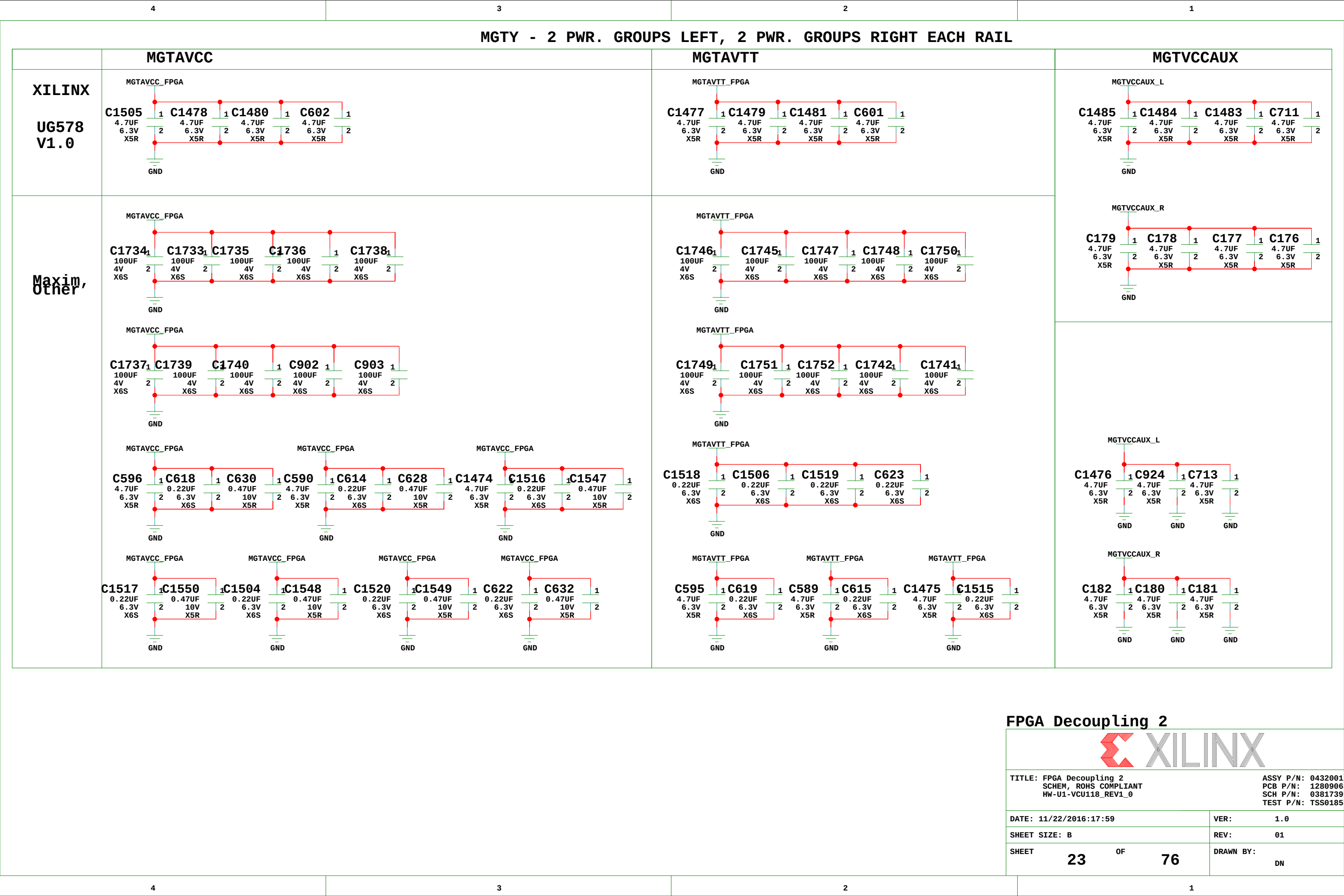




FPGA GND 3

	
TITLE: FPGA GND 3 SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 21 OF 76	DRAWN BY: DN

	4	3	2	1			
	VCCINT		VCCAUX/AUX_IO	VCCINT_IO/VCCBRAM			
D	XILINX UG583 V1.7	C460 C461 C462 C330 VCCINT_FPGA C906 C153 C164 C1486 C907 C163 C165 VCCINT_FPGA C657 C658 C659 C1557 C159 C919 C920 C921 C922 C923 C1755 C1756 C1455 VCCINT_FPGA 	VCC1V8_FPGA 	VCCINTIO_BRAM_FPGA 	VCCINT_FPGA VCC1V8_FPGA VCCINTIO_BRAM_FPGA		
	Maxim, Other	C206 C911 VCCINT_FPGA C908 C909 C914 C916 C917 C918 VCCINT_FPGA C1558 C1559 VCCINT_FPGA C1487 C1488 VCCINT_FPGA C331 C334 VCCINT_FPGA VCC1V8_FPGA C1406 	VCC1V8_FPGA 	VCCINTIO_BRAM_FPGA 	VADJ_1V8_FPGA VCC1V2_FPGA UTIL_1V35 UTIL_3V3		
C	XILINX UG583 V1.7	VCC0_40 / VCC0_41 / VCC0_42 VCC1V2_FPGA 	VCC0_46 / VCC0_47 / VCC0_48 VCC1V2_FPGA 	VCC0_71 / VCC0_72 / VCC0_73 VCC1V2_FPGA 	VCC0_64 / VCC0_65 VCC1V8_FPGA 	VCC0_0 VCC1V8_FPGA 	VCC0_40 1.2V VCC0_41 1.2V VCC0_42 1.2V VCC0_46 1.2V VCC0_47 1.2V VCC0_48 1.2V VCC0_71 1.2V VCC0_72 1.2V VCC0_73 1.2V
B	Maxim, Other	VCC1V2_FPGA C1450 C1451 C1452 	VCC1V2_FPGA C1453 C1454 C1575 	VCC1V2_FPGA C1576 C1577 C1574 	VCC1V8_FPGA C1570 C1571 	VCC1V8_FPGA C1572 	VCC0_64 1.8V VCC0_65 1.8V VCC0_0 1.8V
A	XILINX UG583 V1.7	VADJ_1V8_FPGA C170 	VADJ_1V8_FPGA C169 	VCC0_43 1.8V(Vadj) VCC0_45 1.8V(Vadj) VCC0_66 1.8V(Vadj) VCC0_67 1.8V(Vadj) VCC0_70 1.8V(Vadj)			
	Maxim, Other	VADJ_1V8_FPGA C1573 	VADJ_1V8_FPGA C1568 C1569 C1467 				



FPGA Decoupling 2

XILINX

TITLE: FPGA Decoupling 2

SCHEM, ROHS COMPLIANT

HW-U1-VCU118_REV1_0

ASSY P/N: 0432001

PCB P/N: 1280906

SCH P/N: 0381739

TEST P/N: TSS0185

DATE: 11/22/2016:17:59

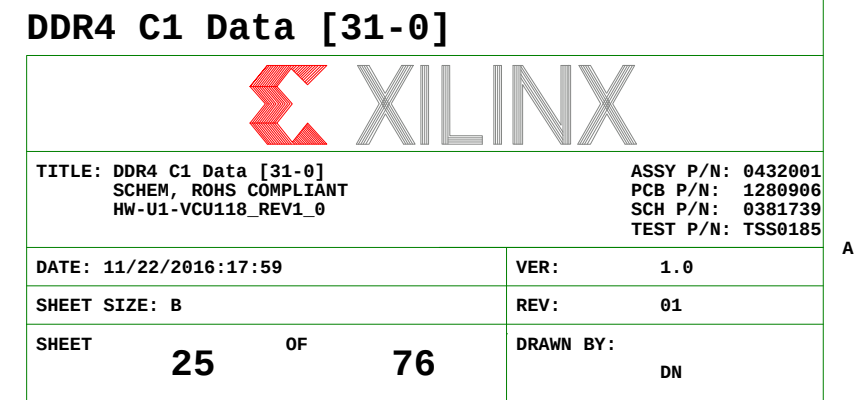
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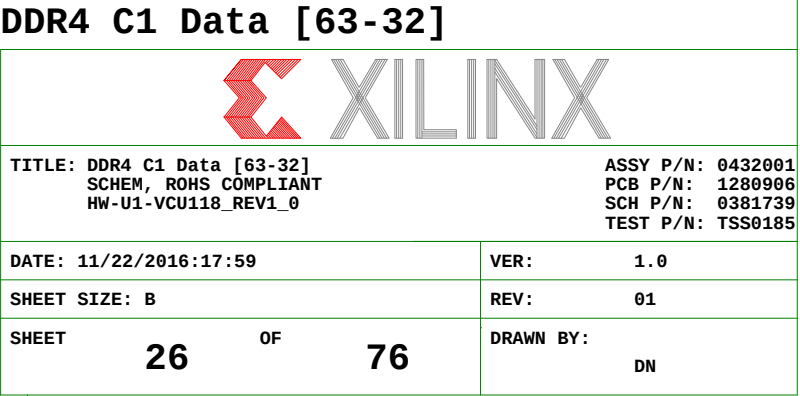
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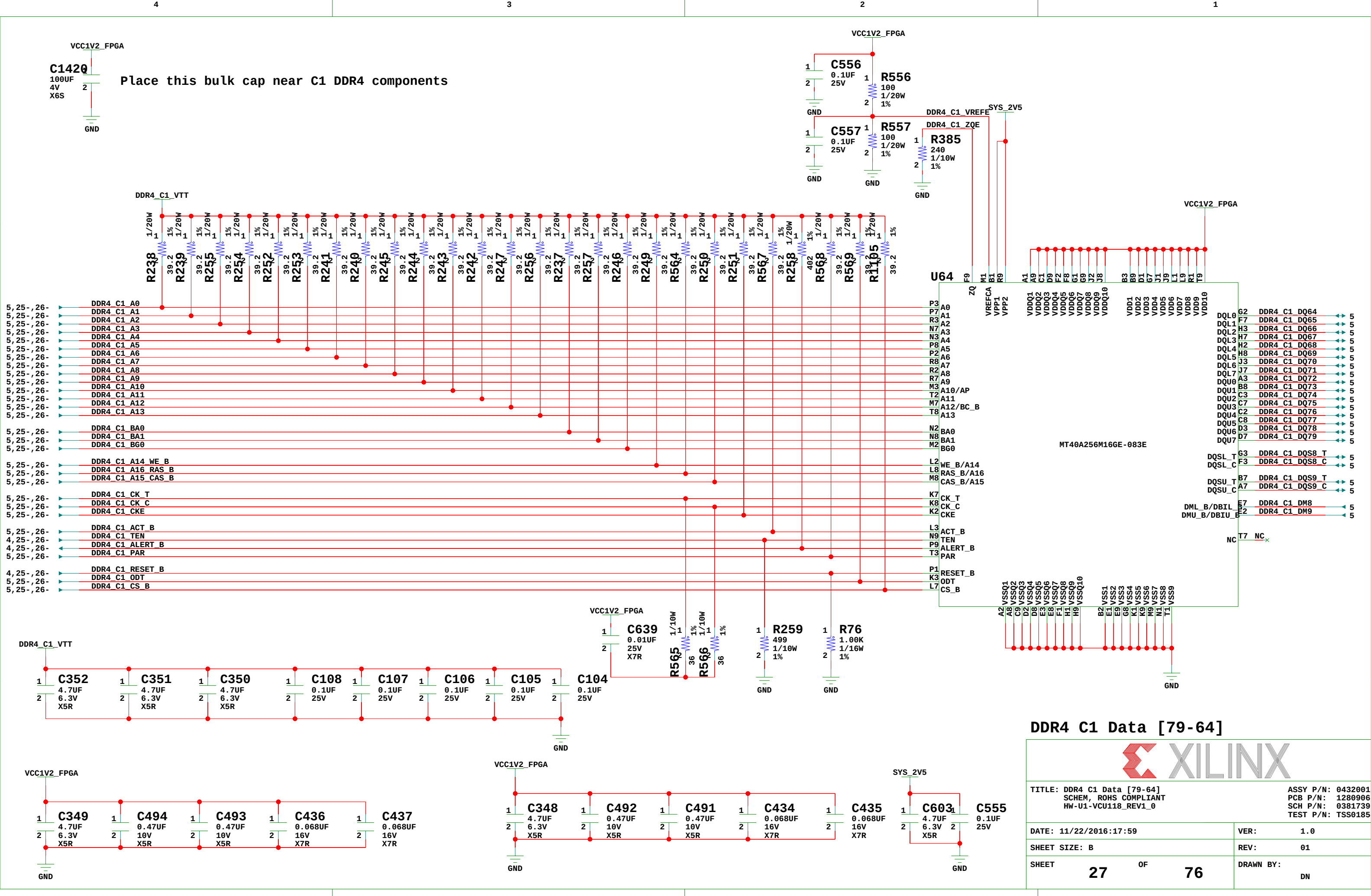
REV: 01

SHEET 23 OF 76

DRAWN BY: DN







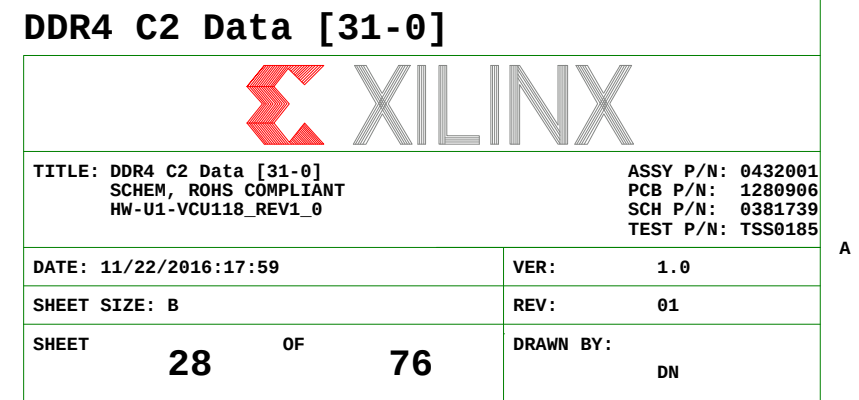
Place this bulk cap near C1 DDR4 components

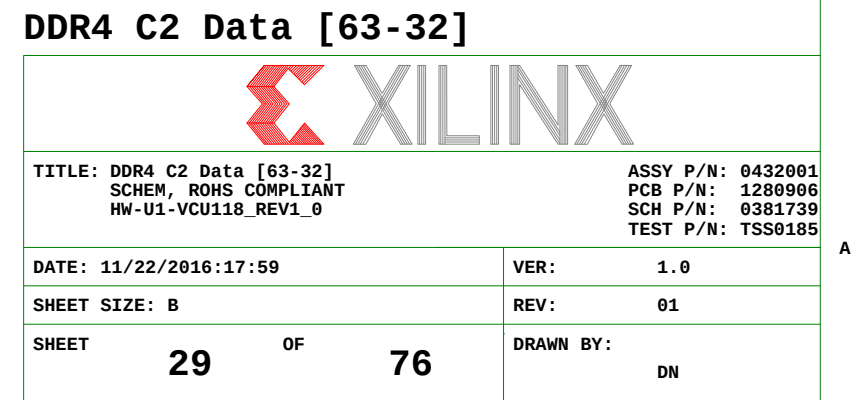
DDR4 C1 Data [79-64]

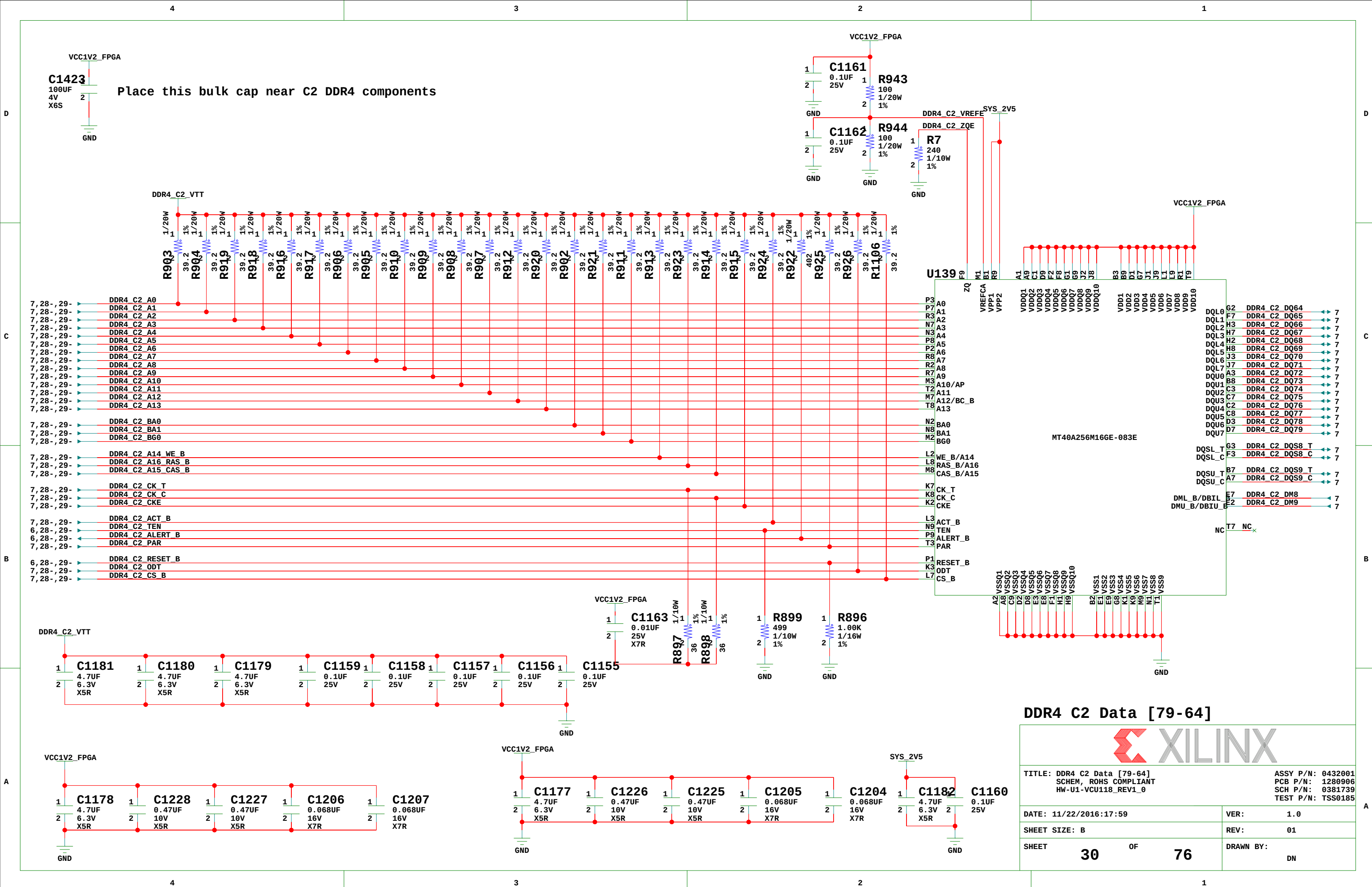


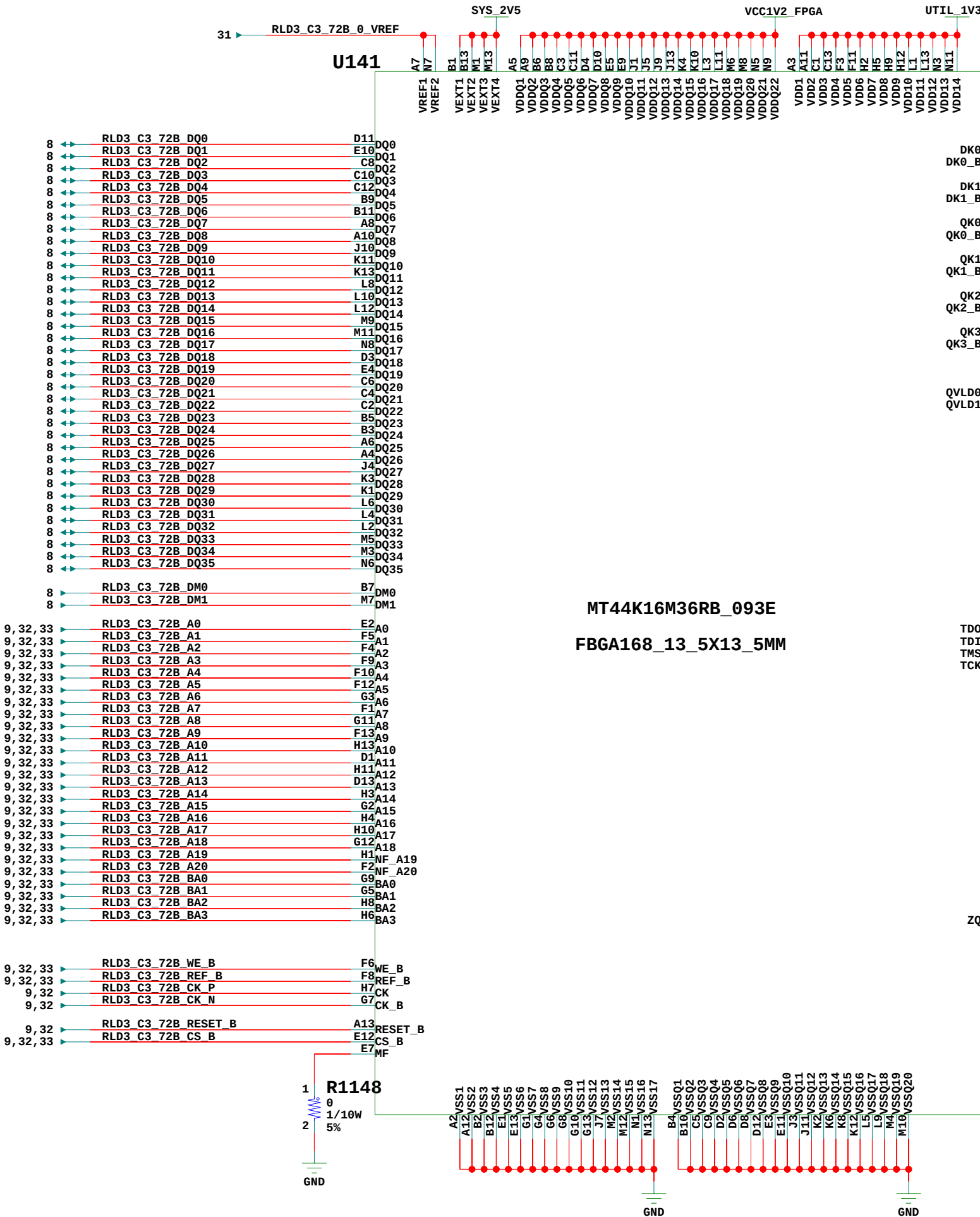
TITLE: DDR4 C1 Data [79-64] SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185
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DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 27 OF 76	DRAWN BY: DN





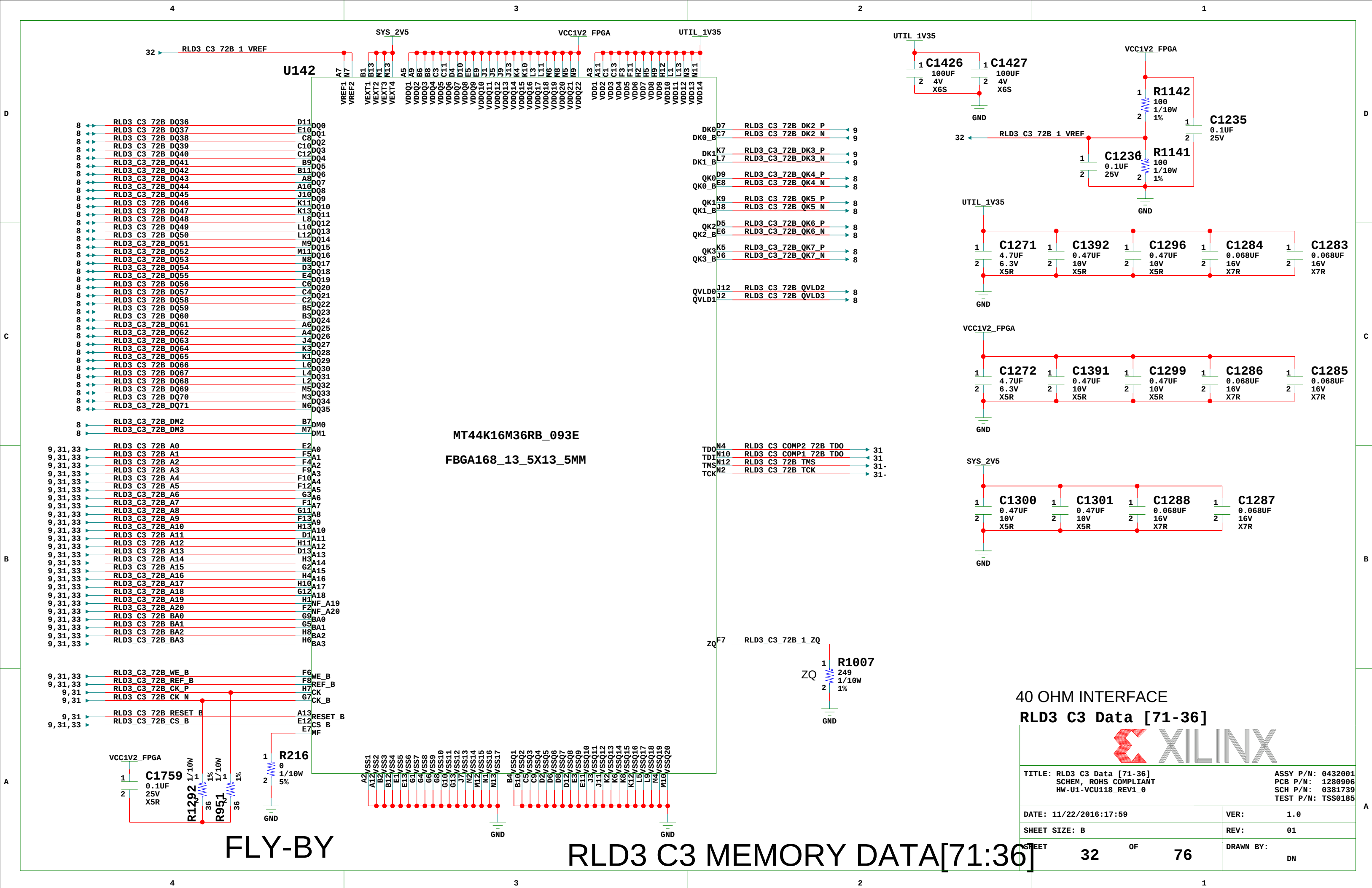


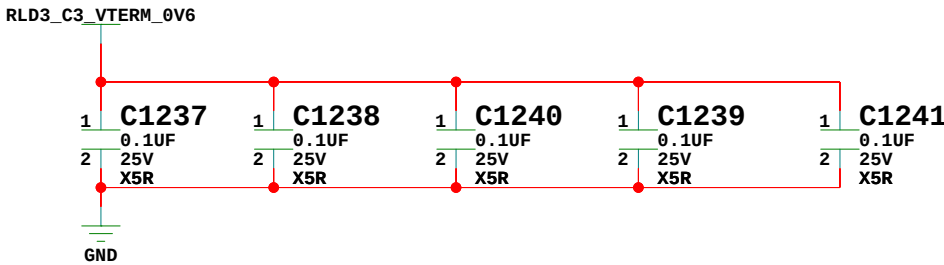
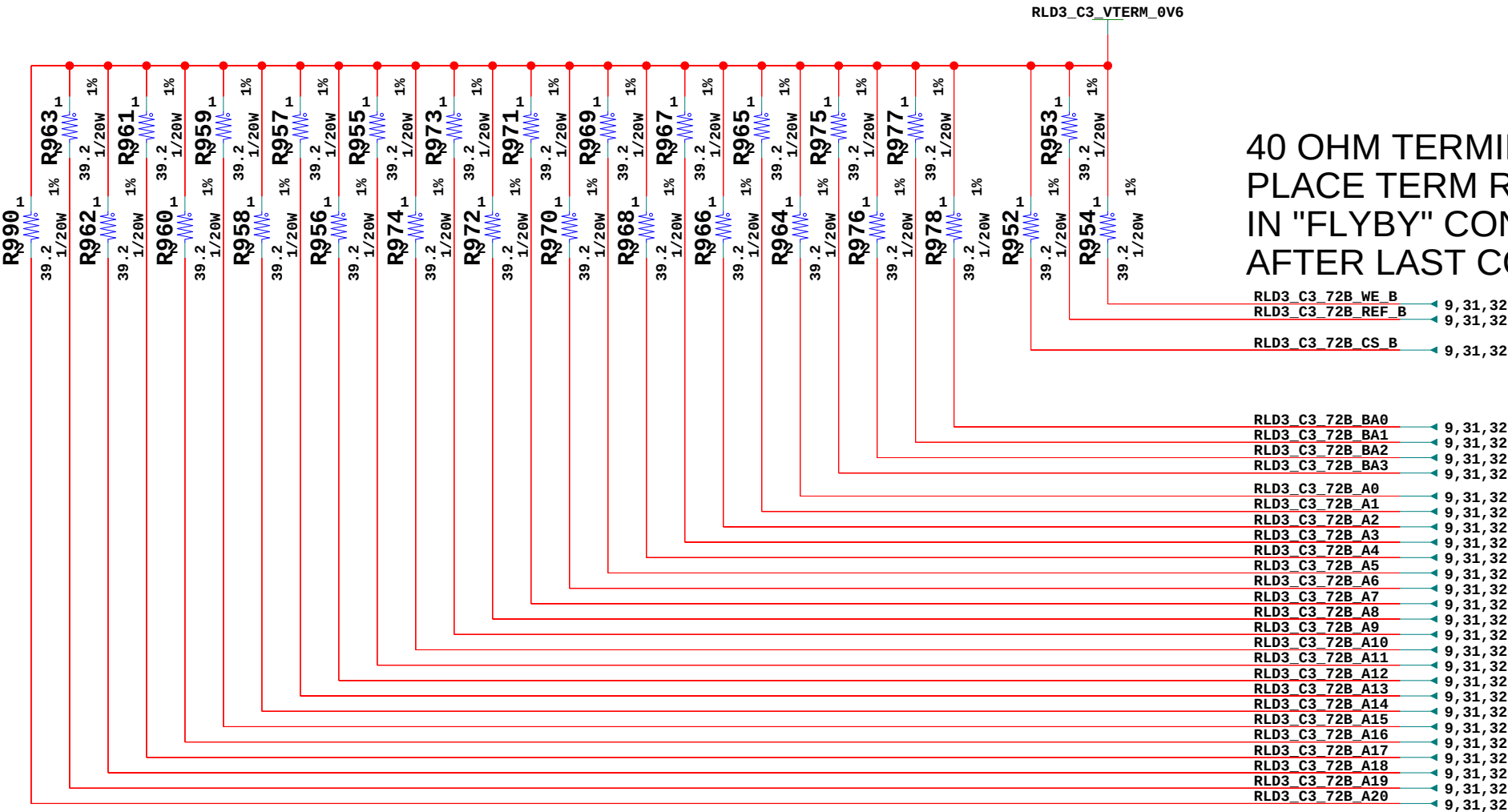


RLD3 C3 MEMORY DATA[35:0]

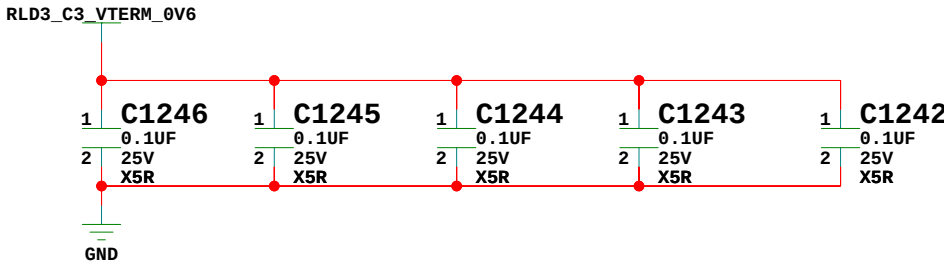
40 OHM INTERFACE RLD3 C3 Data [35-0]

TITLE: RLD3 C3 Data [35-0] SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 31 OF 76	DRAWN BY: DN



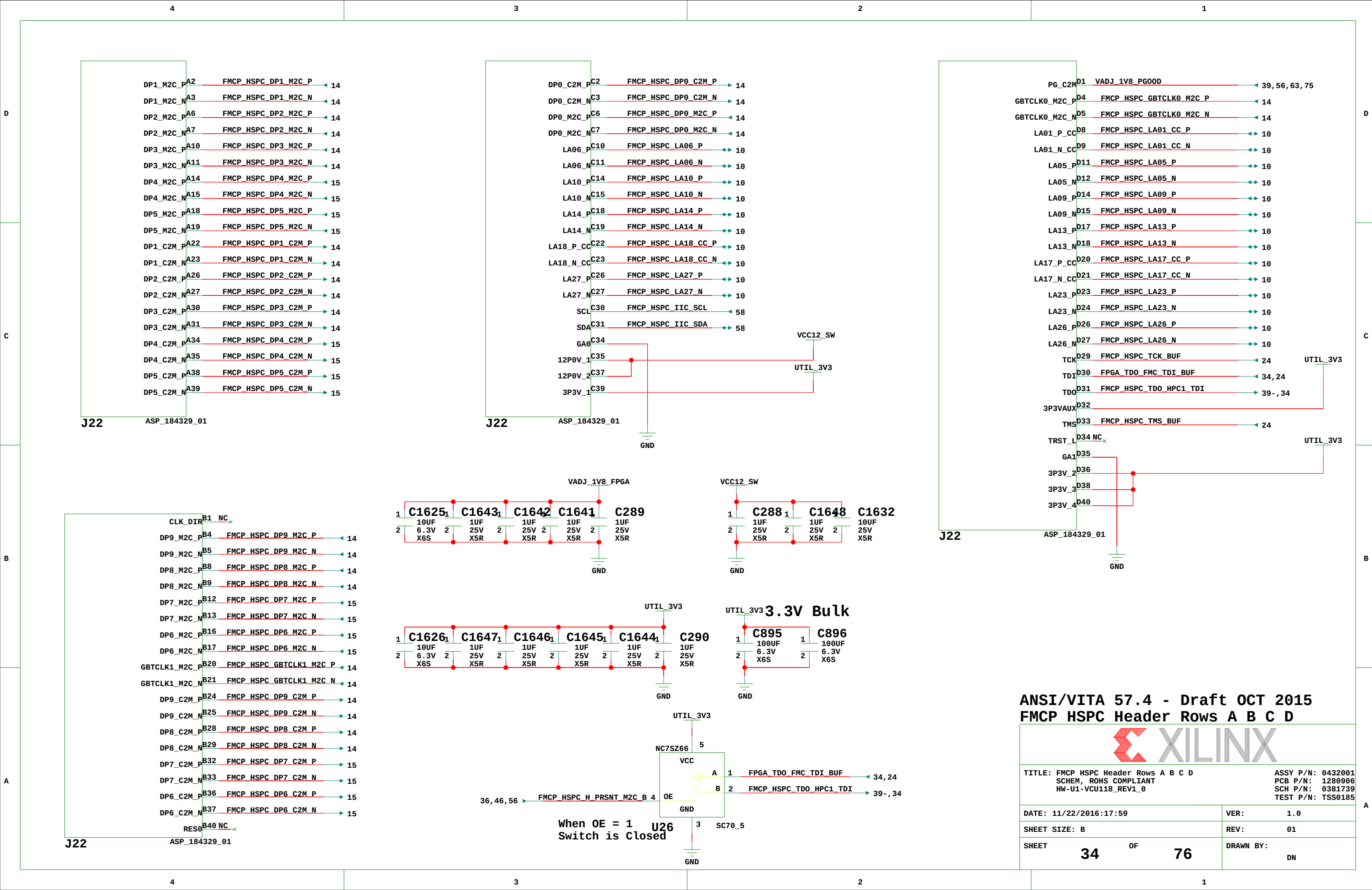


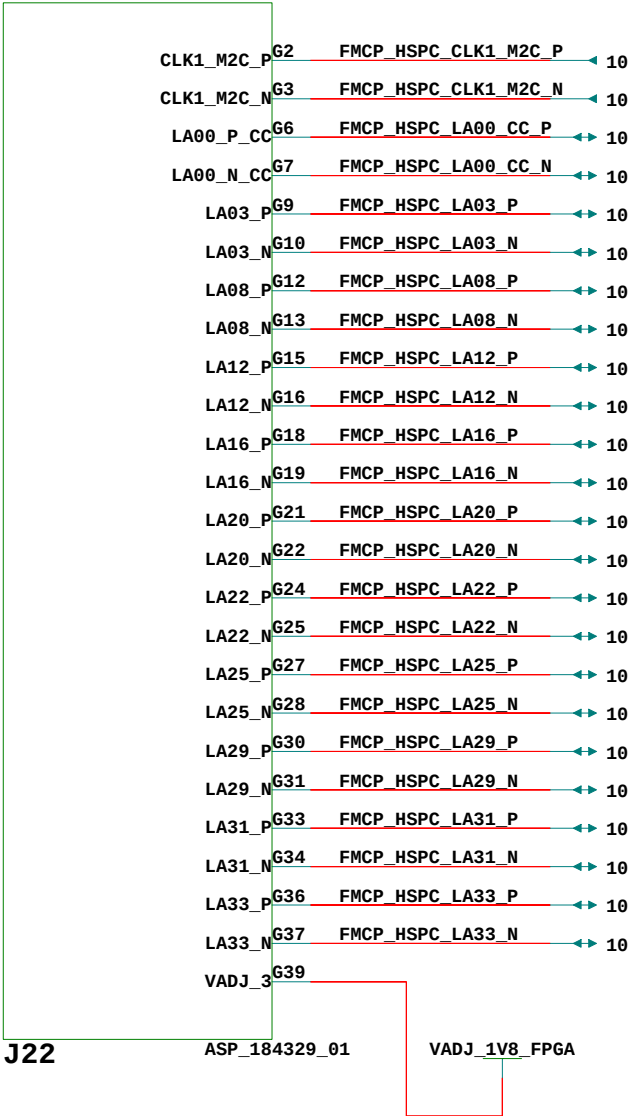
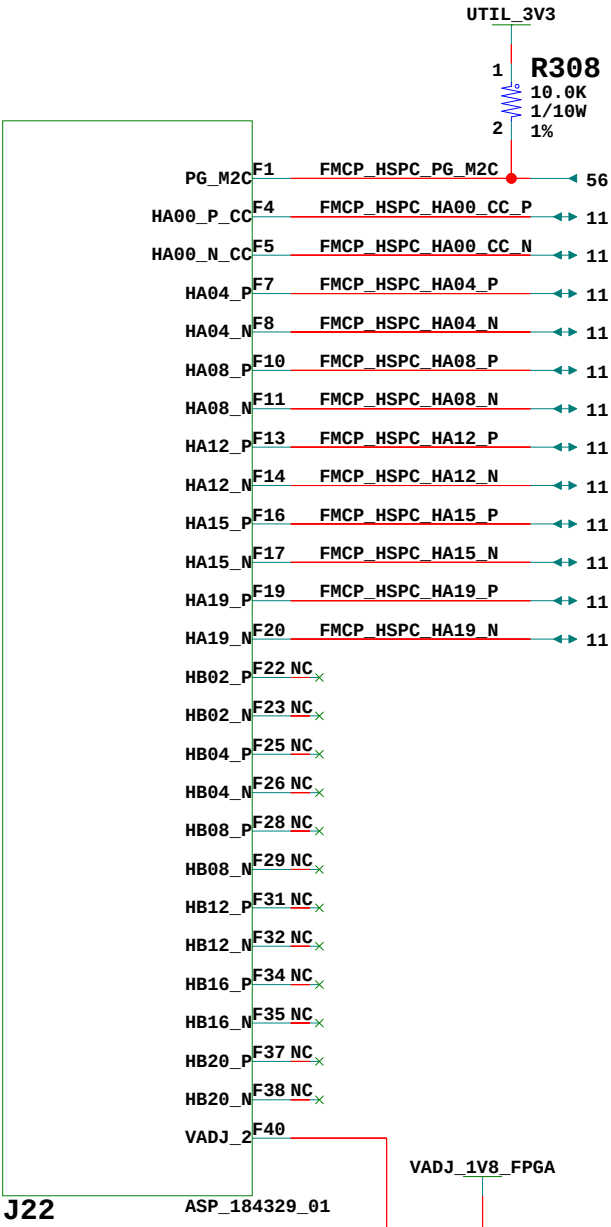
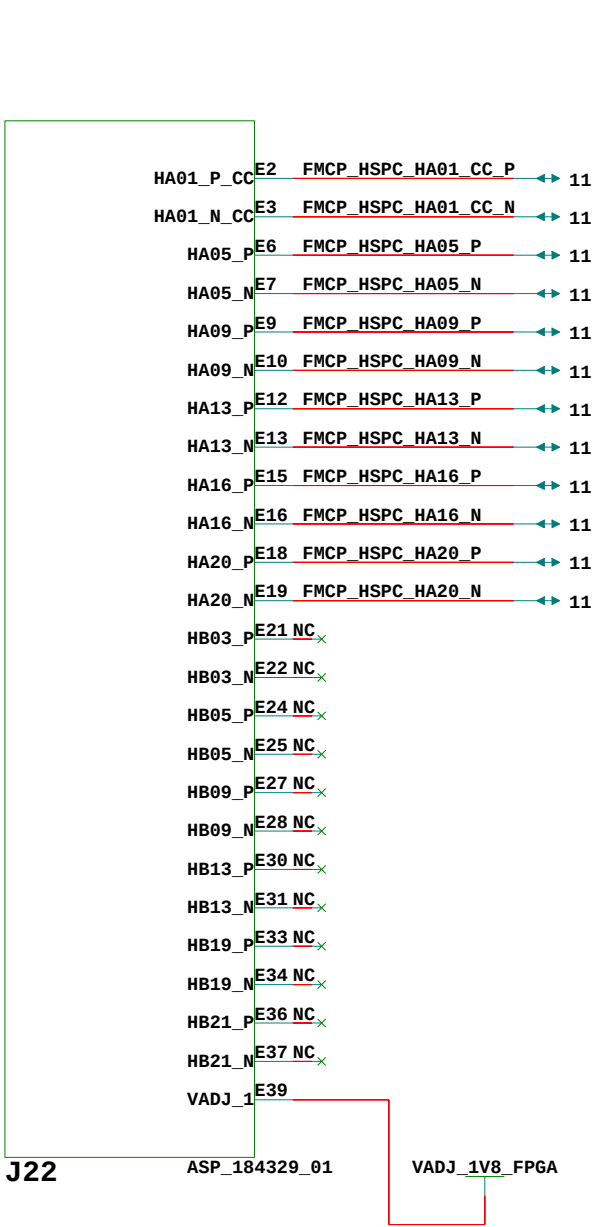
RLD3 C3 72-BIT RTERM DECOUPLING



40 OHM INTERFACE
RLD3 C3 Termination/Decoupling

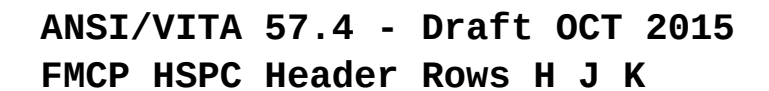
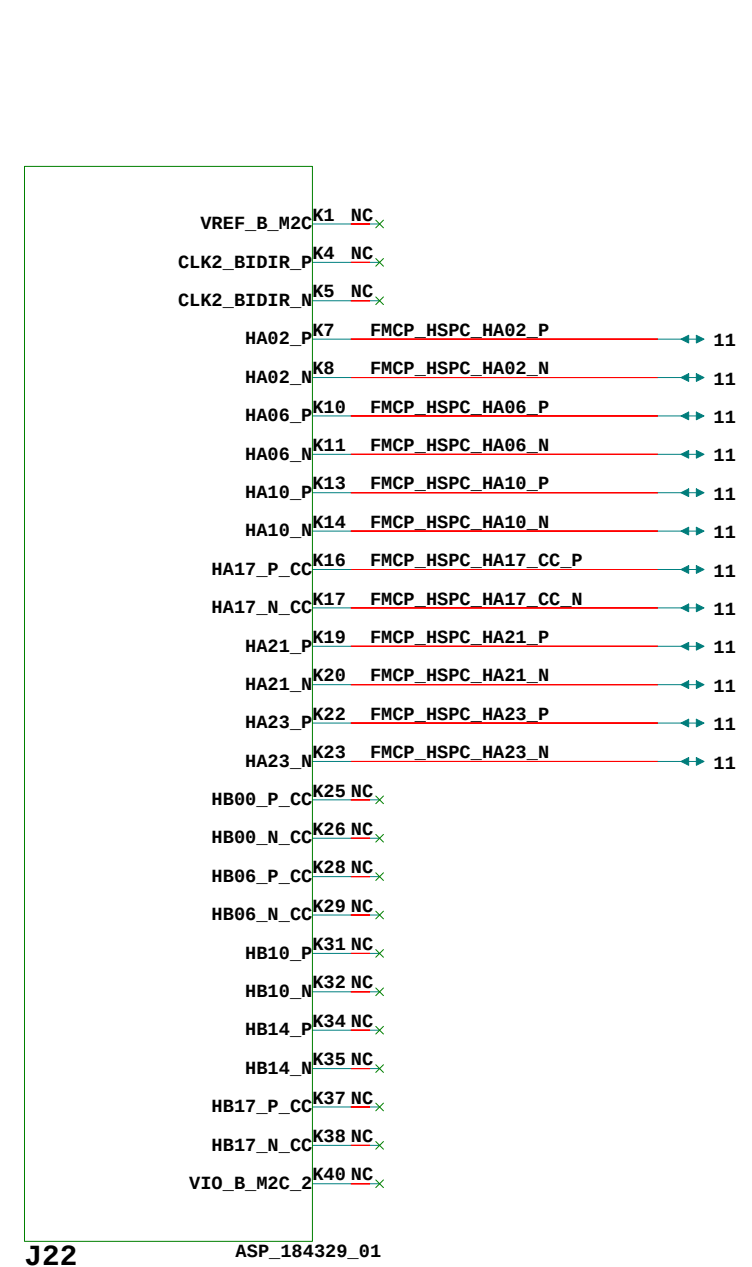
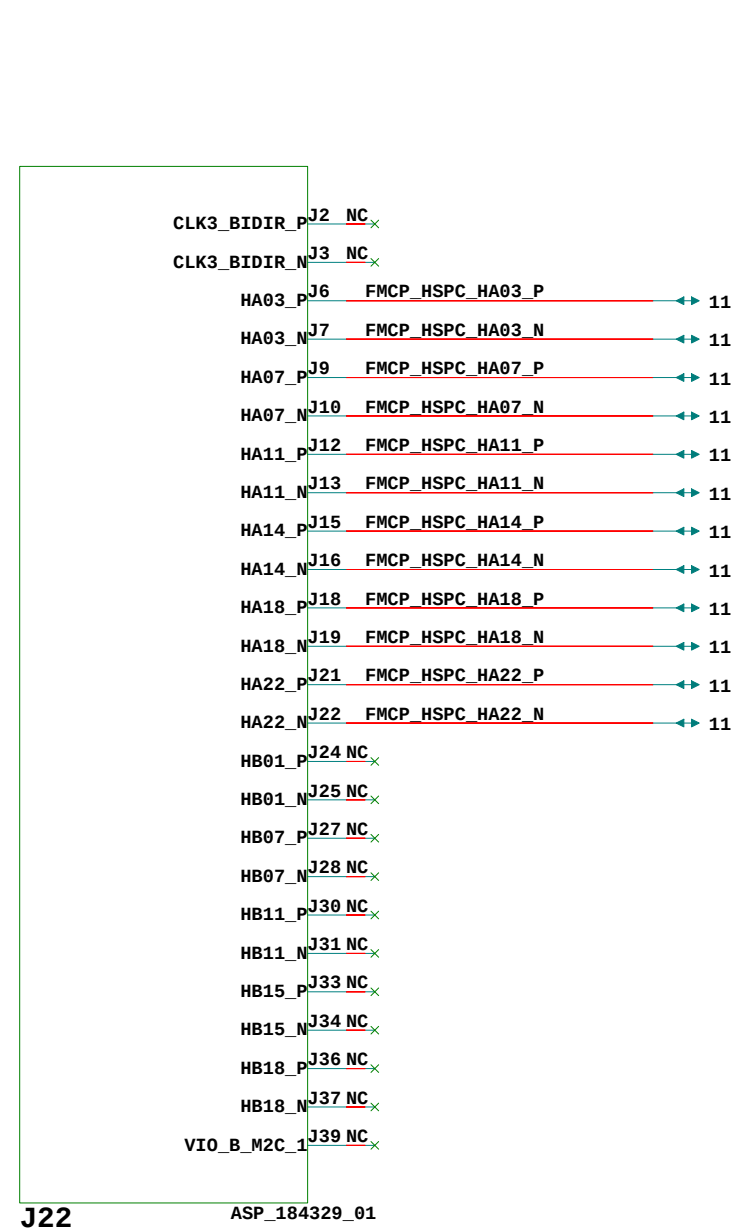
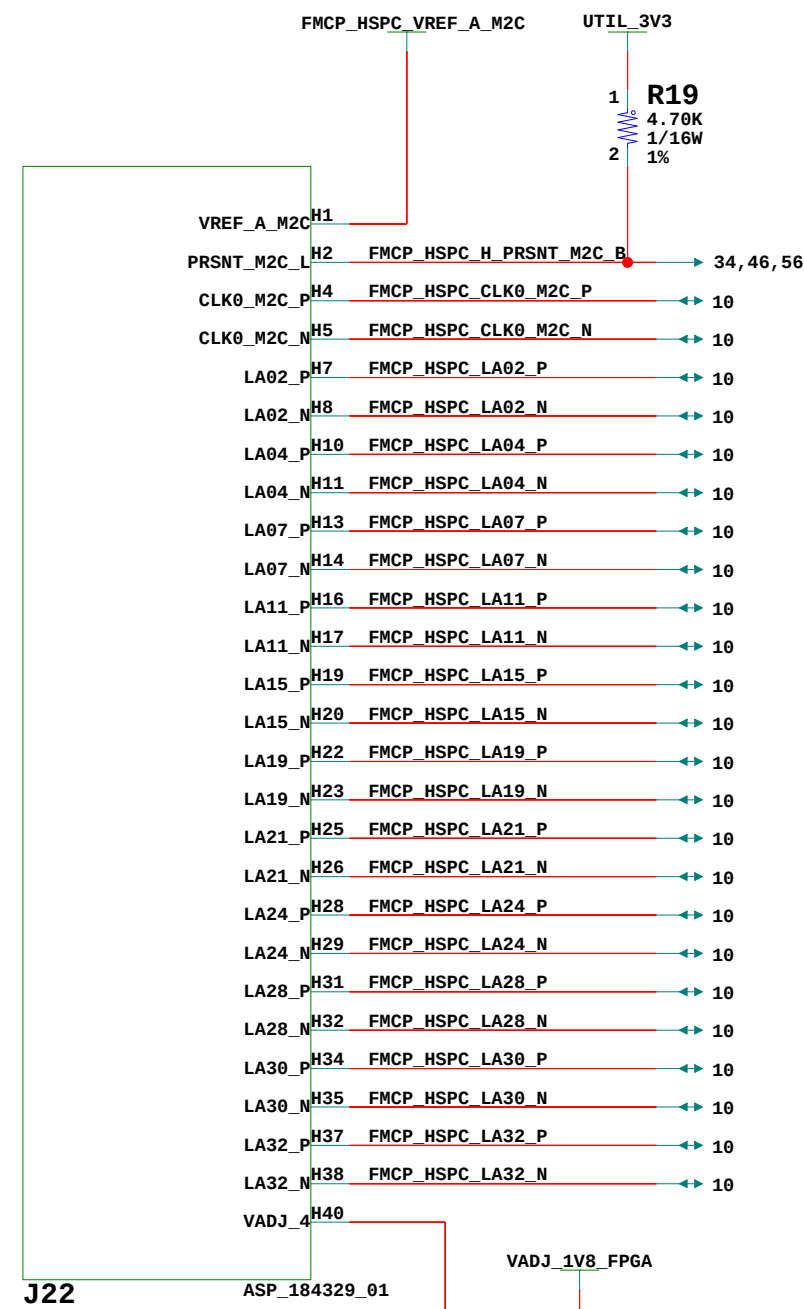
	
TITLE: RLD3 C3 Termination/Decoupling SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 33 OF 76	DRAWN BY: DN





ANSI/VITA 57.4 - Draft OCT 2015
FMCP HSPC Header Rows E F G

TITLE: FMCP HSPC Header Rows E F G SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 35 OF 76	DRAWN BY: DN



TITLE: FMCP HSPC Header Rows H J K
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59

VER:	1.0
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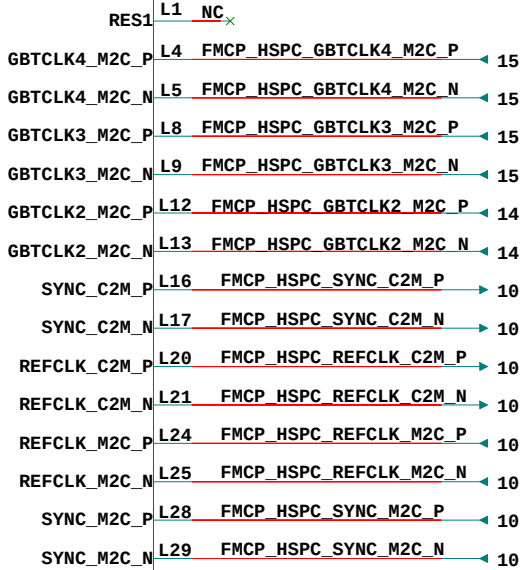
SHEET SIZE: B

REV: 01

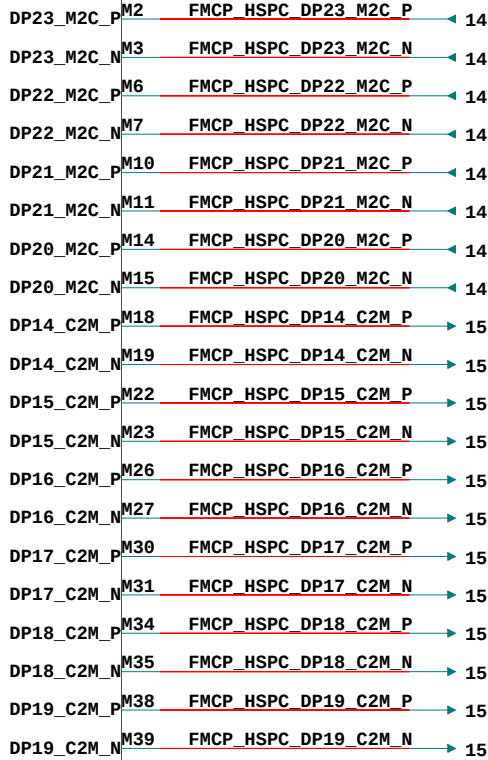
SHEET **36** OF **76**

DRAWN BY:

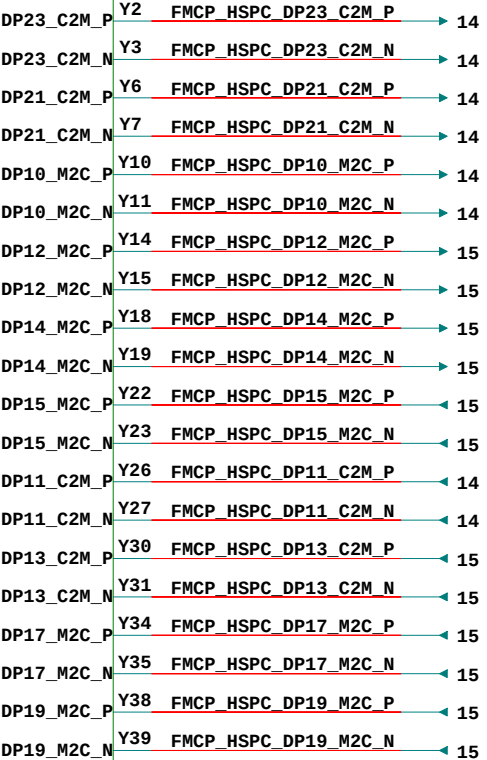
DN



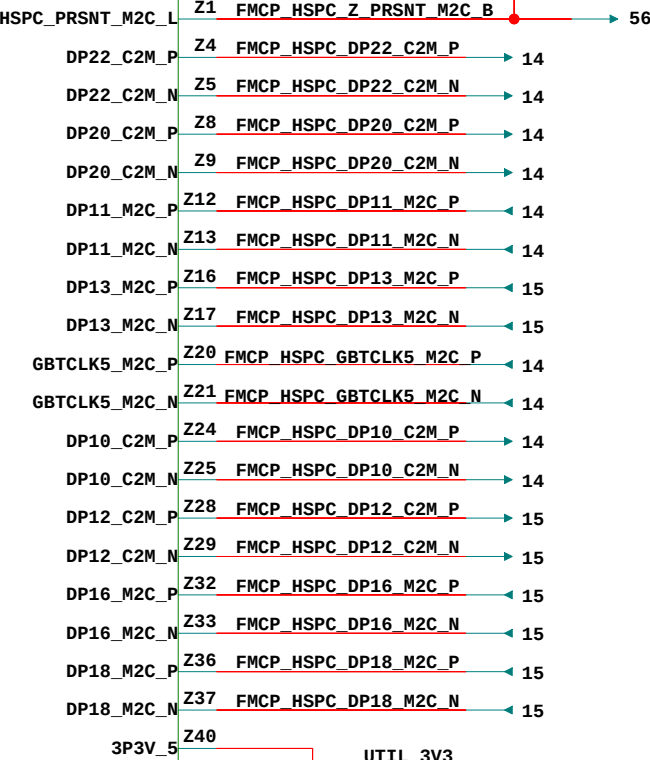
J22 ASP_184329_01



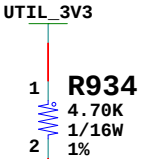
J22 ASP_184329_01



J22 ASP_184329_01



J22 ASP_184329_01



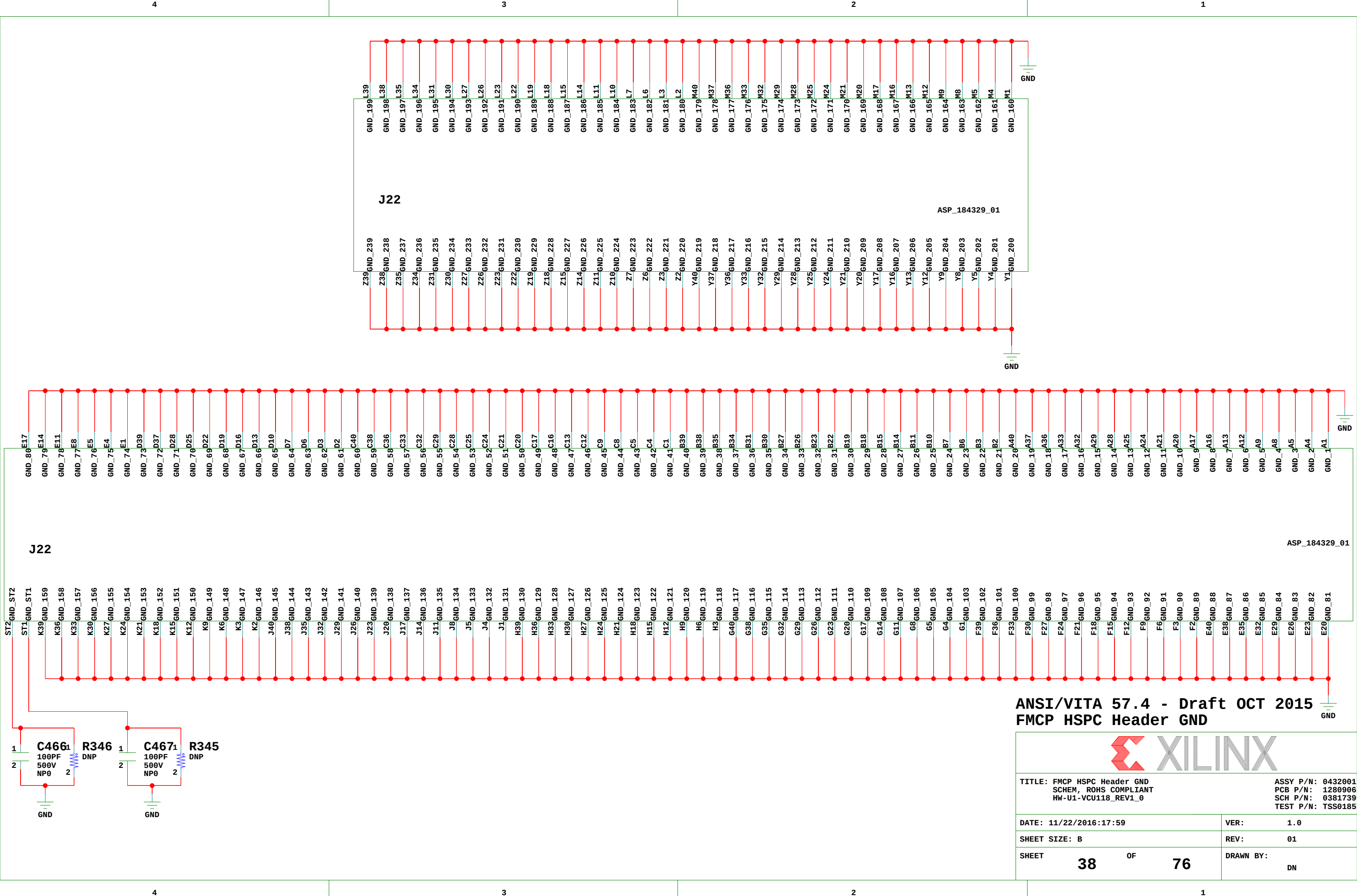
ANSI/VITA 57.4 - Draft OCT 2015
FMCP HSPC Header Rows L M Y Z



TITLE: FMCP HSPC Header Rows L M Y Z
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 37 OF 76	DRAWN BY: DN



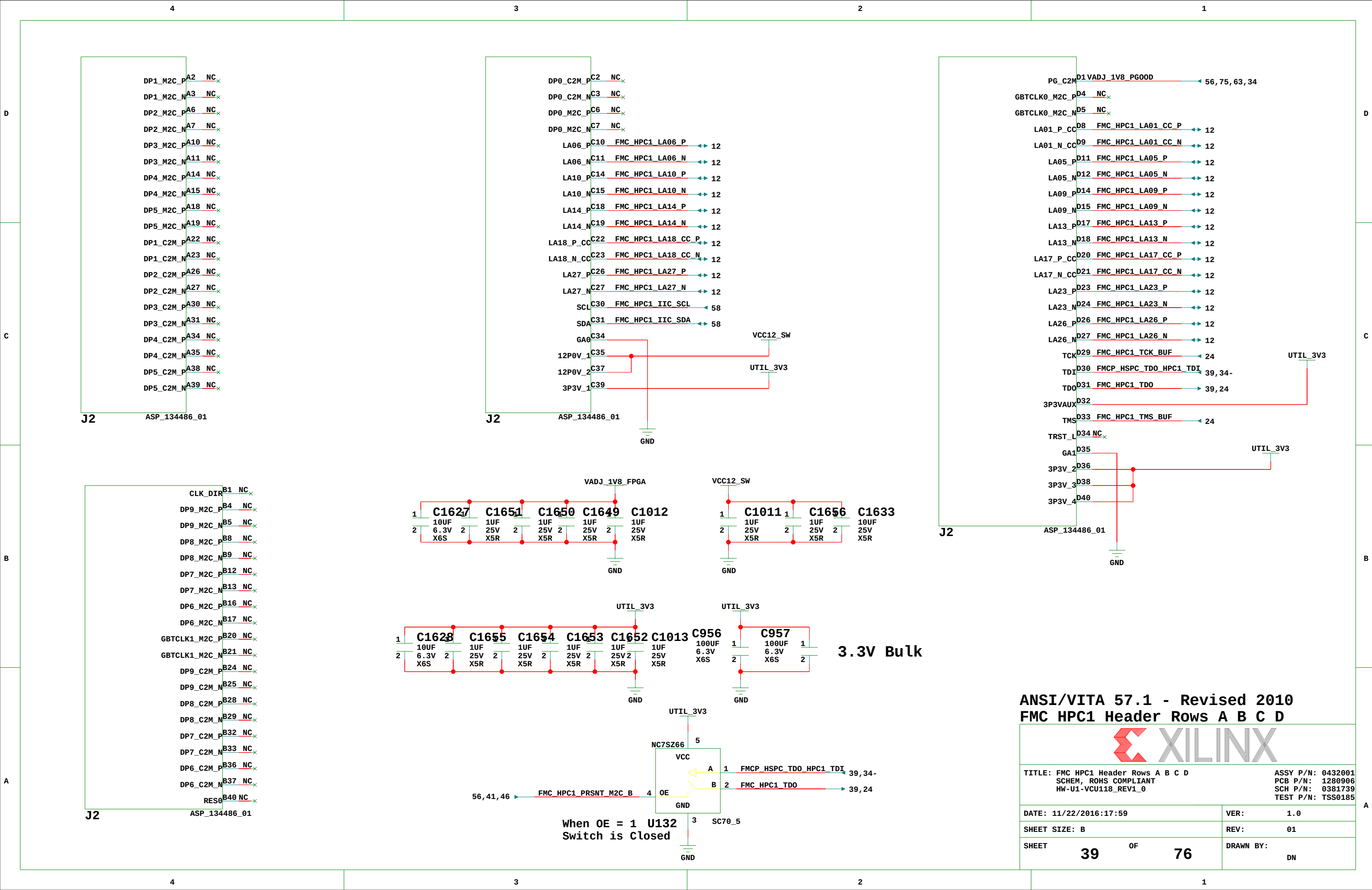
ANSI/VITA 57.4 - Draft OCT 2015
FMCP HSPC Header GND

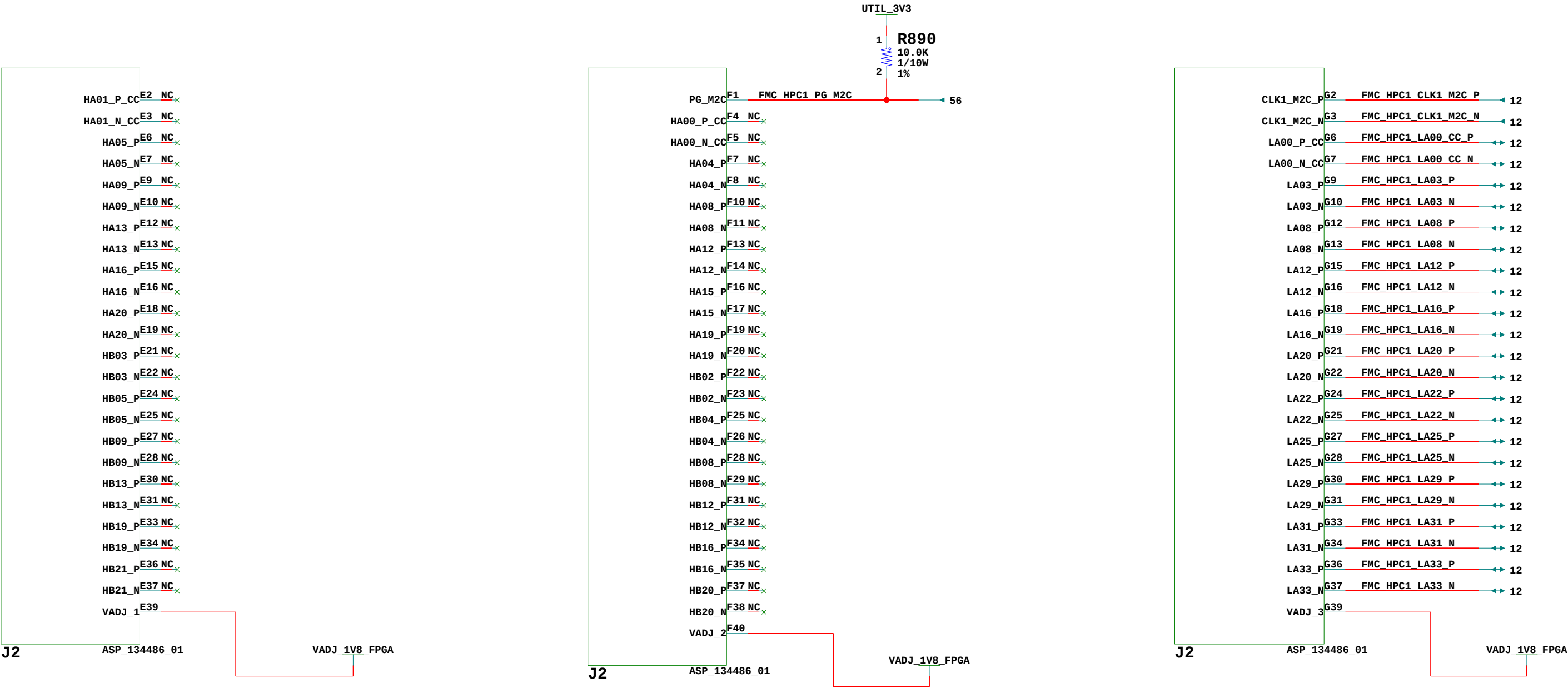


TITLE: FMCP HSPC Header GND
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

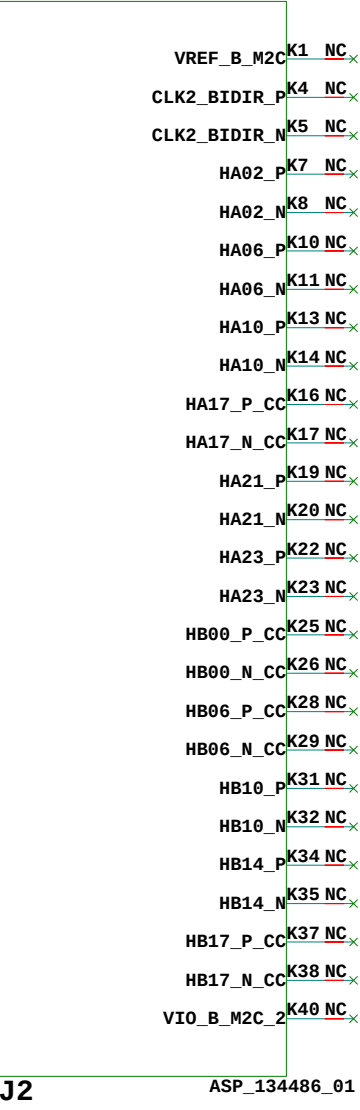
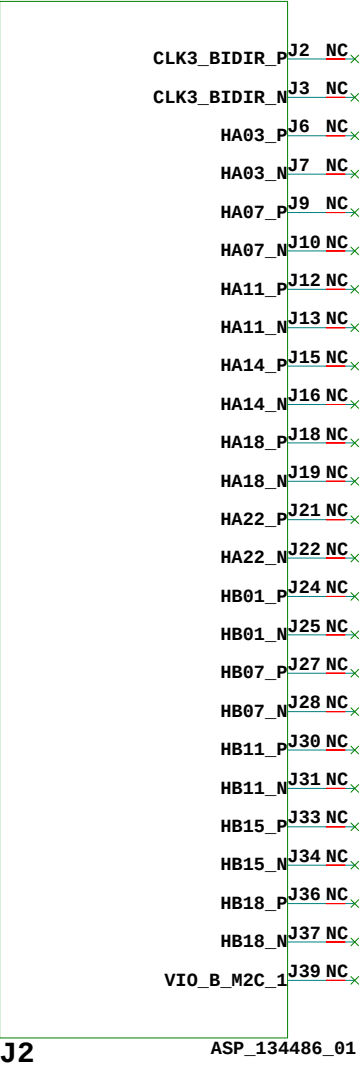
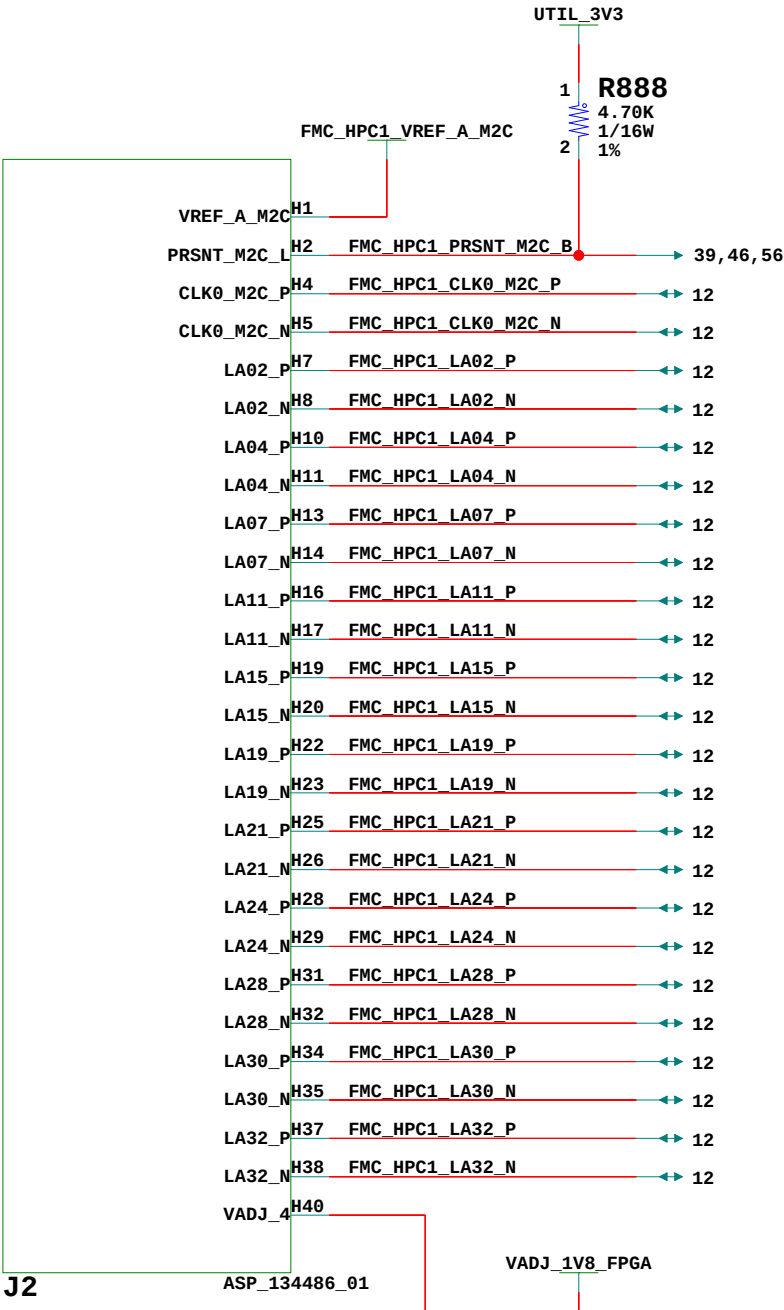
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 38 OF 76	DRAWN BY: DN





ANSI/VITA 57.1 - Revised 2010
FMC HPC1 Header Rows E F G

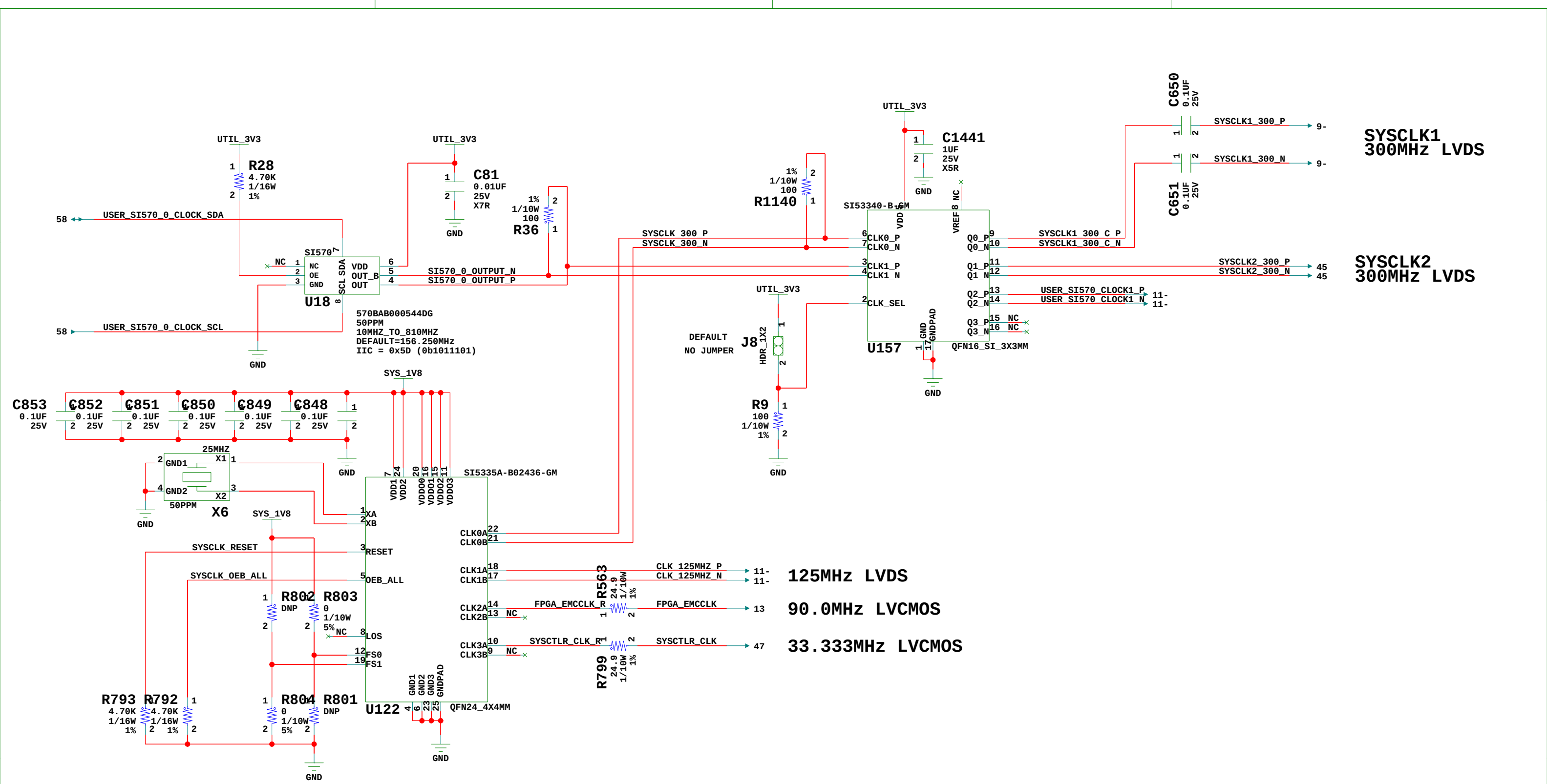
TITLE: FMC HPC1 Header Rows E F G SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 40 OF 76	DRAWN BY: DN



ANSI/VITA 57.1 - Revised 2010
FMC HPC1 Header Rows H J K

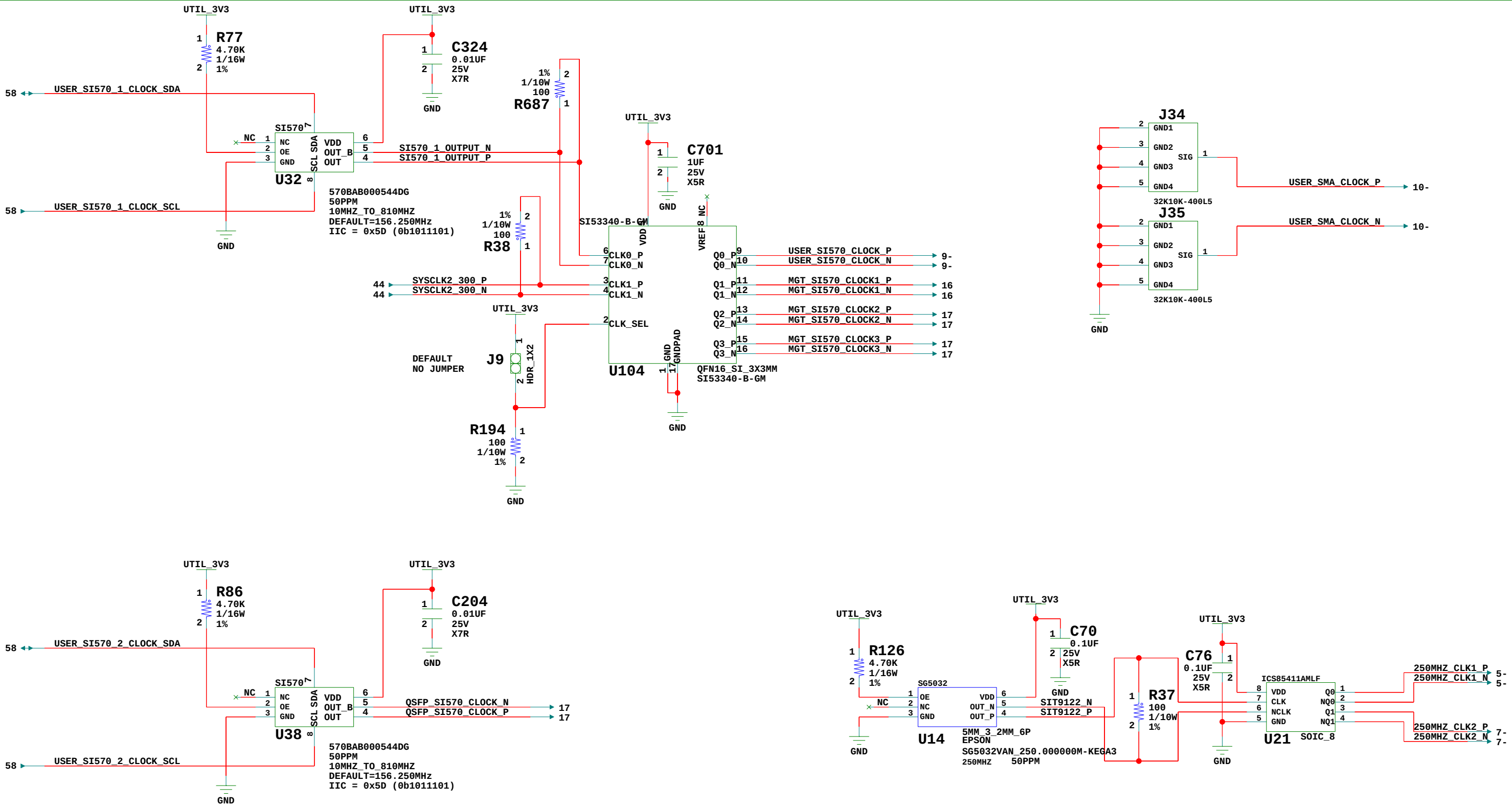
	
TITLE: FMC HPC1 Header Rows H J K SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 41 OF 76	DRAWN BY: DN





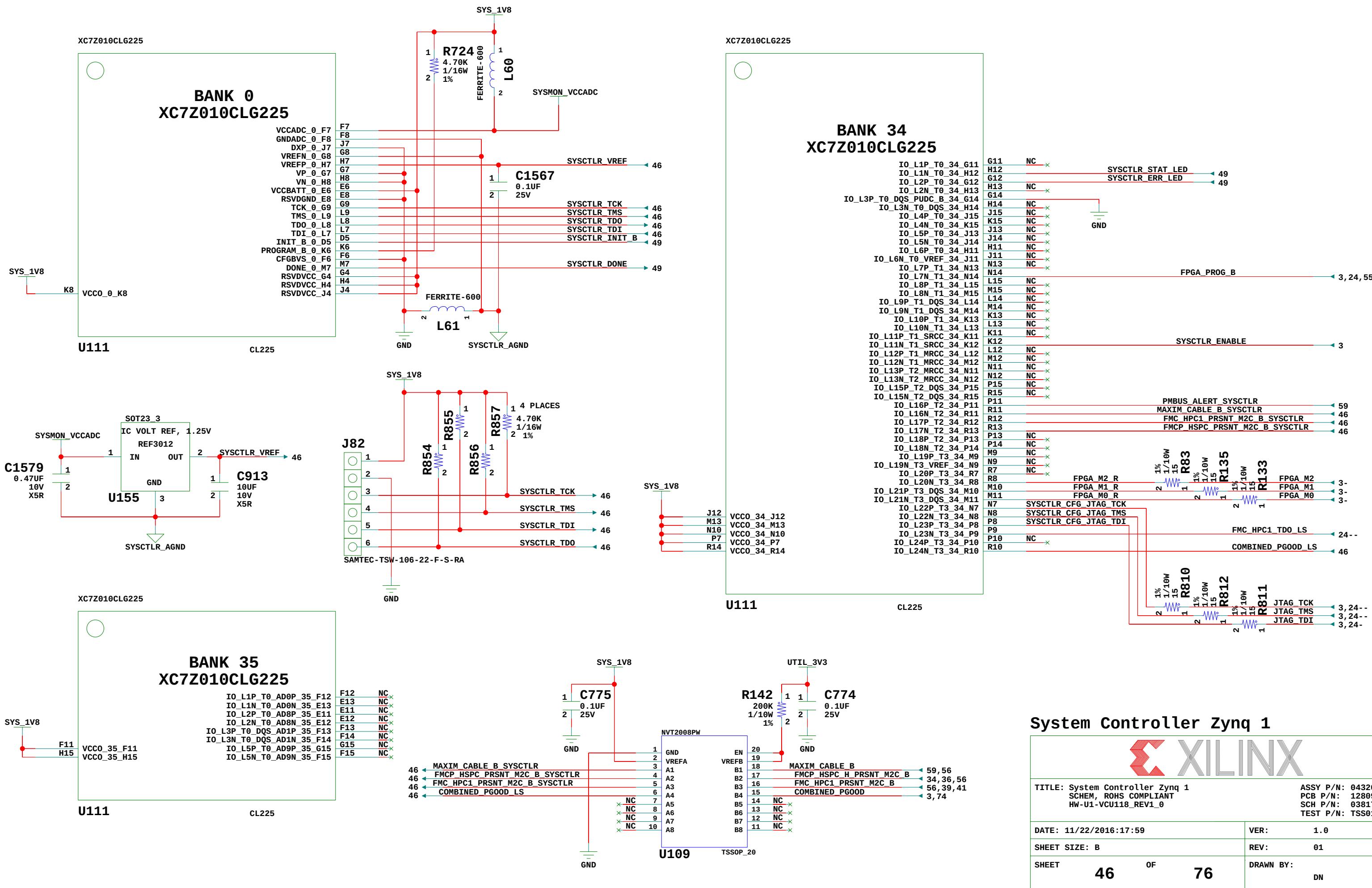
EMCCLK and System Clocks

TITLE: EMCCLK and System Clocks SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 44 OF 76	DRAWN BY: DN



SMA Clk and Buffered Prog Clock

	
TITLE: SMA Clk and Buffered Prog Clock SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 45 OF 76	DRAWN BY: DN



System Controller Zynq 1

TITLE: System Controller Zynq 1 SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 46 OF 76	DRAWN BY: DN

XC7Z010CLG225

BANK 500
XC7Z010CLG225

PS_CLK_500_C7
PS_POR_B_500_C9
PS_MIO15_500_D10
PS_MIO14_500_B9
PS_MIO13_500_C6
PS_MIO12_500_B7
PS_MIO11_500_B10
PS_MIO10_500_D6
PS_MIO9_500_B5
PS_MIO8_500_B6
PS_MIO7_500_D9
PS_MIO6_500_A10
PS_MIO5_500_A9
PS_MIO4_500_C8
PS_MIO3_500_A7
PS_MIO2_500_A8
PS_MIO1_500_A5
PS_MIO0_500_D8

U111

CL225

SYS_1V8

1 R1197 1 R1196 1 R776
20K 20K 20K
1/10W 1/10W 1/10W
1% 1% 1%

SYSCTLR_CLK
SYSCTLR_POR_B
SDIO_DATA3
SDIO_DATA2
SDIO_DATA1
SDIO_CLK
SDIO_CMD
SDIO_DATA0

SYSCTLR_QSPI_SCLK
SYSCTLR_QSPI_IO3
SYSCTLR_QSPI_IO2
SYSCTLR_QSPI_IO1
SYSCTLR_QSPI_IO0
SYSCTLR_QSPI_SS_B
SDIO_CD

1 R1124 1 J110 1 R773
4.70K 1 20K
1/16W 2 HDR_1X2 1/10W
1% 1% 1%

1 R774
20K
1/10W
1%

GND

XC7Z010CLG225

BANK 501
XC7Z010CLG225

PS_MIO29_501_D11
PS_MIO31_501_C12
PS_MIO33_501_C11
PS_MIO35_501_A14
PS_MIO38_501_A13
PS_MIO48_501_B12
PS_MIO52_501_A12
PS_SRST_B_501_B11
PS_MIO28_501_A15
PS_MIO30_501_B15
PS_MIO32_501_E15
PS_MIO34_501_D15
PS_MIO36_501_B14
PS_MIO37_501_C14
PS_MIO39_501_D14
PS_MIO49_501_D13
PS_MIO53_501_C13

U111

CL225

SYS_1V8

1 R826
4.70K
1/16W
1%

IIC_MAIN_SDA
SYSCTLR_UART_TX
SYSCTLR_MODE1
SYSCTLR_MODE3
GPIO_SW_N
GPIO_SW_C
SYSCTLR_GPIO_6
IIC_MAIN_SCL
SYSCTLR_UART_RX
SYSCTLR_MODE0
SYSCTLR_MODE2
GPIO_SW_E
GPIO_SW_S
GPIO_SW_W
SYSCTLR_GPIO_5
SYSCTLR_GPIO_7

XC7Z010CLG225

BANK 502
XC7Z010CLG225

PS_DDR_DRST_B_502_L4
PS_DDR_DQ0_502_D4
PS_DDR_DQ1_502_A2
PS_DDR_DQ2_502_C4
PS_DDR_DQ3_502_C1
PS_DDR_DM0_502_B1
PS_DDR_DQS_P0_502_C2
PS_DDR_DQS_N0_502_B2
PS_DDR_DQ4_502_B4
PS_DDR_DQ5_502_A4
PS_DDR_DQ6_502_C3
PS_DDR_DQ7_502_A3
PS_DDR_DQ8_502_E1
PS_DDR_DQ9_502_D1
PS_DDR_DQ10_502_E2
PS_DDR_DQ11_502_E3
PS_DDR_DM1_502_D3
PS_DDR_DQS_P1_502_G2
PS_DDR_DQS_N1_502_F2
PS_DDR_DQ12_502_F3
PS_DDR_DQ13_502_G1
PS_DDR_DQ14_502_H1
PS_DDR_DQ15_502_H2
PS_DDR_A14_502_K1
PS_DDR_A13_502_K2
PS_DDR_A12_502_M2
PS_DDR_A11_502_L2
PS_DDR_A10_502_J1
PS_DDR_A9_502_N4
PS_DDR_A8_502_P6
PS_DDR_A7_502_M5
PS_DDR_A6_502_P5
PS_DDR_A5_502_P4
PS_DDR_A4_502_P3
PS_DDR_A3_502_M4
PS_DDR_VRN_502_J3
PS_DDR_VRP_502_H3
PS_DDR_CKP_502_N3
PS_DDR_CKN_502_N2
PS_DDR_A2_502_M1
PS_DDR_A1_502_N1
PS_DDR_A0_502_P1
PS_DDR_BA2_502_N6
PS_DDR_BA1_502_R1
PS_DDR_BA0_502_M6
PS_DDR_ODT_502_K3
PS_DDR_CS_B_502_R2
PS_DDR_CKE_502_L3
PS_DDR_WE_B_502_R3
PS_DDR_CAS_B_502_R5
PS_DDR_RAS_B_502_R6
PS_DDR_VREF0_502_F4

U111

CL225

SYS_1V8

B3 VCCO_DDR_502_B3
E4 VCCO_DDR_502_E4
F1 VCCO_DDR_502_F1
J2 VCCO_DDR_502_J2
M3 VCCO_DDR_502_M3
R4 VCCO_DDR_502_R4

System Controller Zynq 2



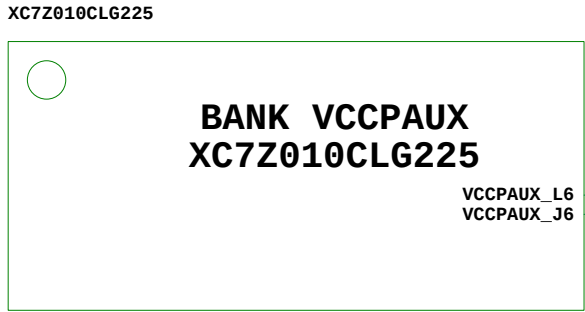
TITLE: System Controller Zynq 2
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 47 OF 76	DRAWN BY: DN



U111

CL225



U111

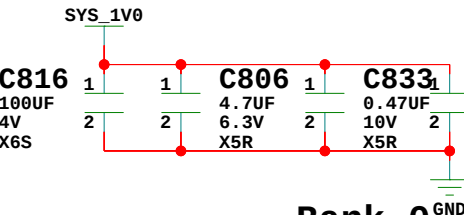
CL225



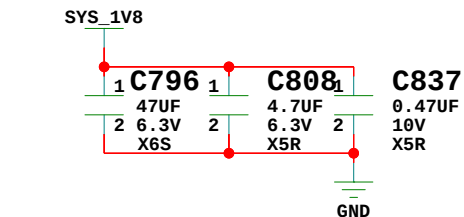
U111

CL225

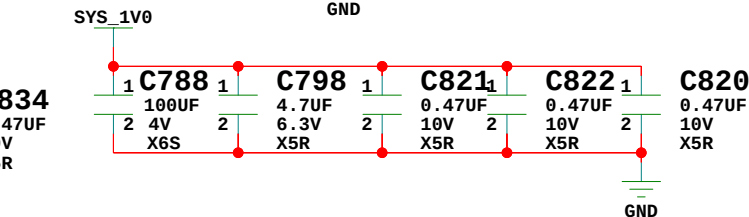
VCCINT/VCCBRAM



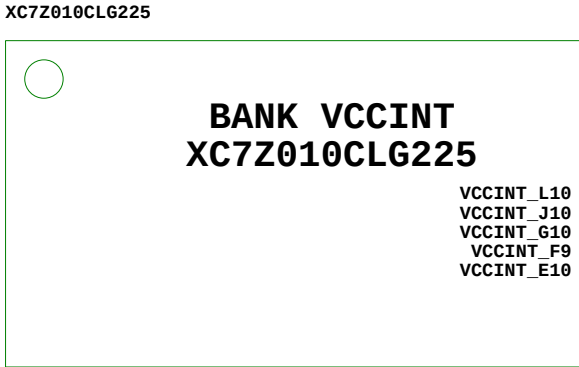
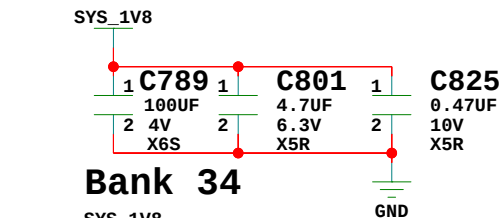
VCCAUX



VCCPINT

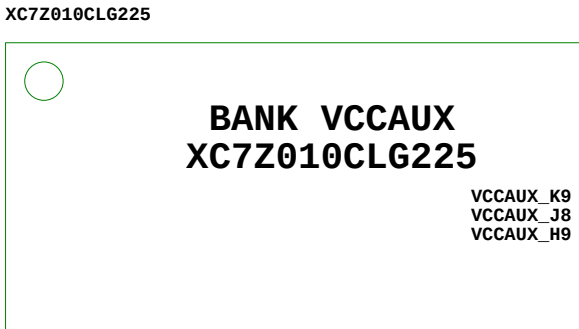


VCCPAUX



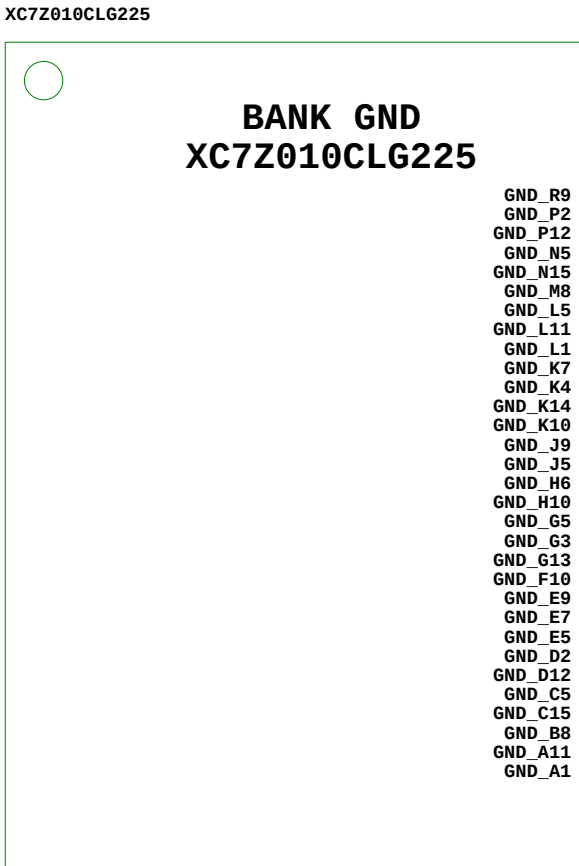
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CL225



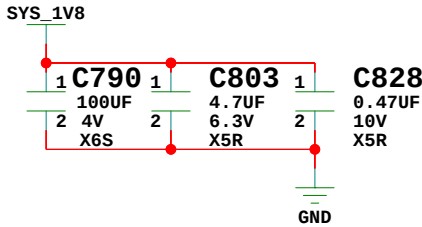
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CL225

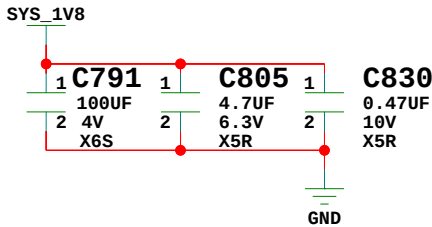


U111

VCCO_MI00_BANK500



VCCO_MI01_BANK501



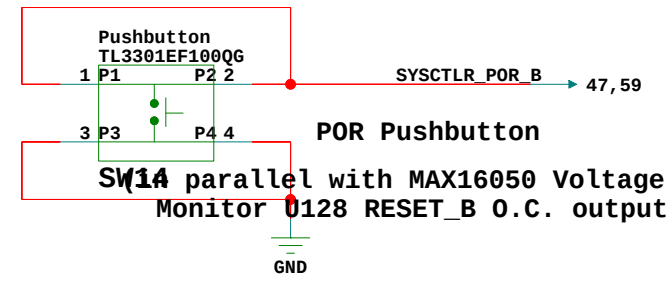
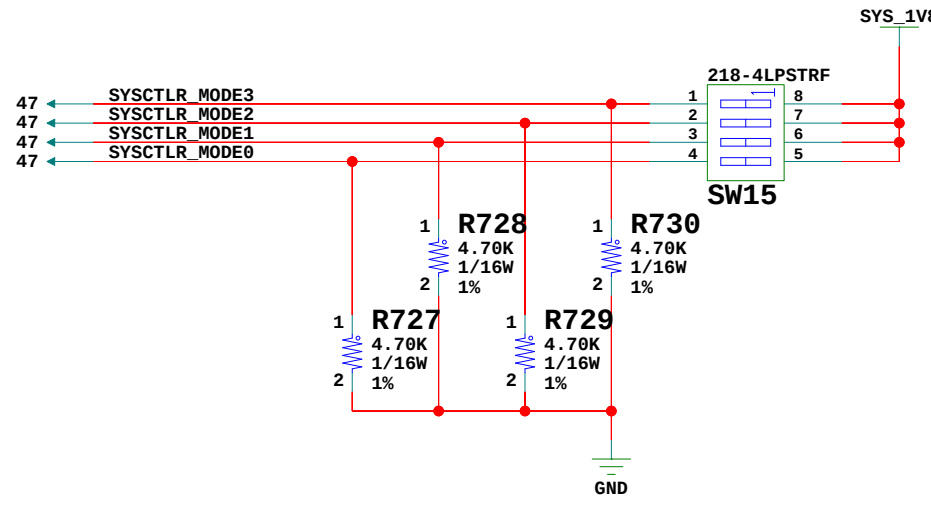
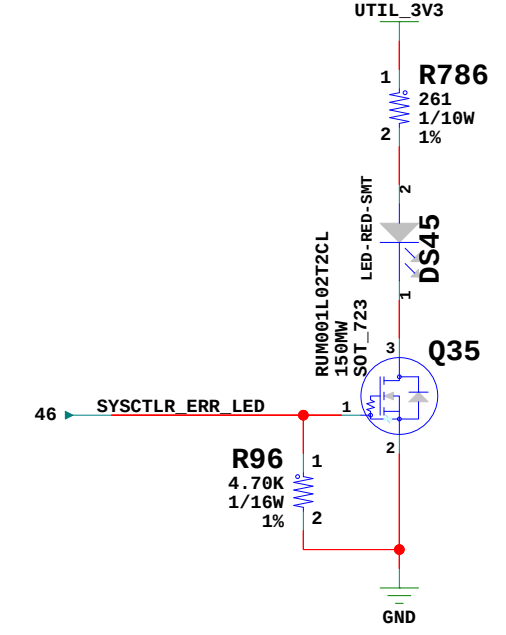
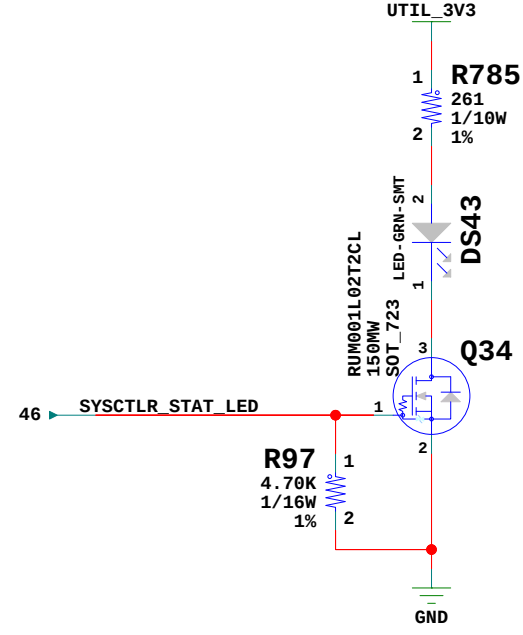
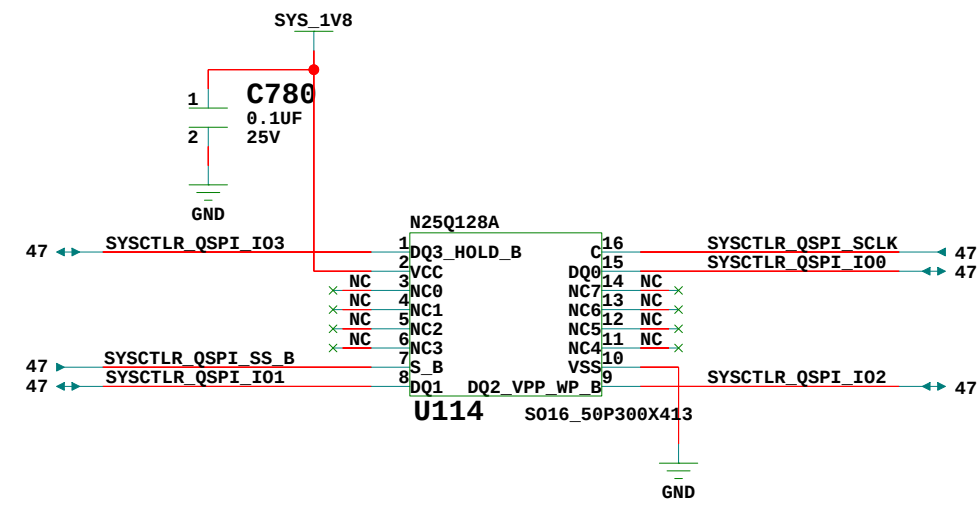
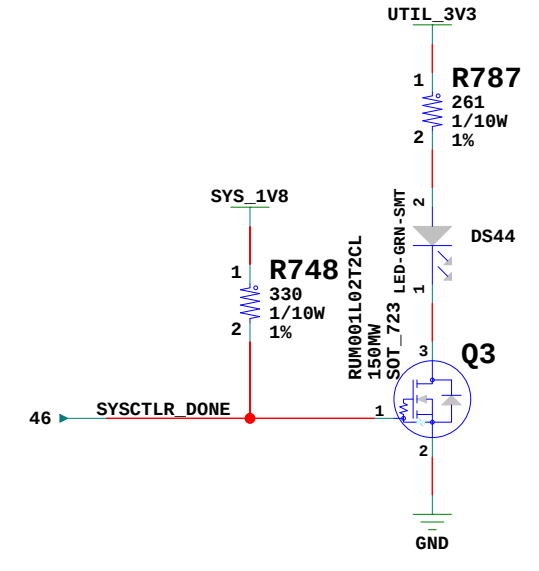
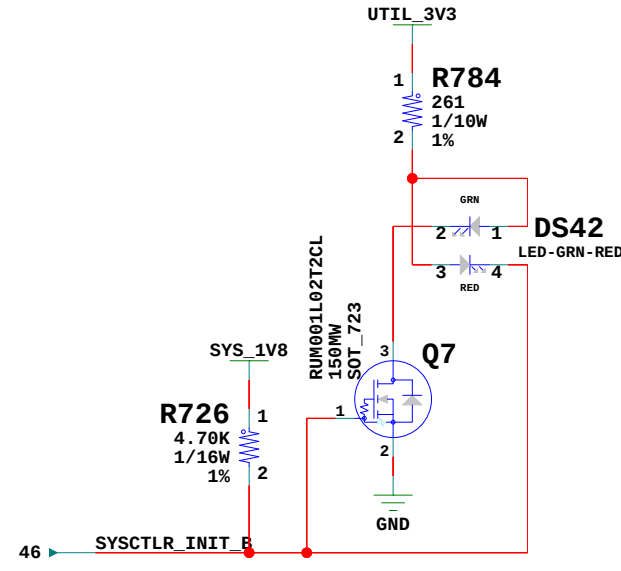
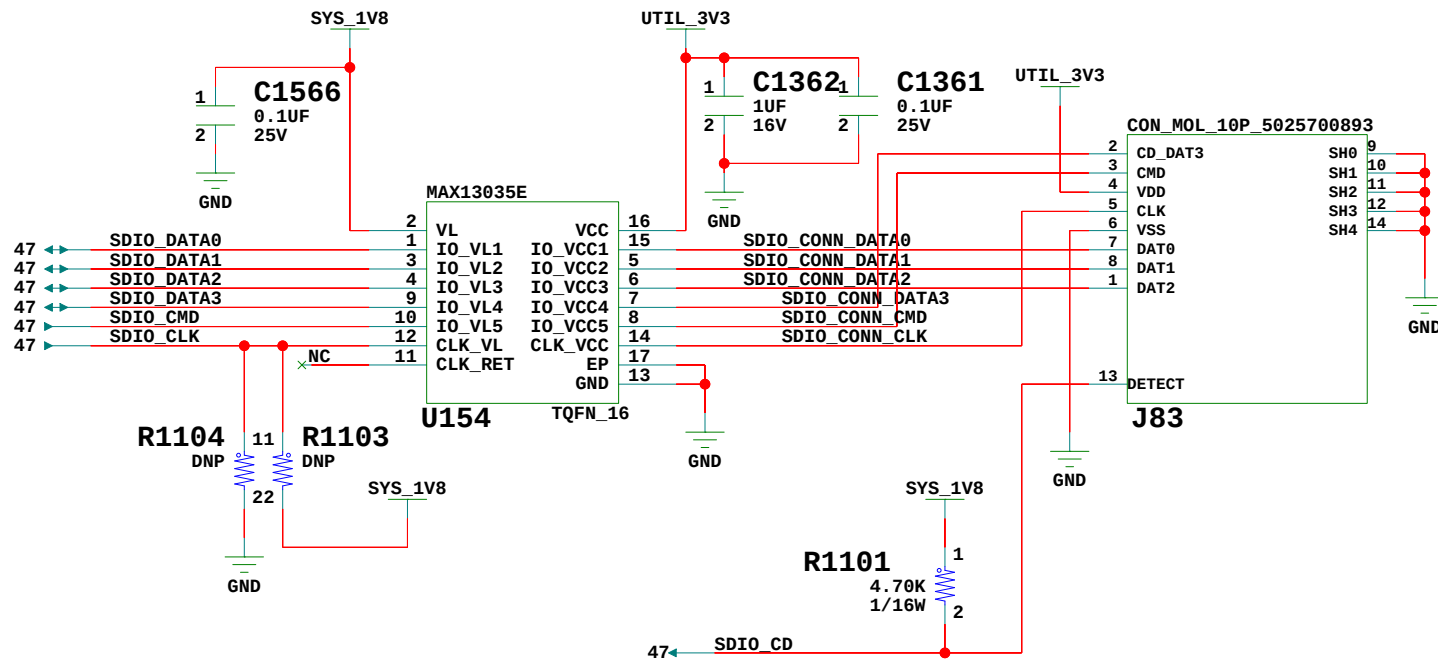
System Controller Zynq 3



TITLE: System Controller Zynq 3
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

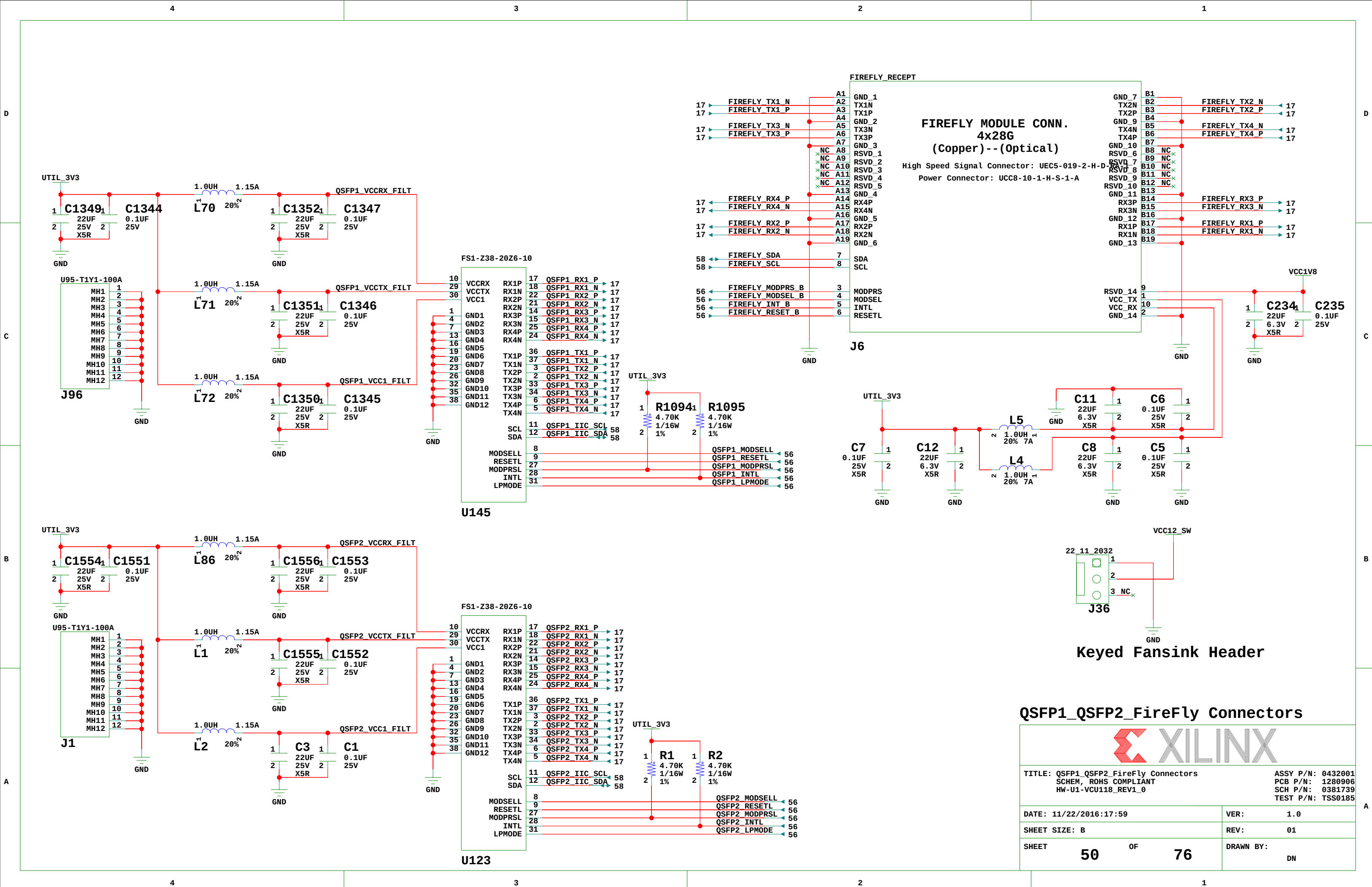
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

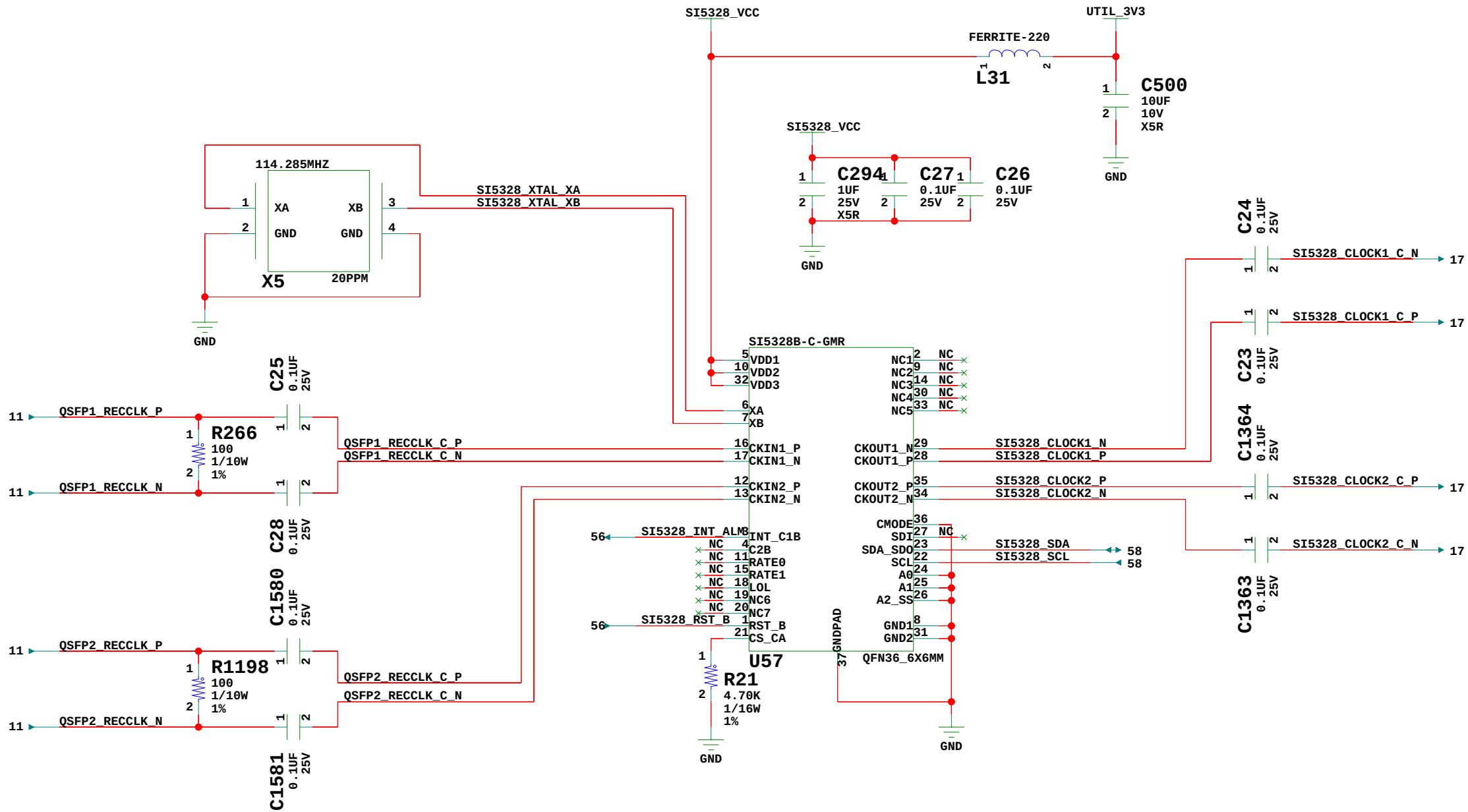
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 48 OF 76	DRAWN BY: DN



System Controller Zynq 4

TITLE: System Controller Zynq 4 SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 49 OF 76	DRAWN BY: DN





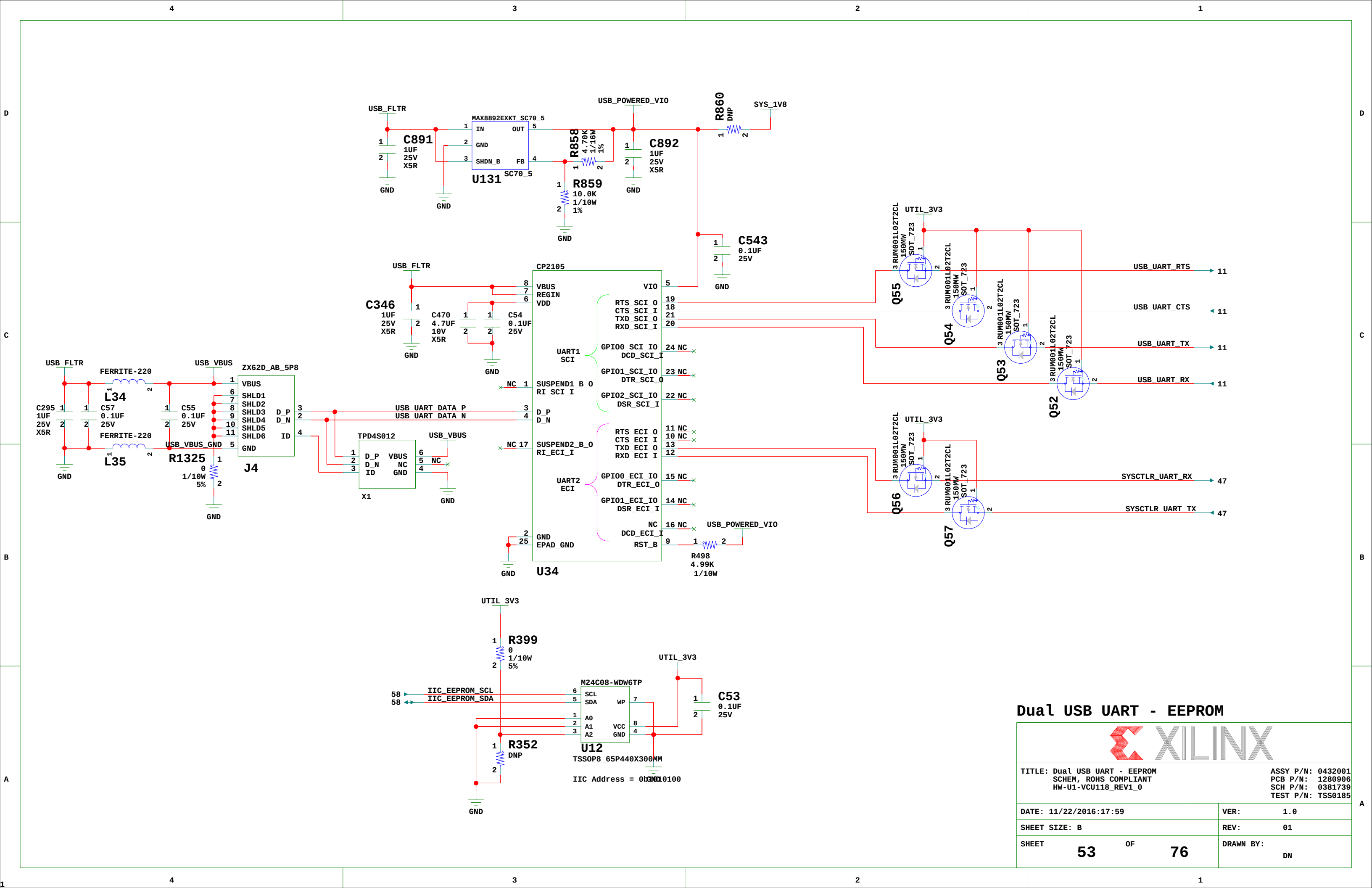
QSFPP+ 5328 Jitter Recovery



TITLE: QSFPP+ 5328 Jitter Recovery SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185
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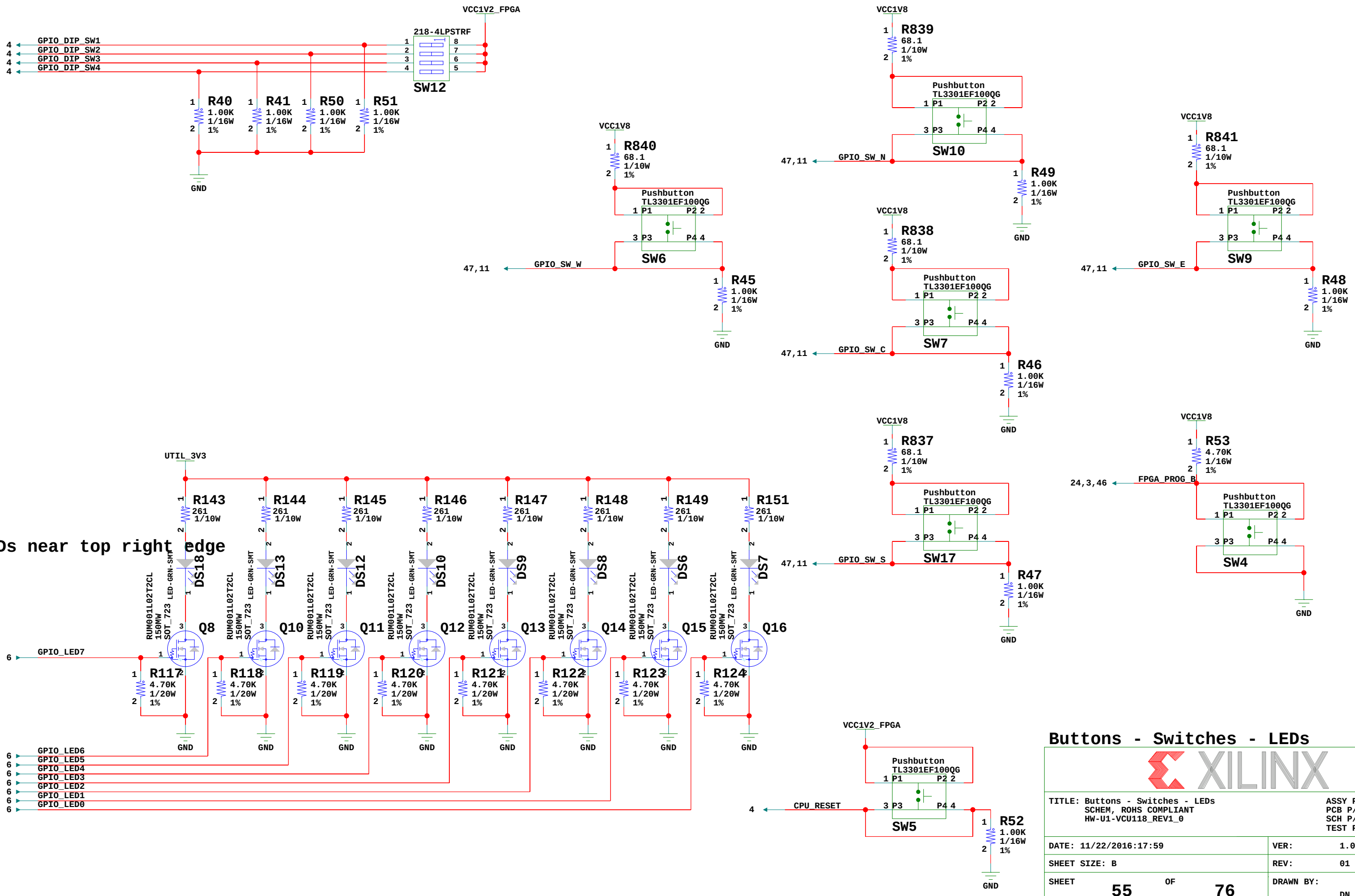
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 51 OF 76	DRAWN BY: DN

10/100/1000 Ethernet PHY - SGMII



Dual USB UART - EEPROM

TITLE: Dual USB UART - EEPROM SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 53 OF 76	DRAWN BY: DN



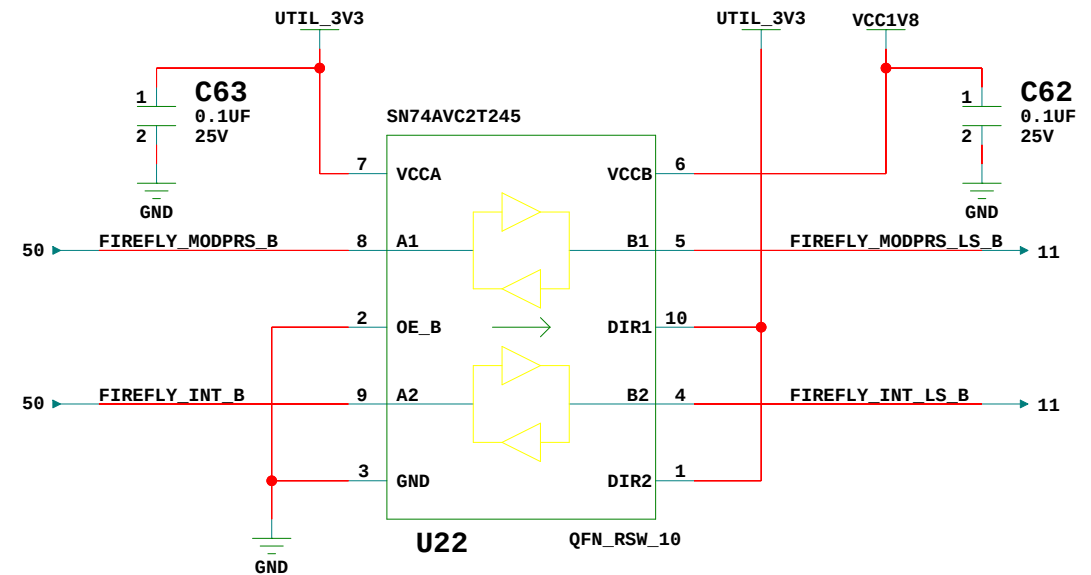
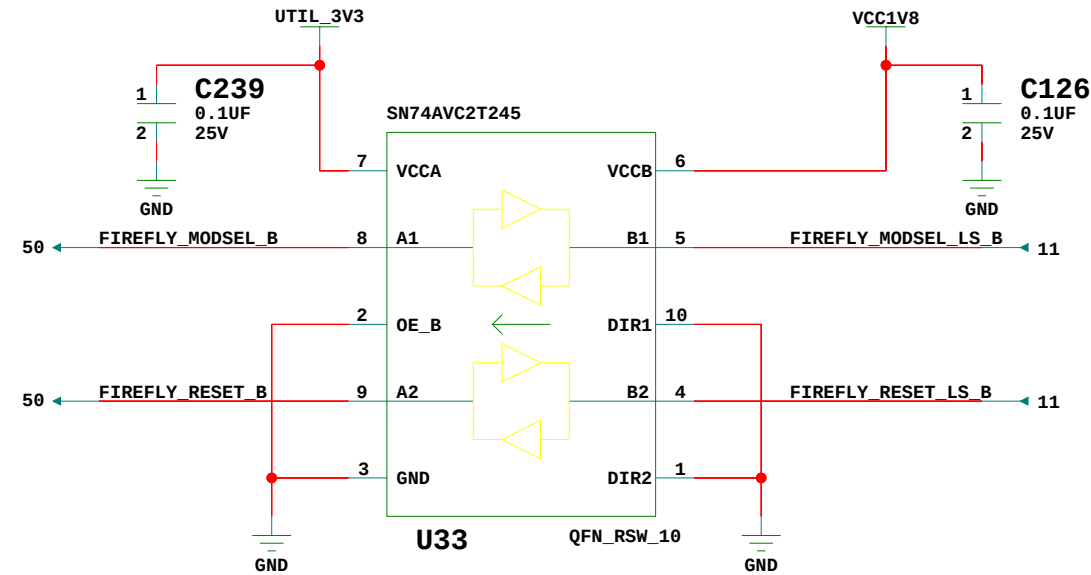
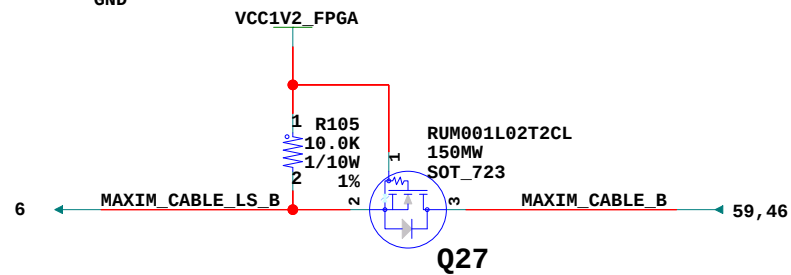
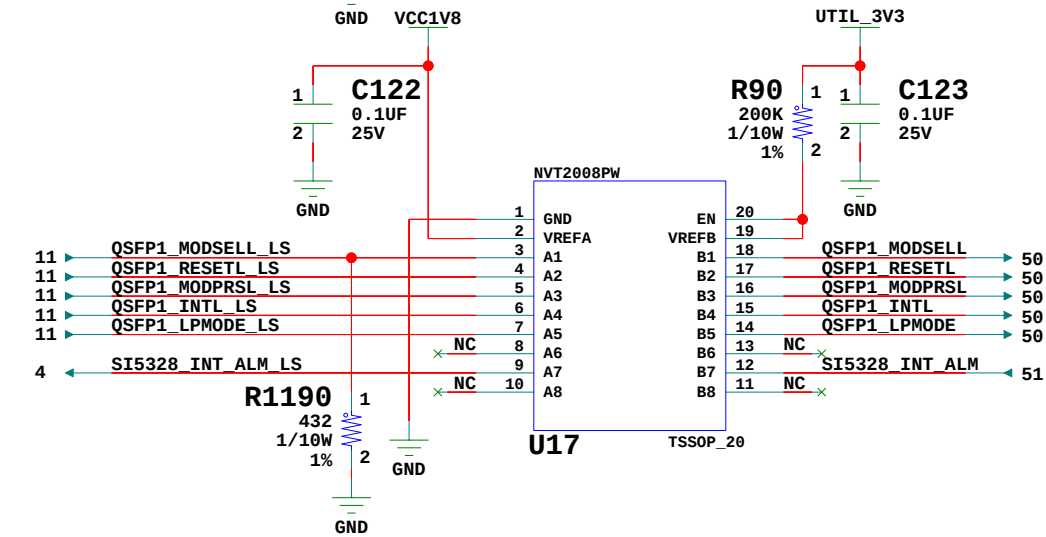
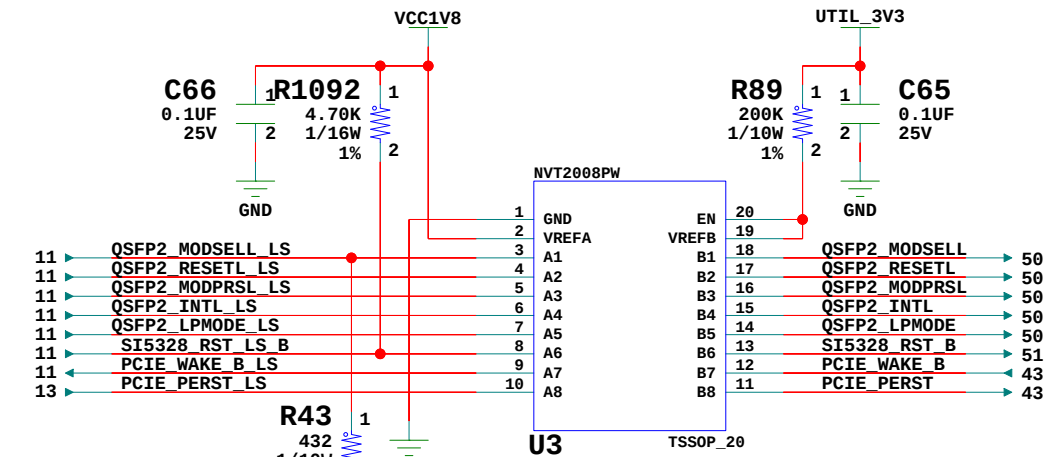
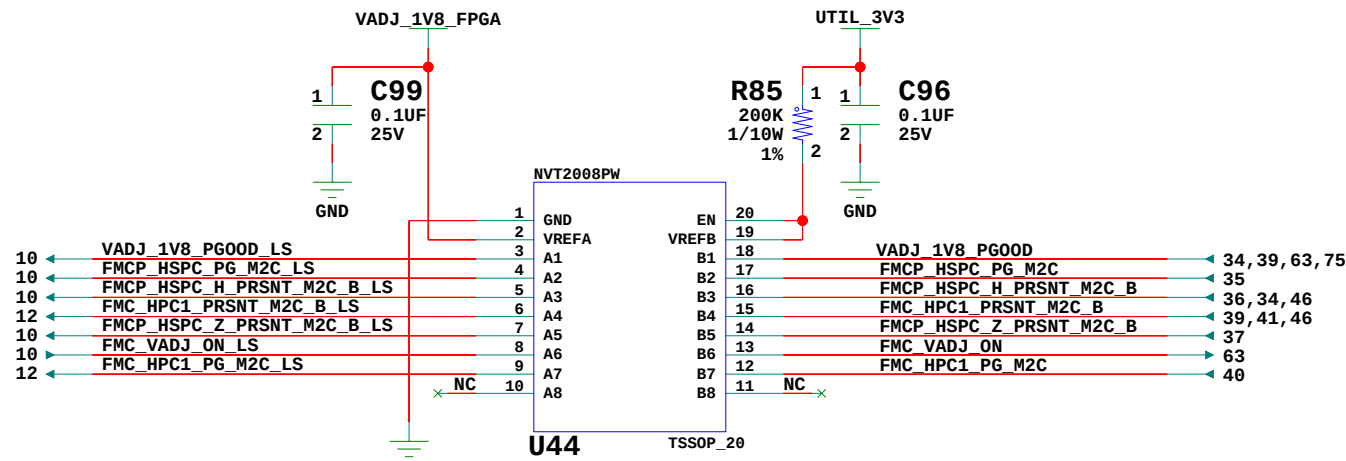
LEDs near top right edge

Buttons - Switches - LEDs

TITLE: Buttons - Switches - LEDs
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

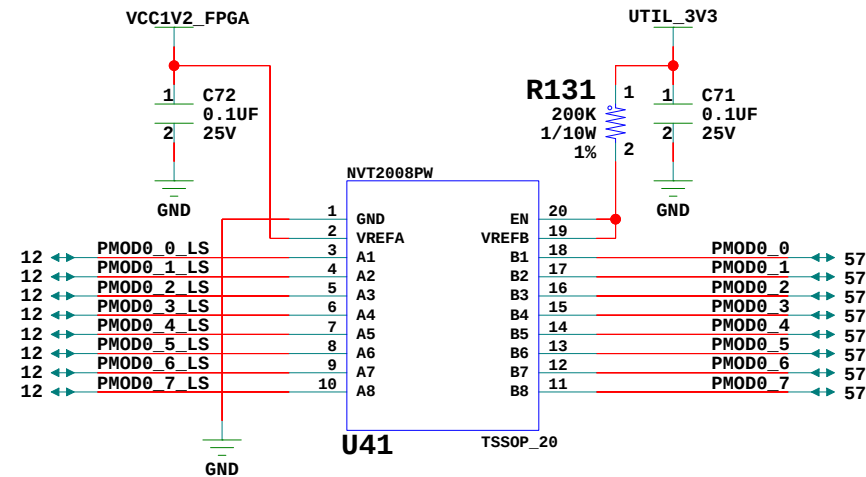
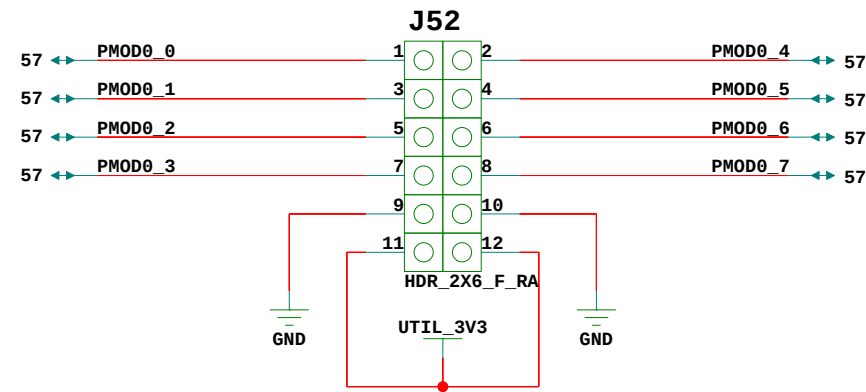
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 55 OF 76	DRAWN BY: DN



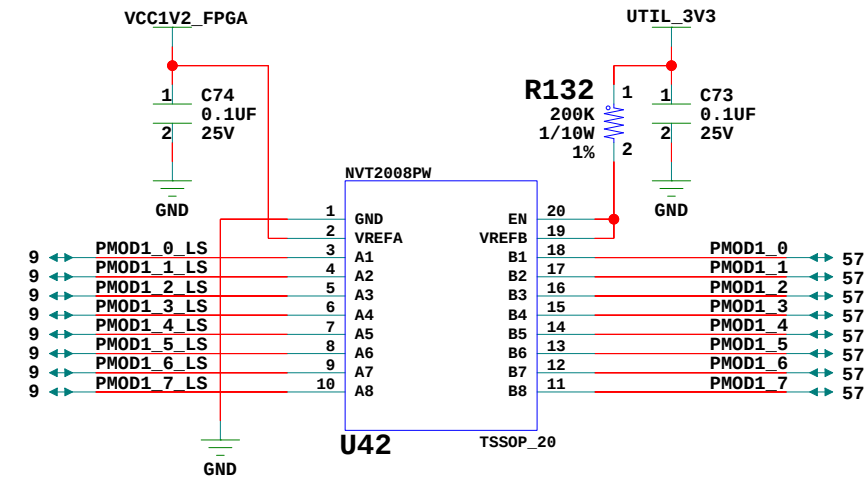
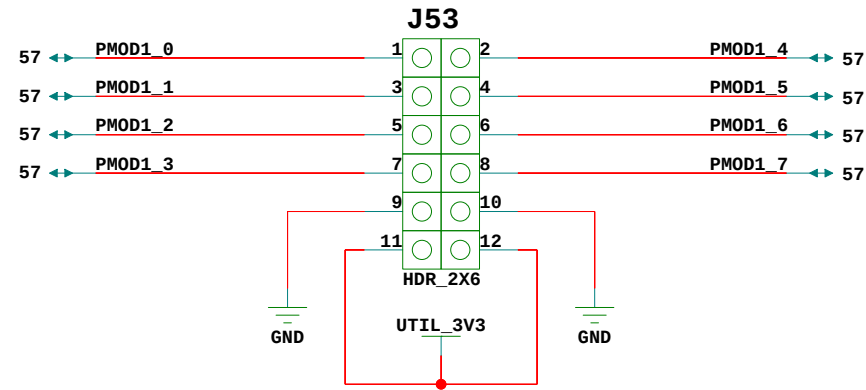
Level Shifters

TITLE: Level Shifters SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 56 OF 76	DRAWN BY: DN

PMOD Rt-Angle Female Receptacle

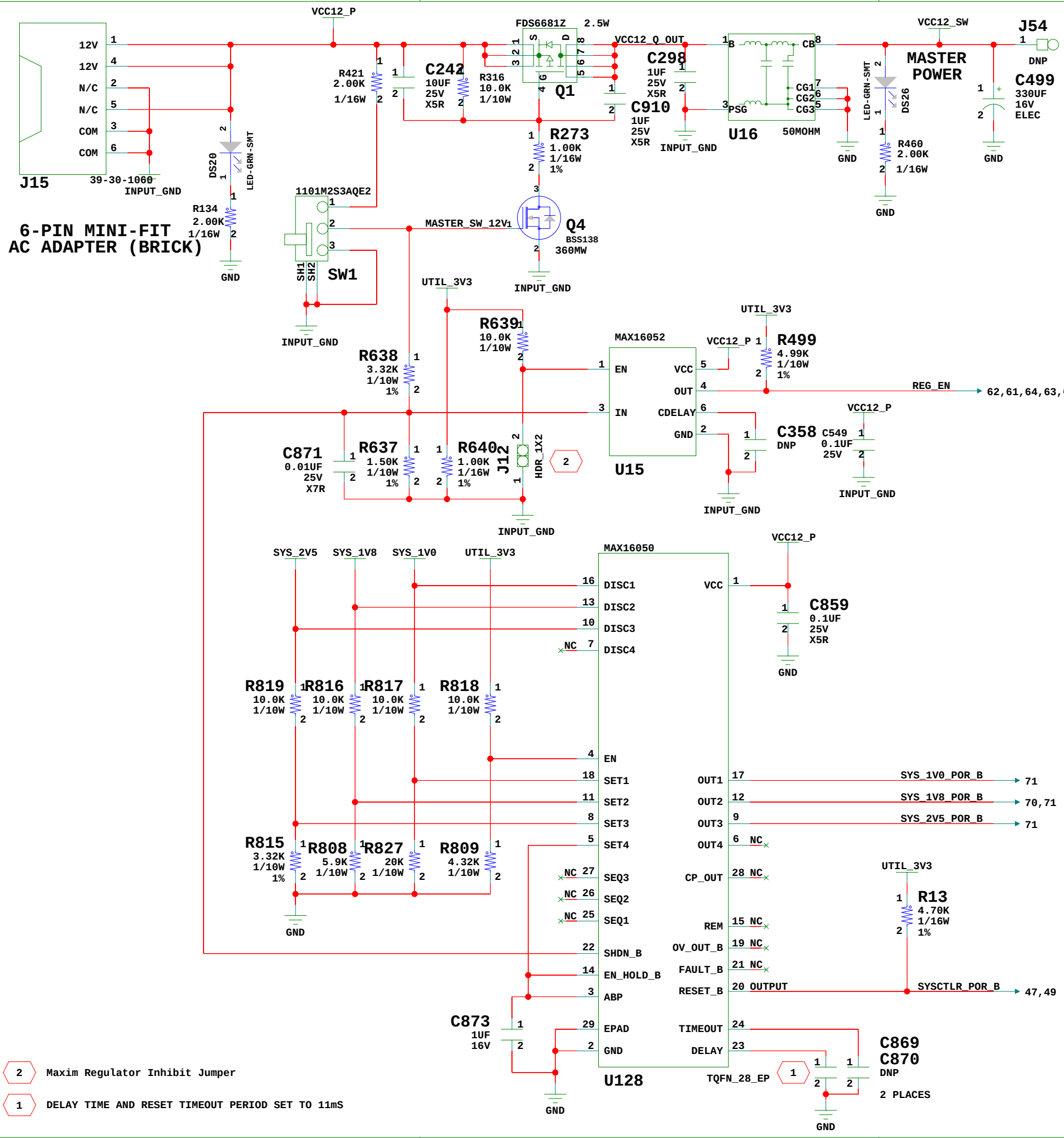


PMOD Male Pin Header



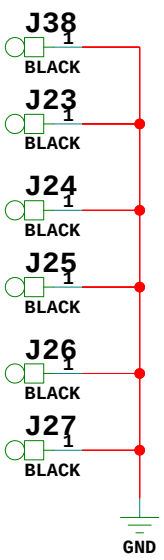
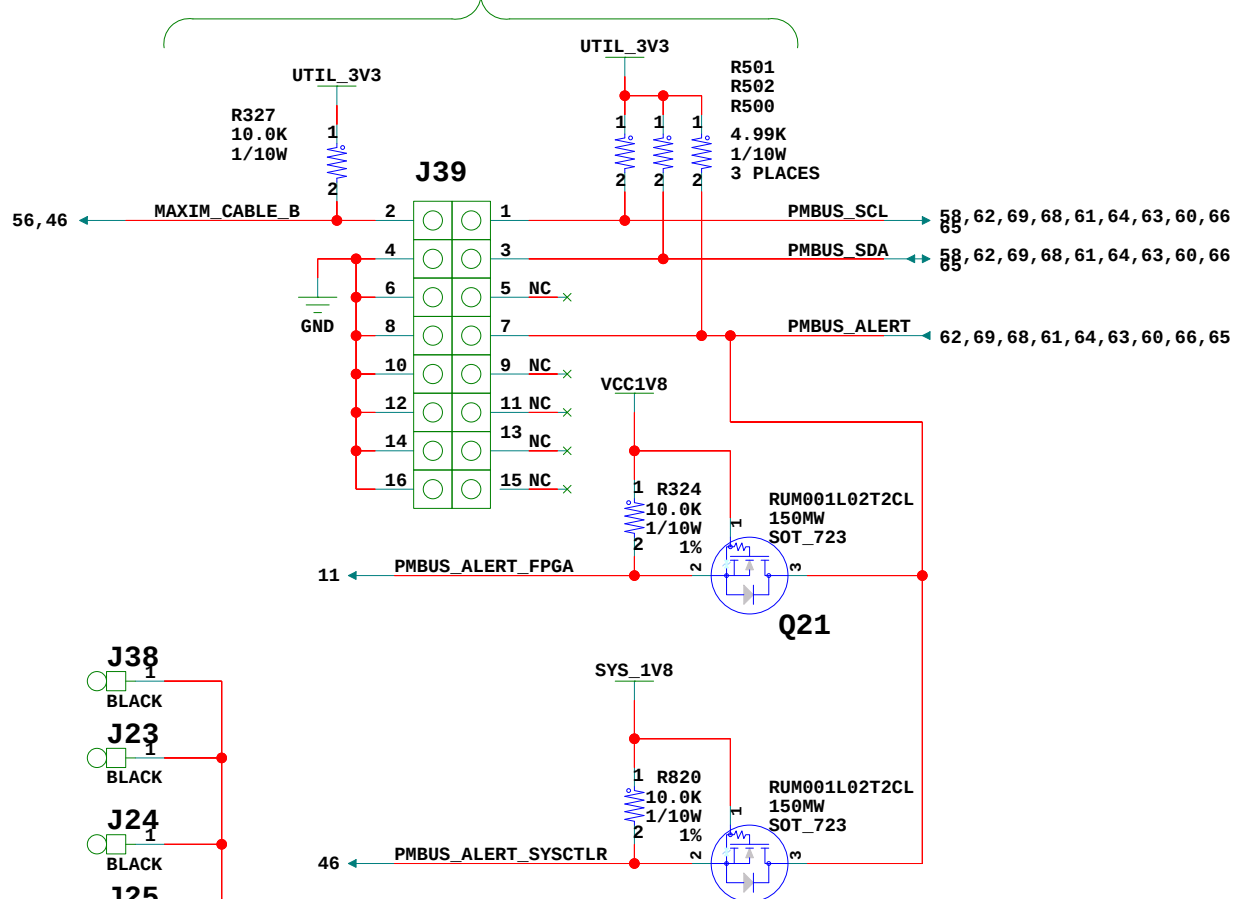
PMODs and Level Shifters

TITLE: PMODs and Level Shifters SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0		ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59		VER:	1.0
SHEET SIZE: B		REV:	01
SHEET	57	OF	76
		DRAWN BY:	DN



Fan Control circuit on page 13

MAXIM PMBUS PROGRAMMING CABLE

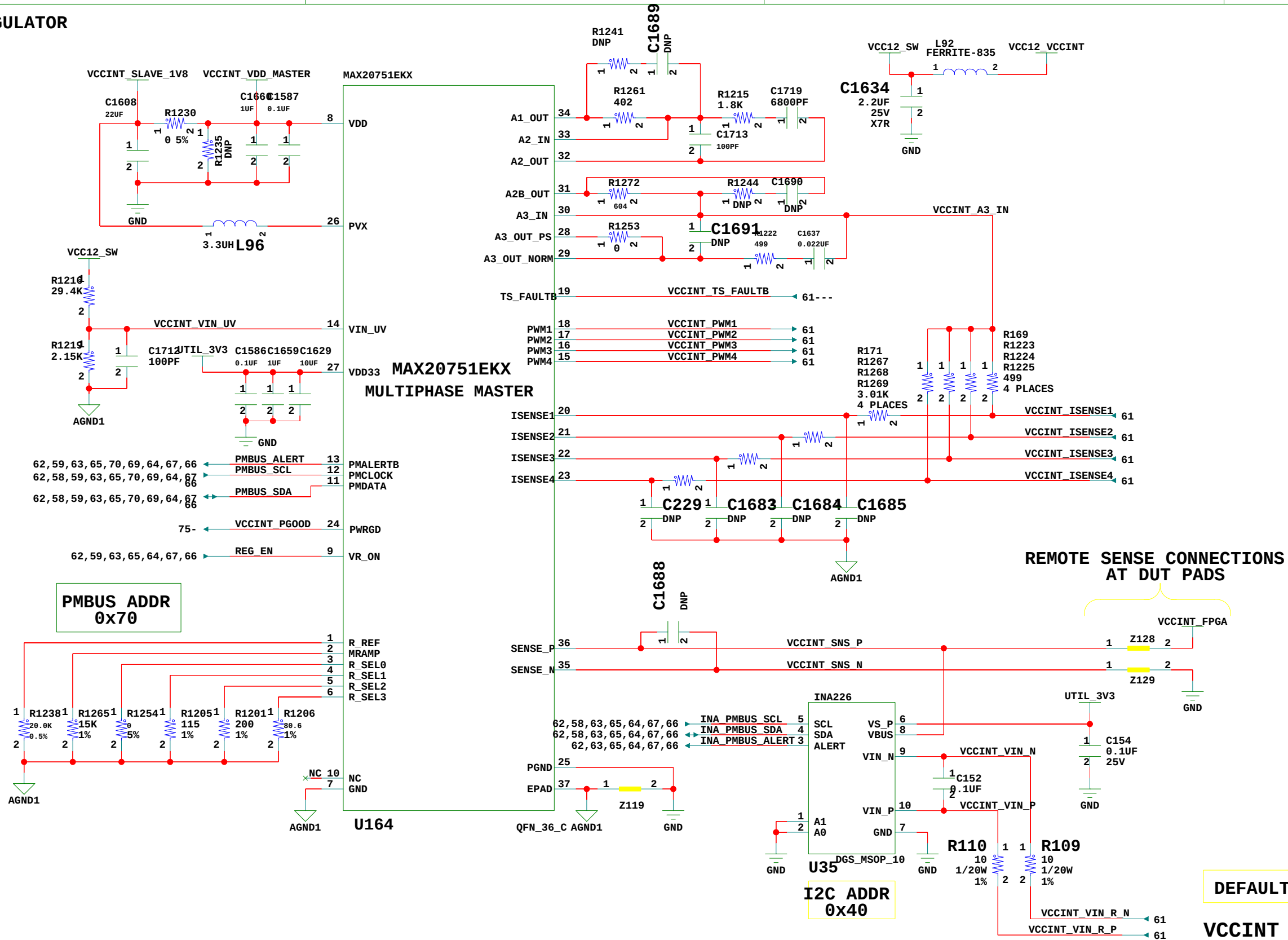


GND Test points

Pwr Connector - Switch - Pwr Supervisor

TITLE: Pwr Connector - Switch - Pwr Supervisor	
SCHEM, ROHS COMPLIANT	
HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001	
PCB P/N: 1280906	
SCH P/N: 0381739	
TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 59 OF 76	DRAWN BY: DN

VCCINT REGULATOR



DEFAULT = 0.72VDC

VCCINT 80A Regulator Master



**TITLE: VCCINT 80A Regulator Master
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0**

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59

VER:	1.0
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SHEET SIZE: B

REV:	01
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SHEET **60** OF **76**

DRAWN BY:

DN

D

C

B

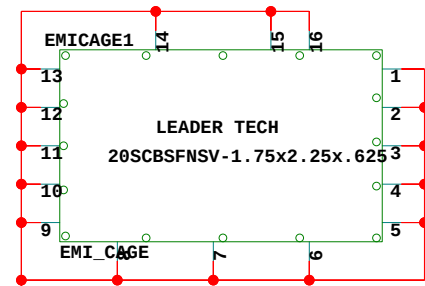
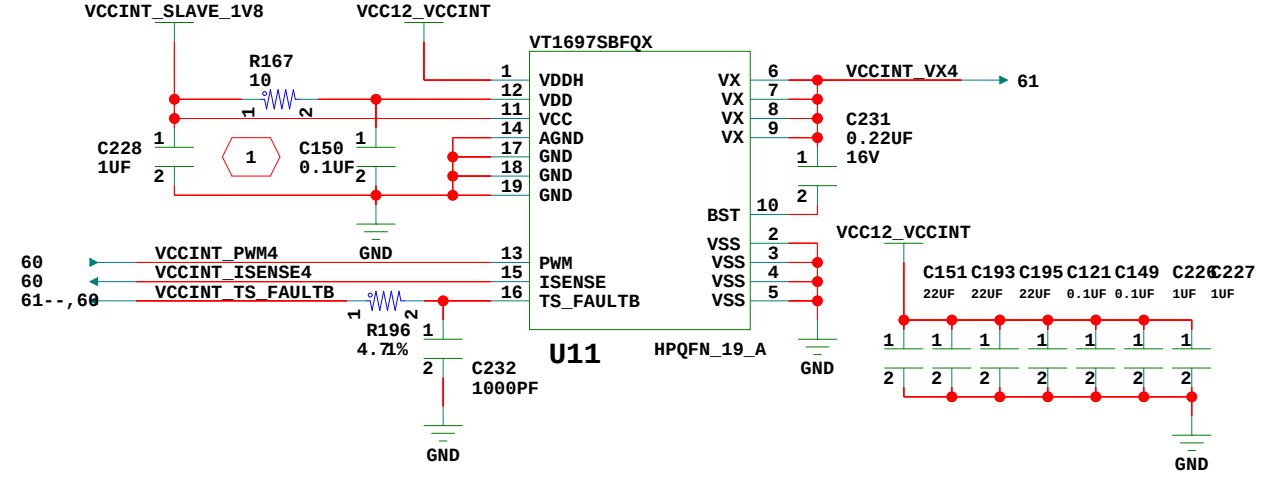
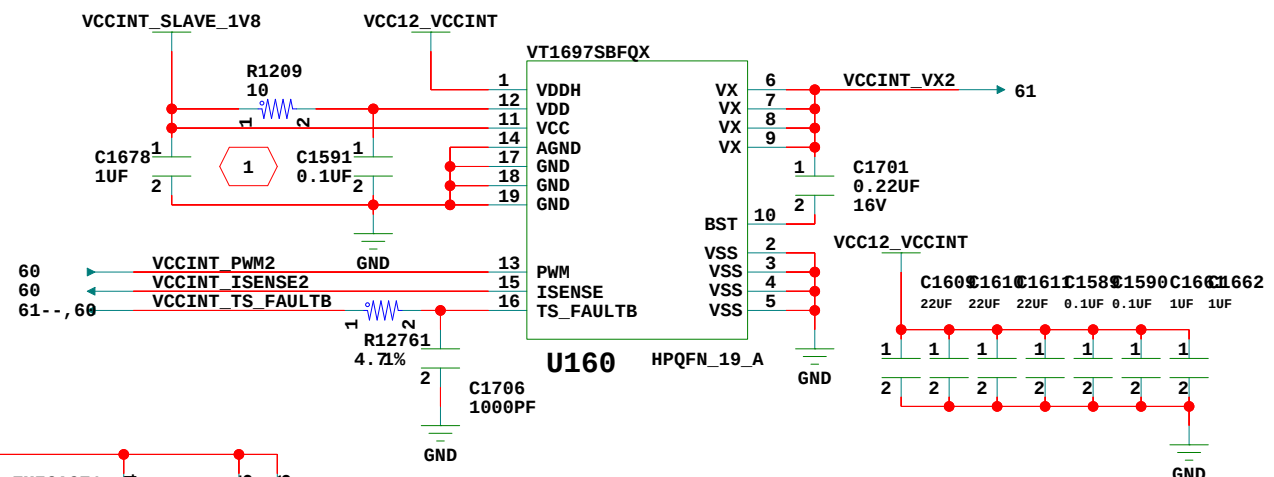
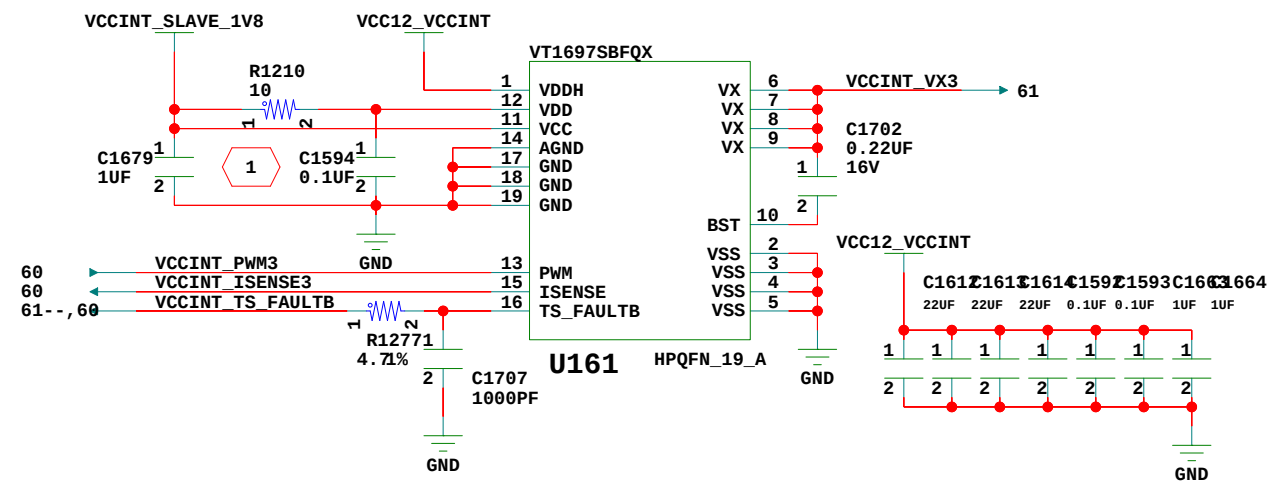
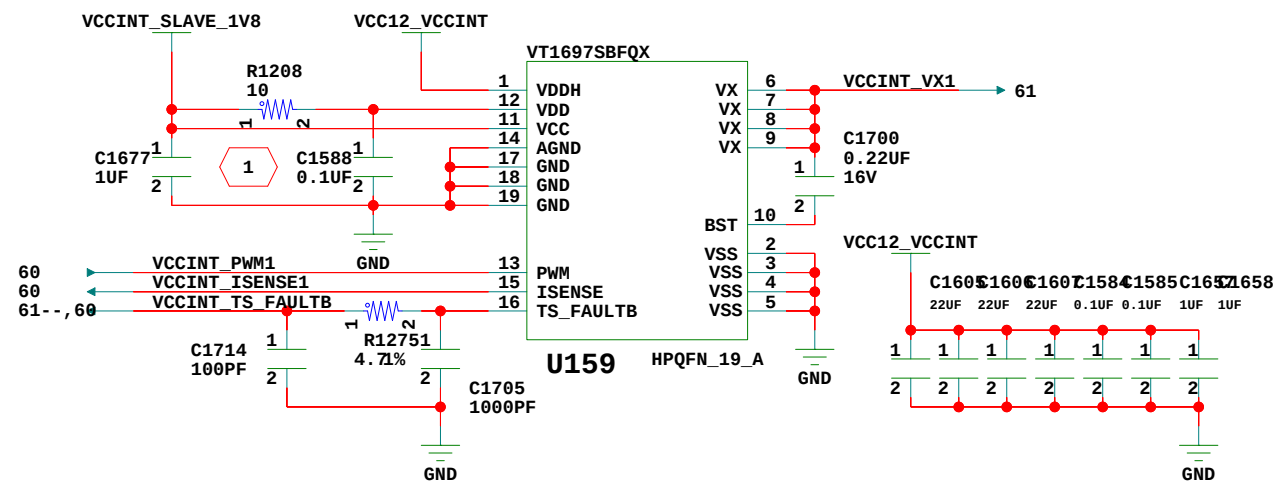
A

D

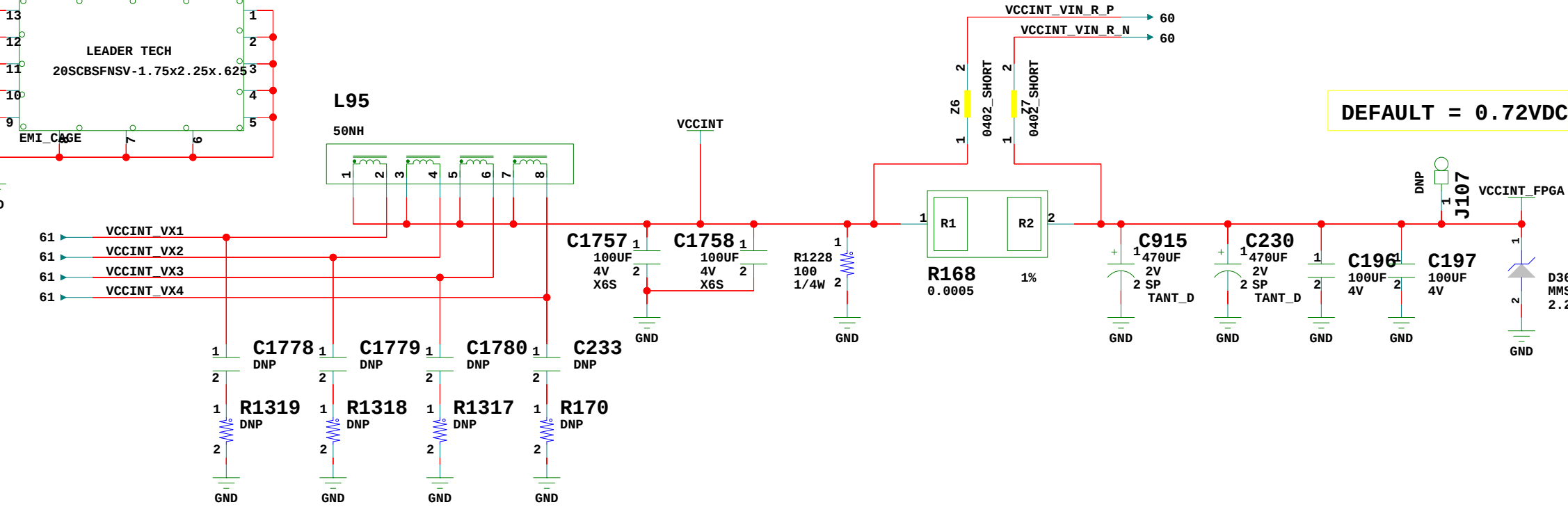
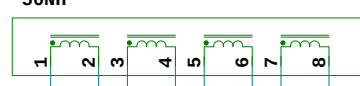
C

B

A



L95



NOTES

- 1 Capacitors should be placed as close as possible to Pin12 and 14

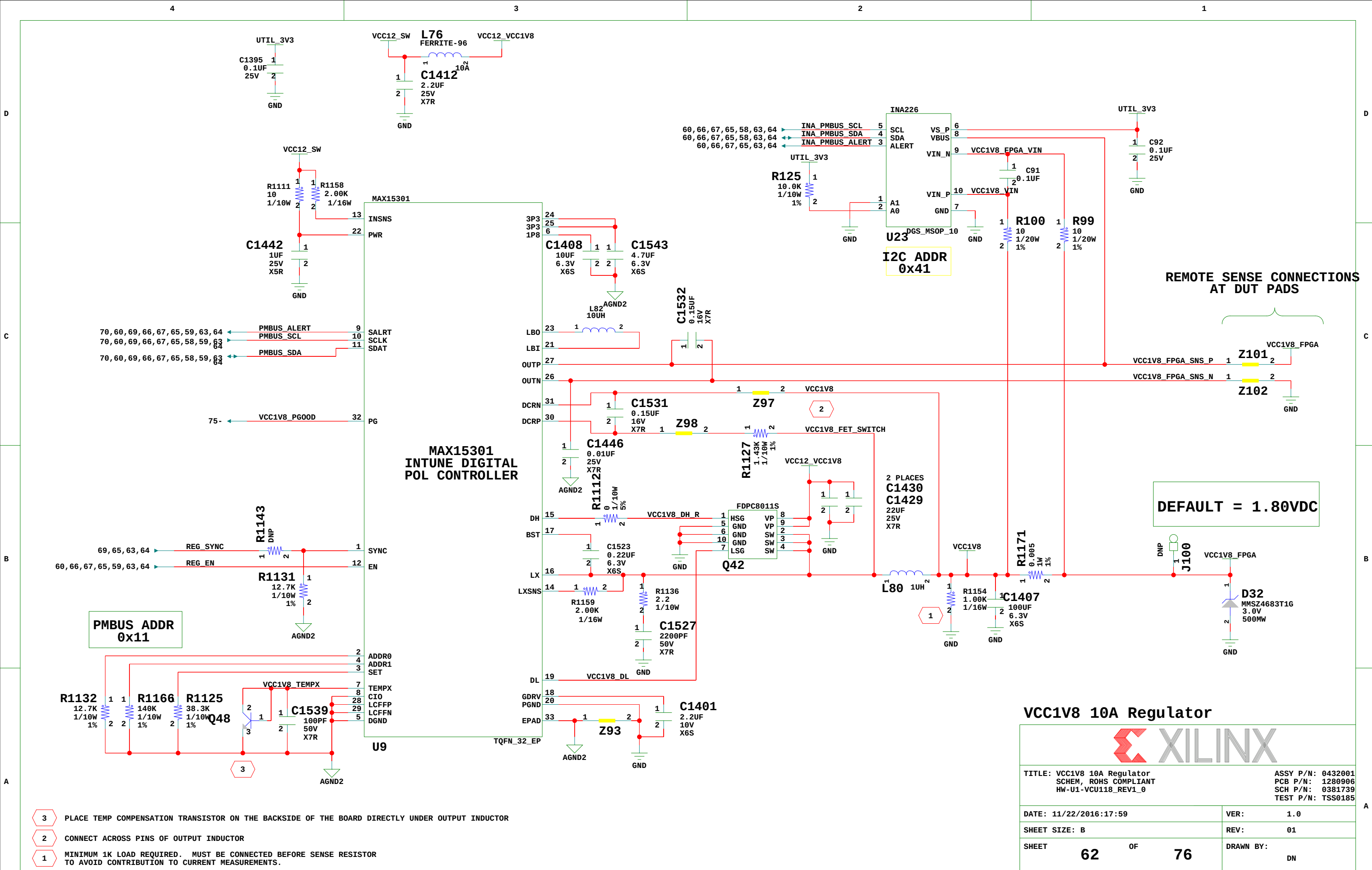
VCCINT 80A Regulator Slave

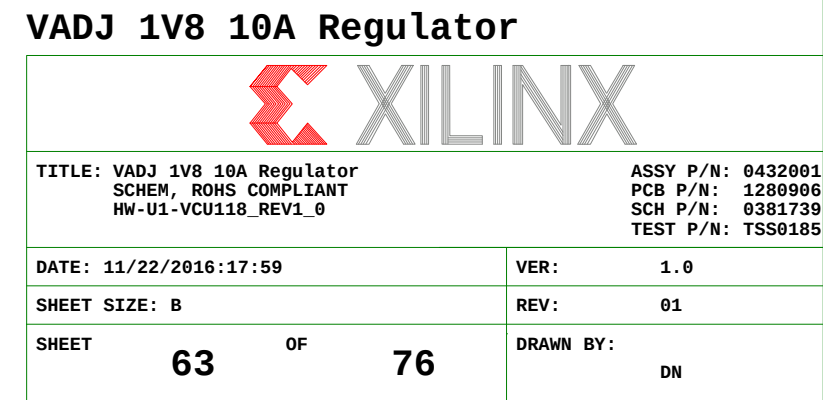


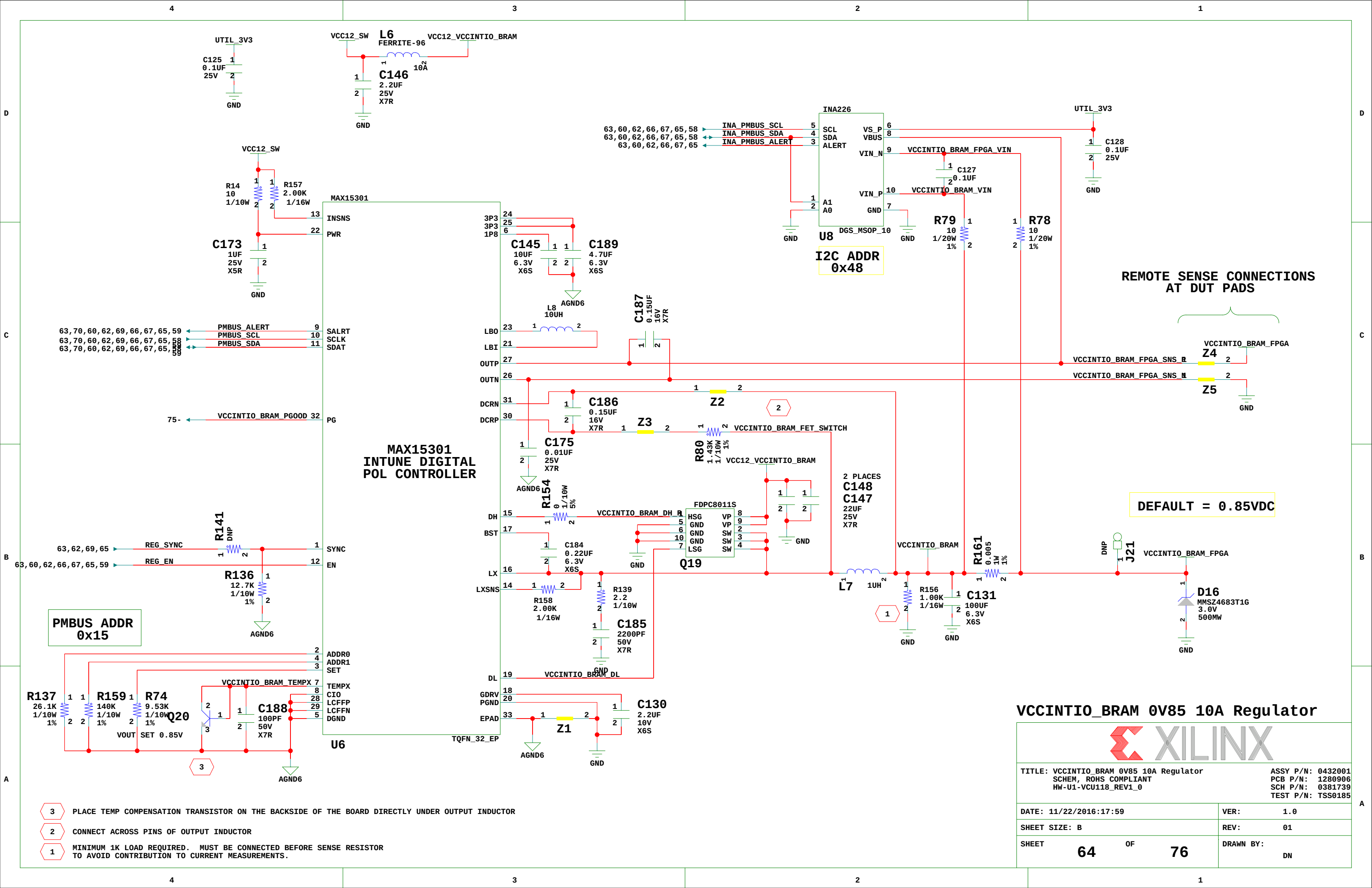
TITLE: VCCINT 80A Regulator Slave
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0

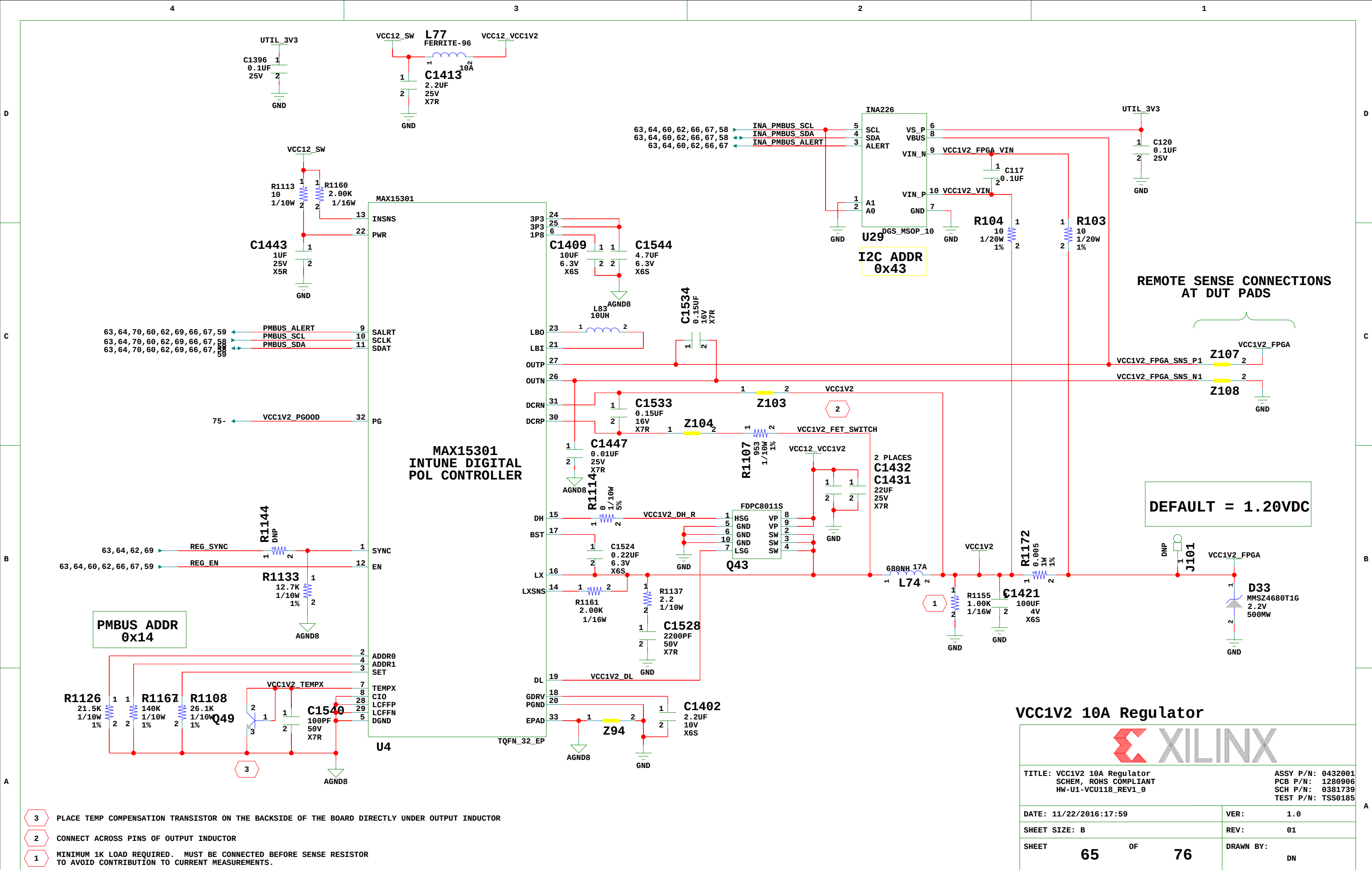
ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

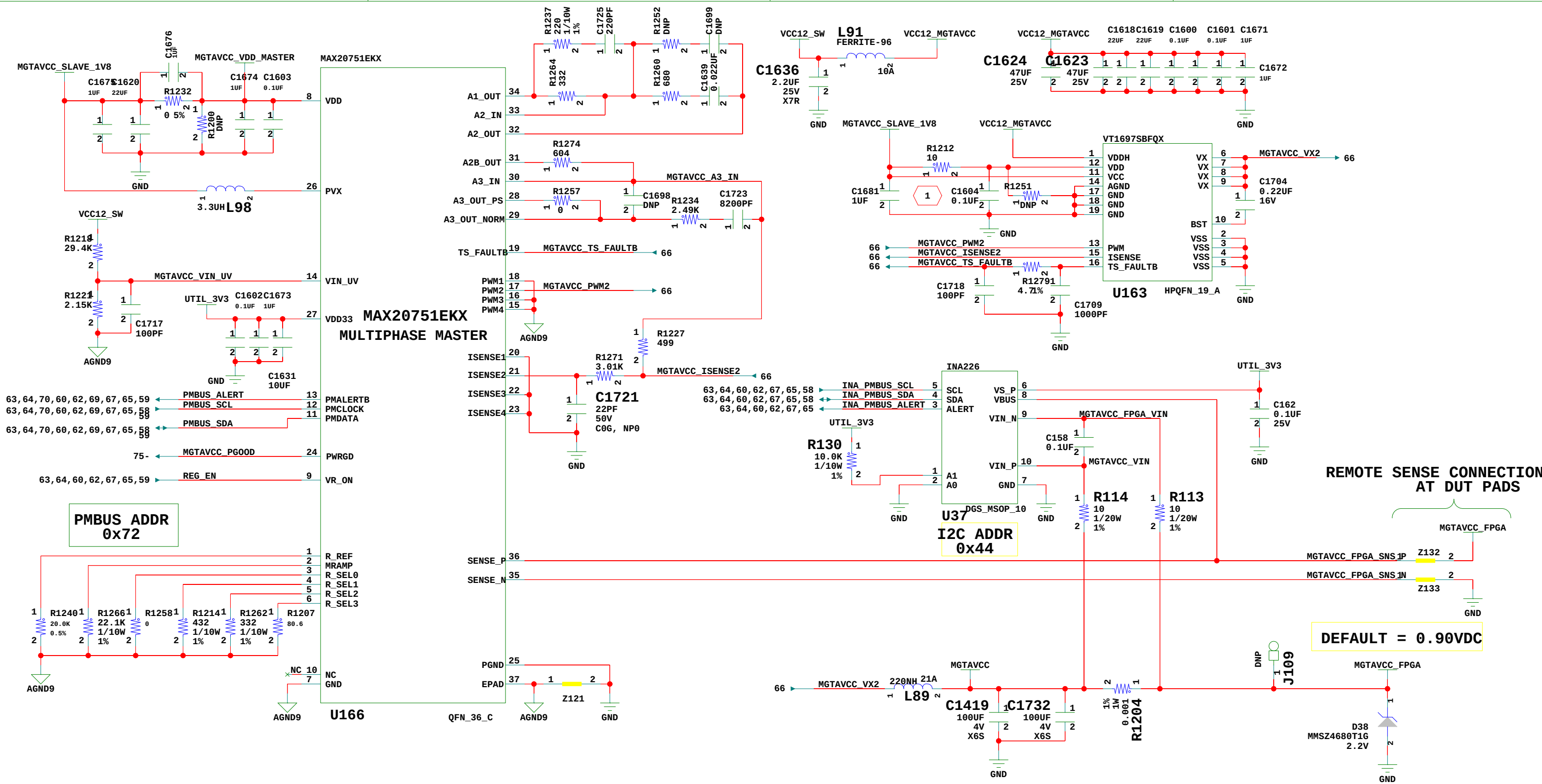
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 61 OF 76	DRAWN BY: DN









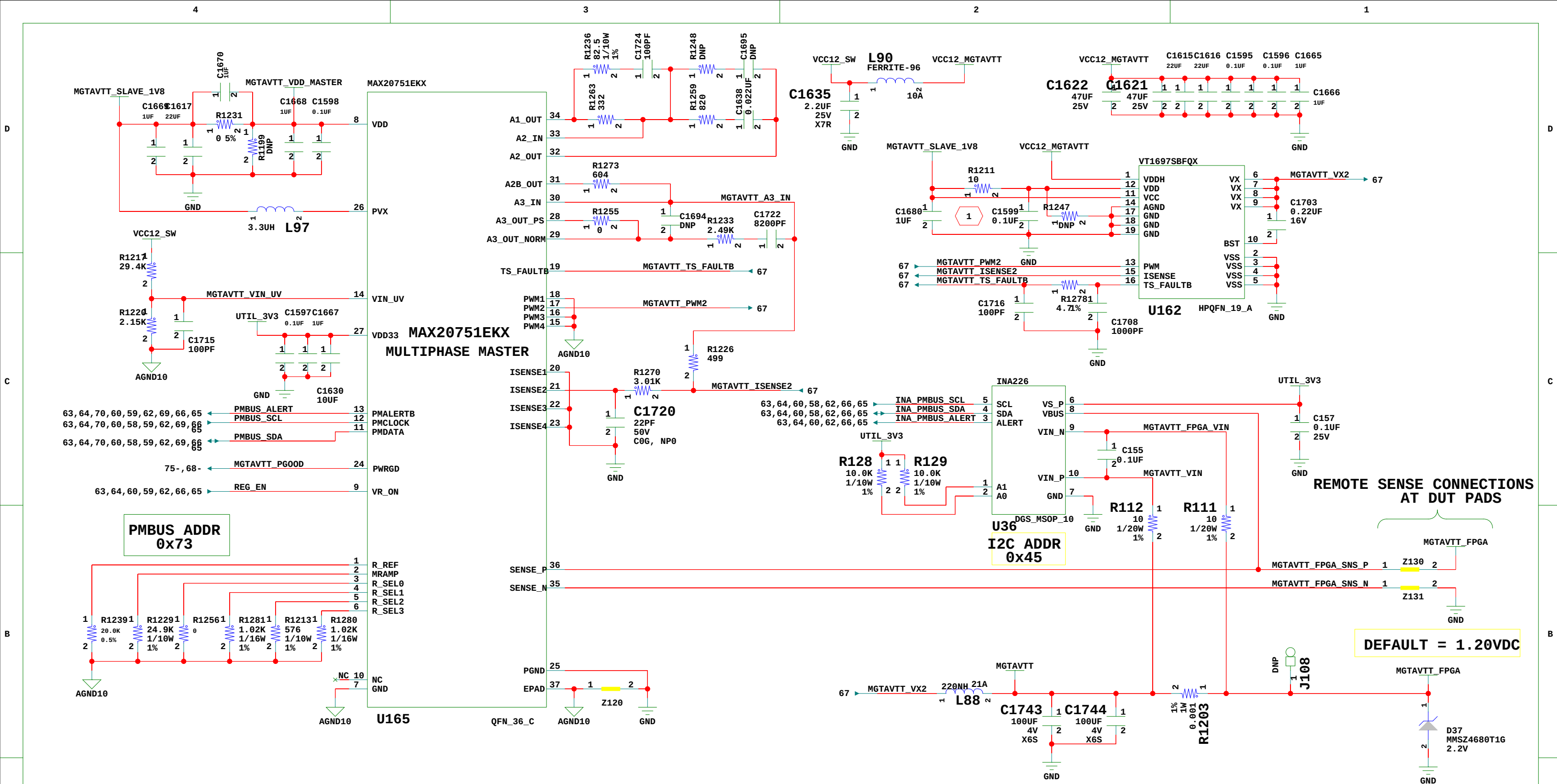


NOTES

1 Capacitors should be placed as close as possible to Pin12 and 14

MGTAVCC 17A Regulator

TITLE: MGTAVCC 17A Regulator SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 66 OF 76	DRAWN BY: DN



NOTES

1 Capacitors should be placed as close as possible to Pin12 and 14

MGTAVTT 17A Regulator



**TITLE: MGTAVTT 17A Regulator
SCHEM, ROHS COMPLIANT
HW-U1-VCU118_REV1_0**

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59

VER:	1.0
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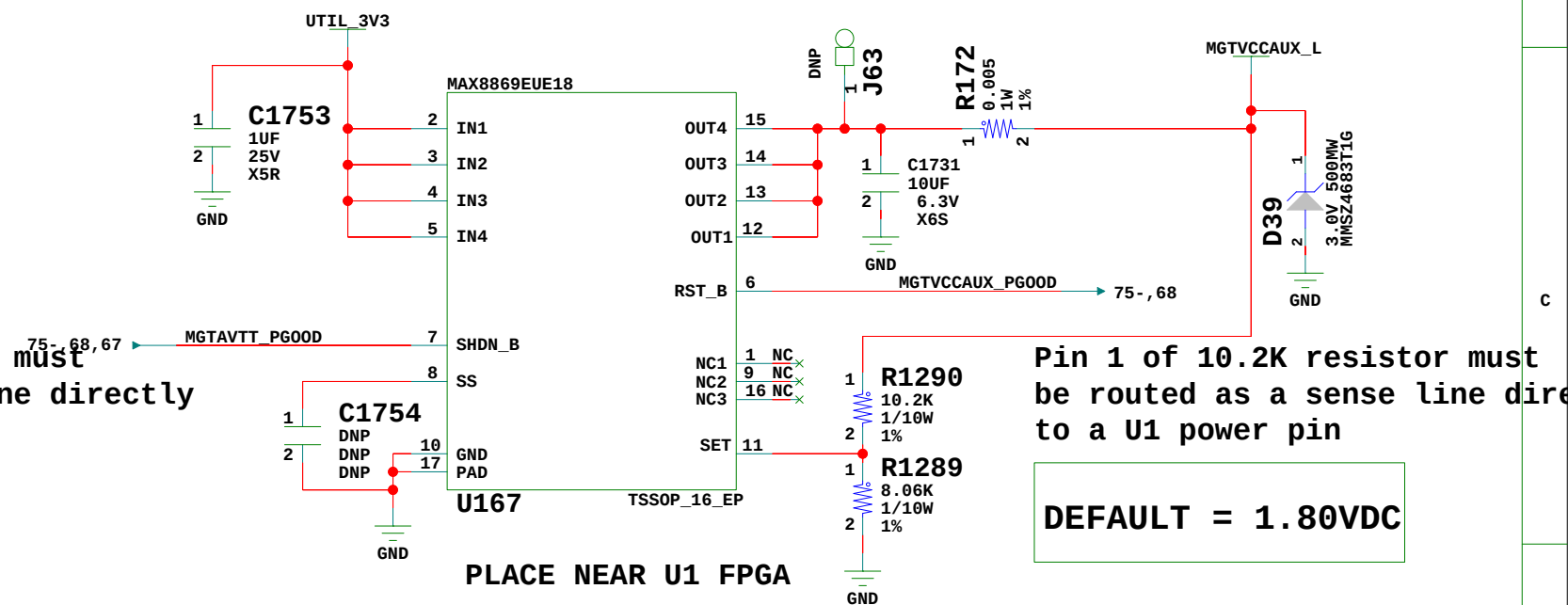
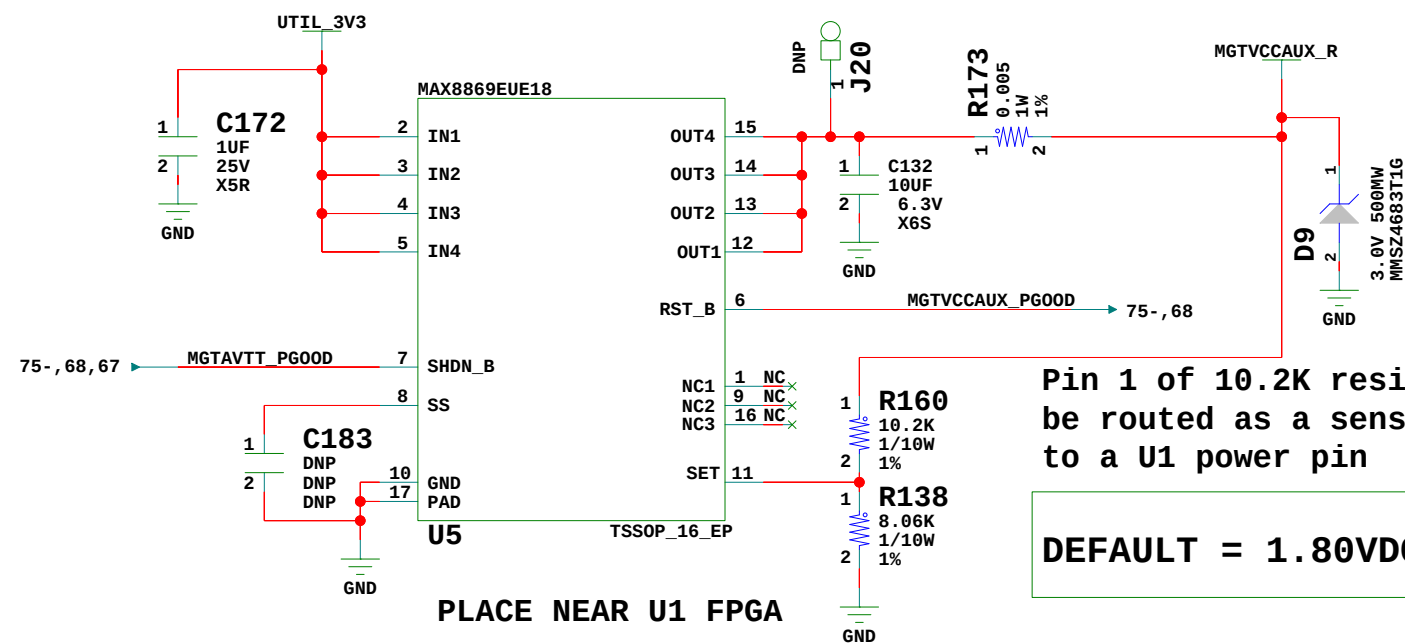
SHEET SIZE: B

REV:	01
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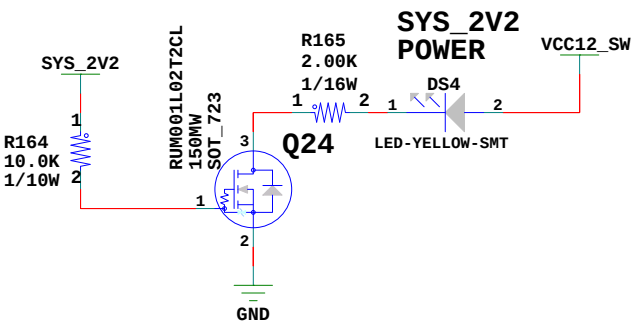
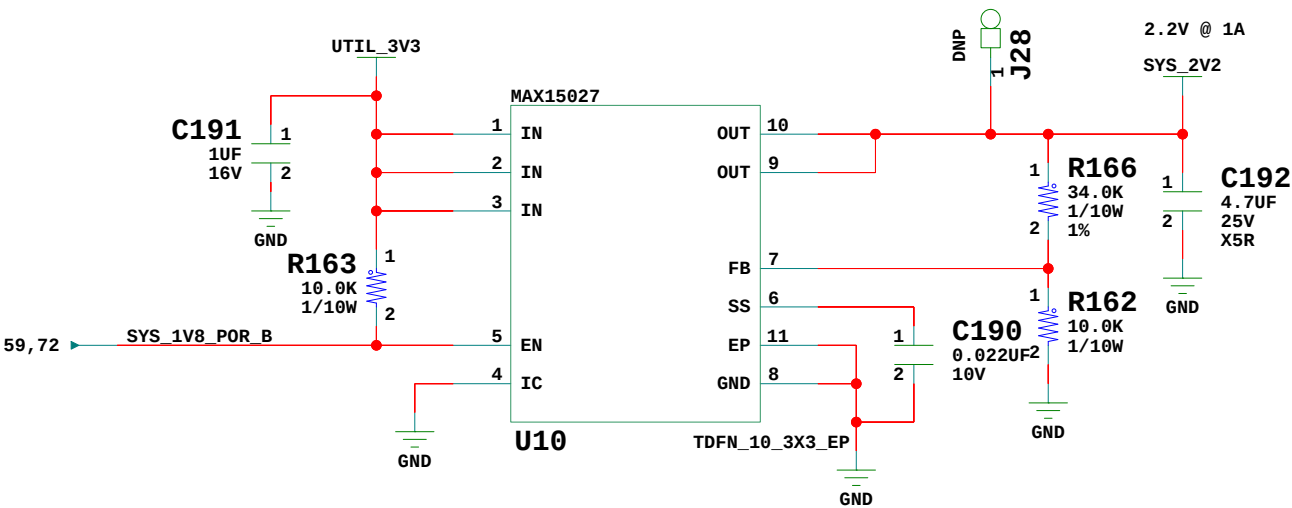
SHEET

DRAWN BY:

SHEET	67	OF	76	DRAWN BY:	DN
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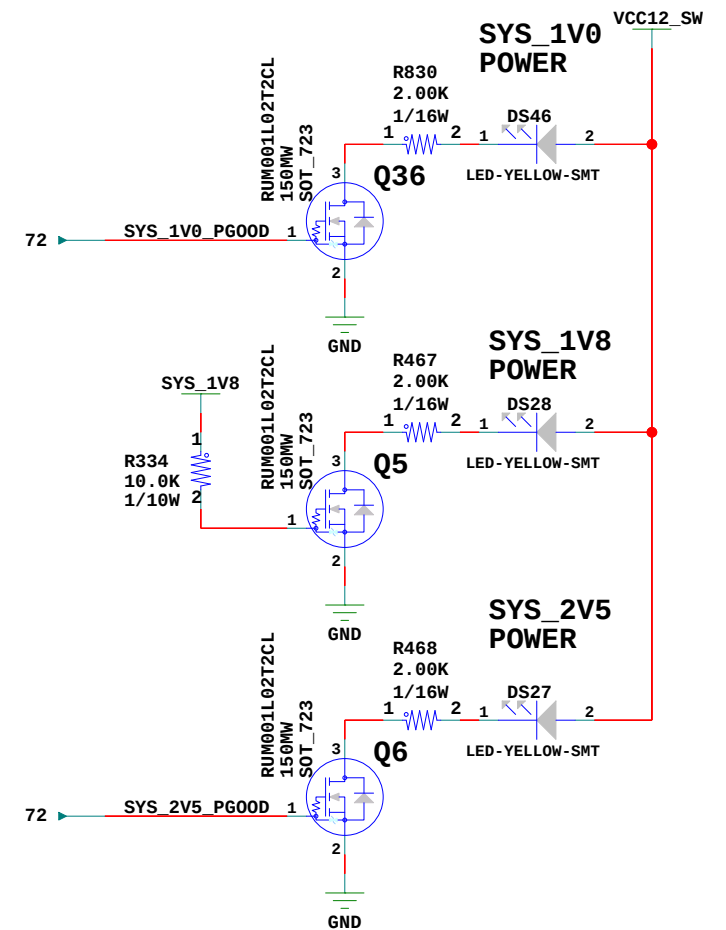
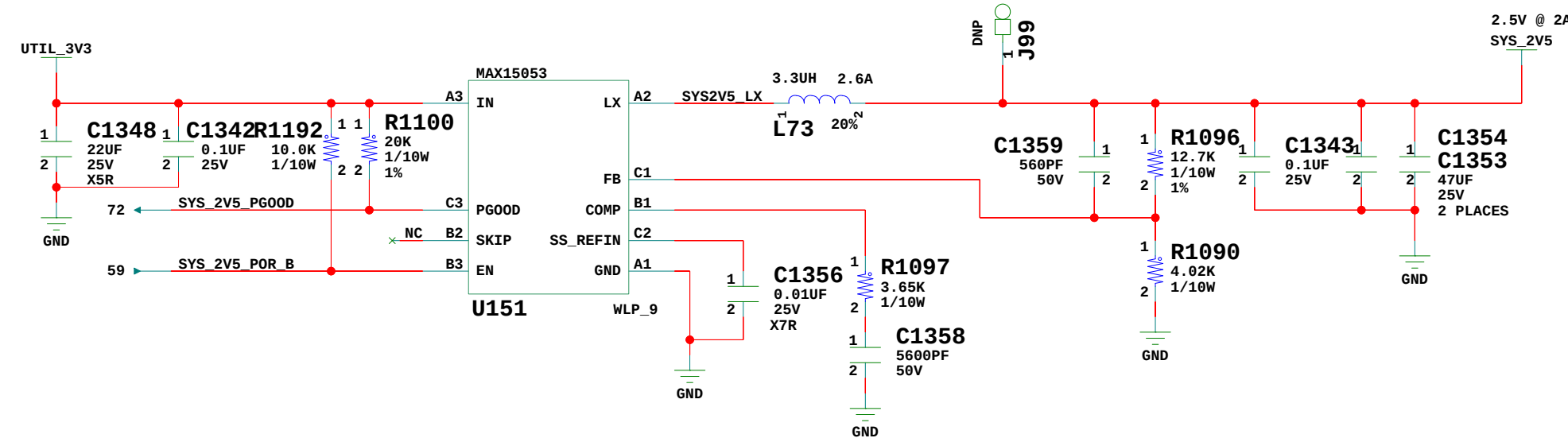
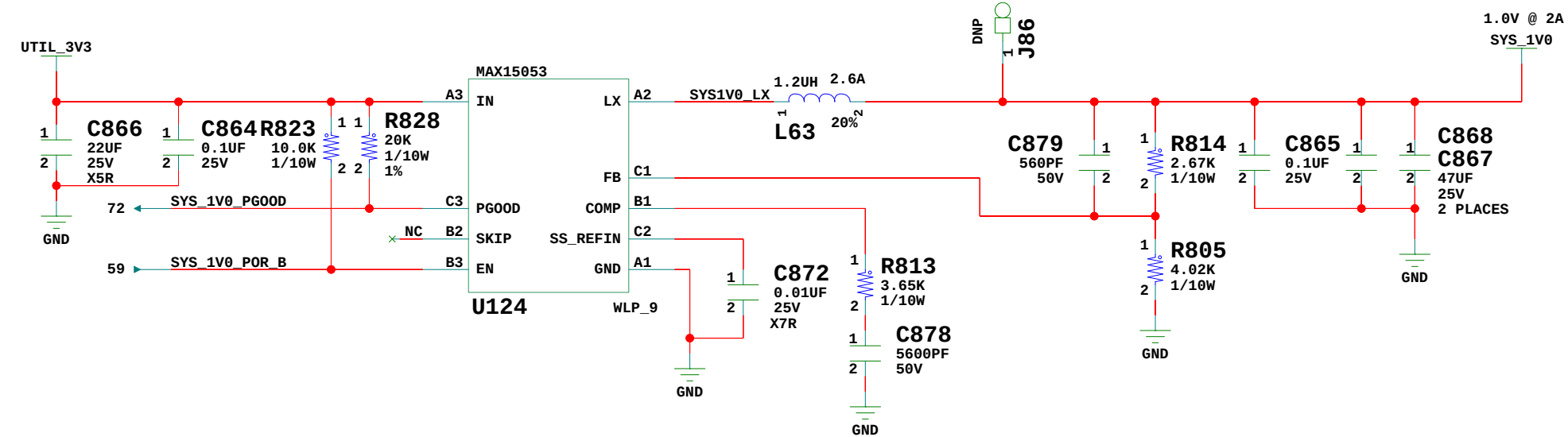
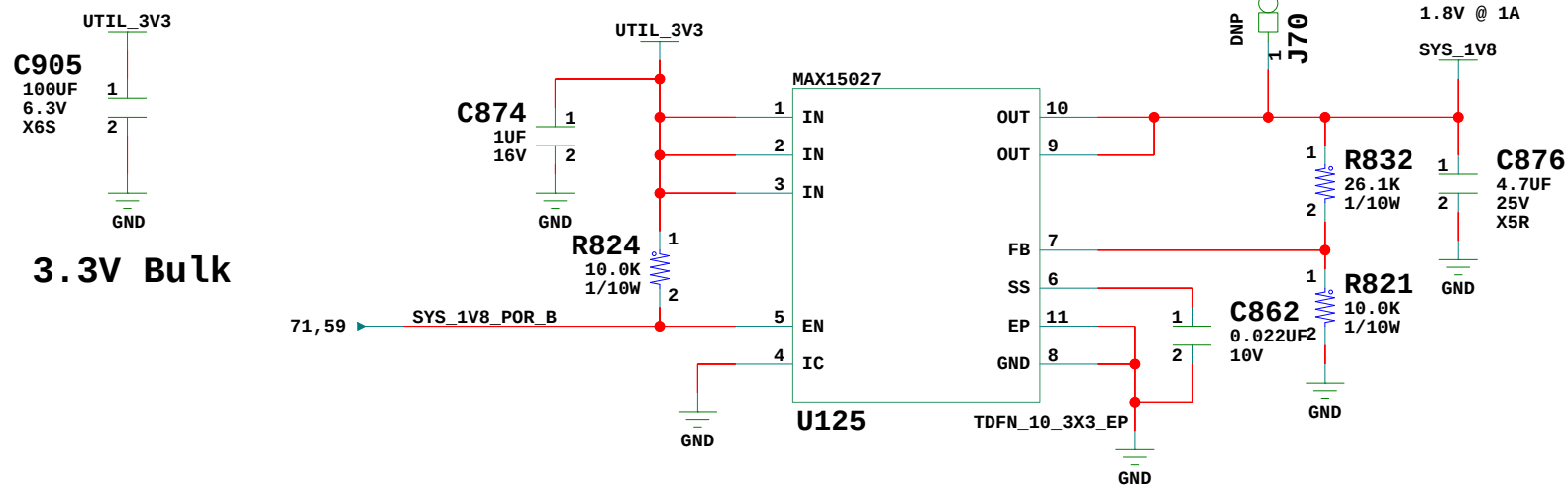


DRAWN BY: **DN**



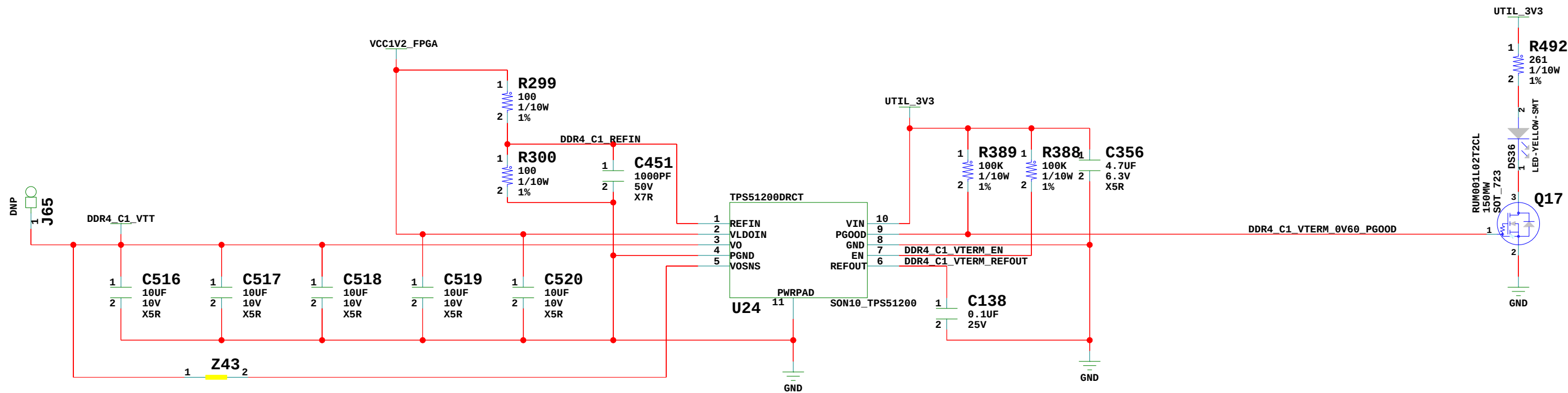
SYS_ 2V2 Regulator

TITLE: SYS_ 2V2 Regulator SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0		ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59		VER:	1.0
SHEET SIZE: B		REV:	01
SHEET	71	OF	76
		DRAWN BY:	DN

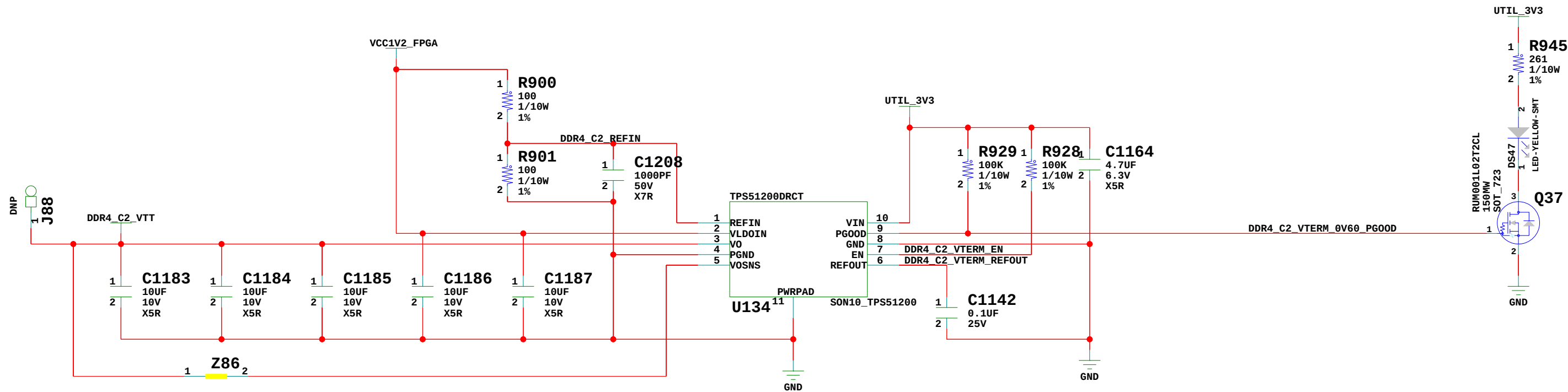


SYS_2V5 SYS_1V8 SYS_1V0 Regulators

TITLE: SYS_2V5 SYS_1V8 SYS_1V0 Regulators SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 72 OF 76	DRAWN BY: DN



Place sense near DDR4 C1 components

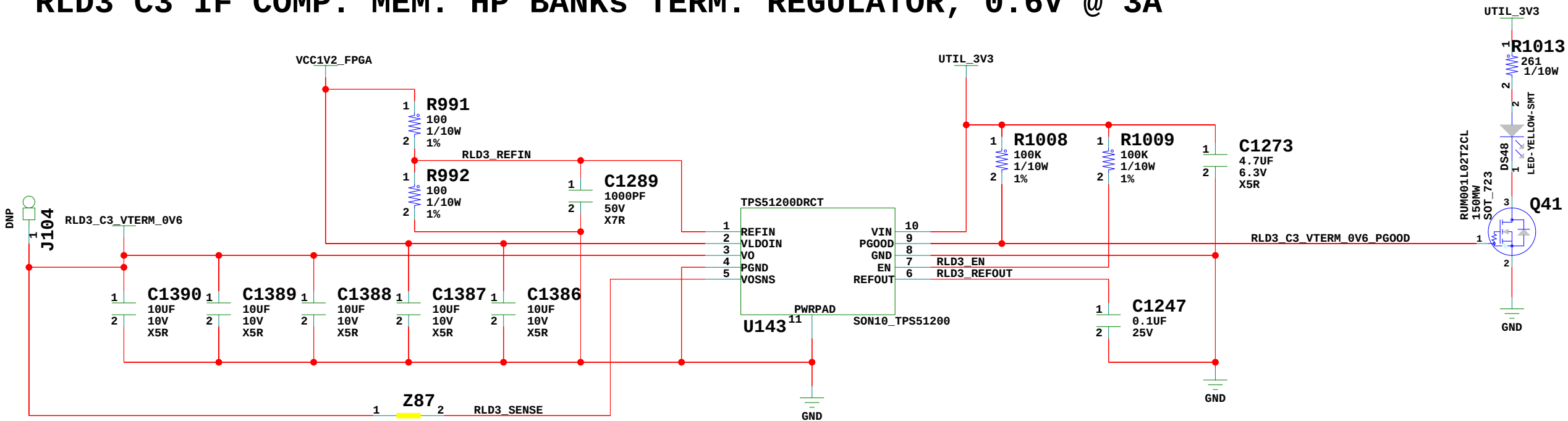


Place sense near DDR4 C2 components

DDR4 Termination Supplies

TITLE: DDR4 Termination Supplies SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 73 OF 76	DRAWN BY: DN

RLD3 C3 IF COMP. MEM. HP BANKs TERM. REGULATOR, 0.6V @ 3A

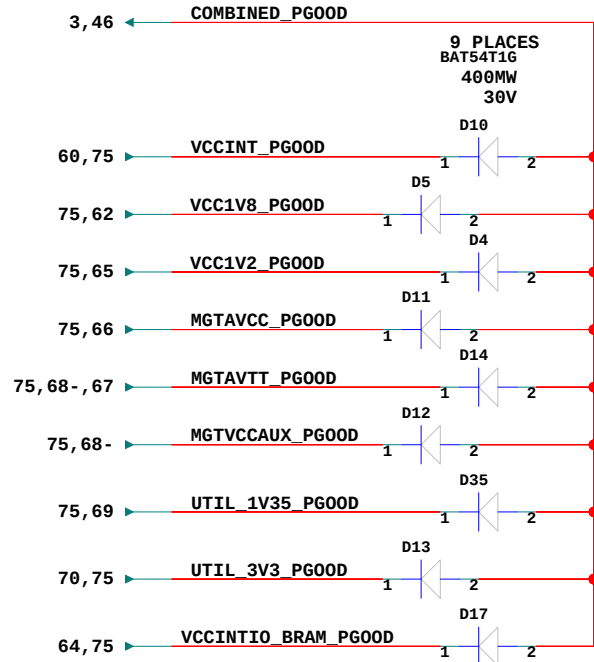
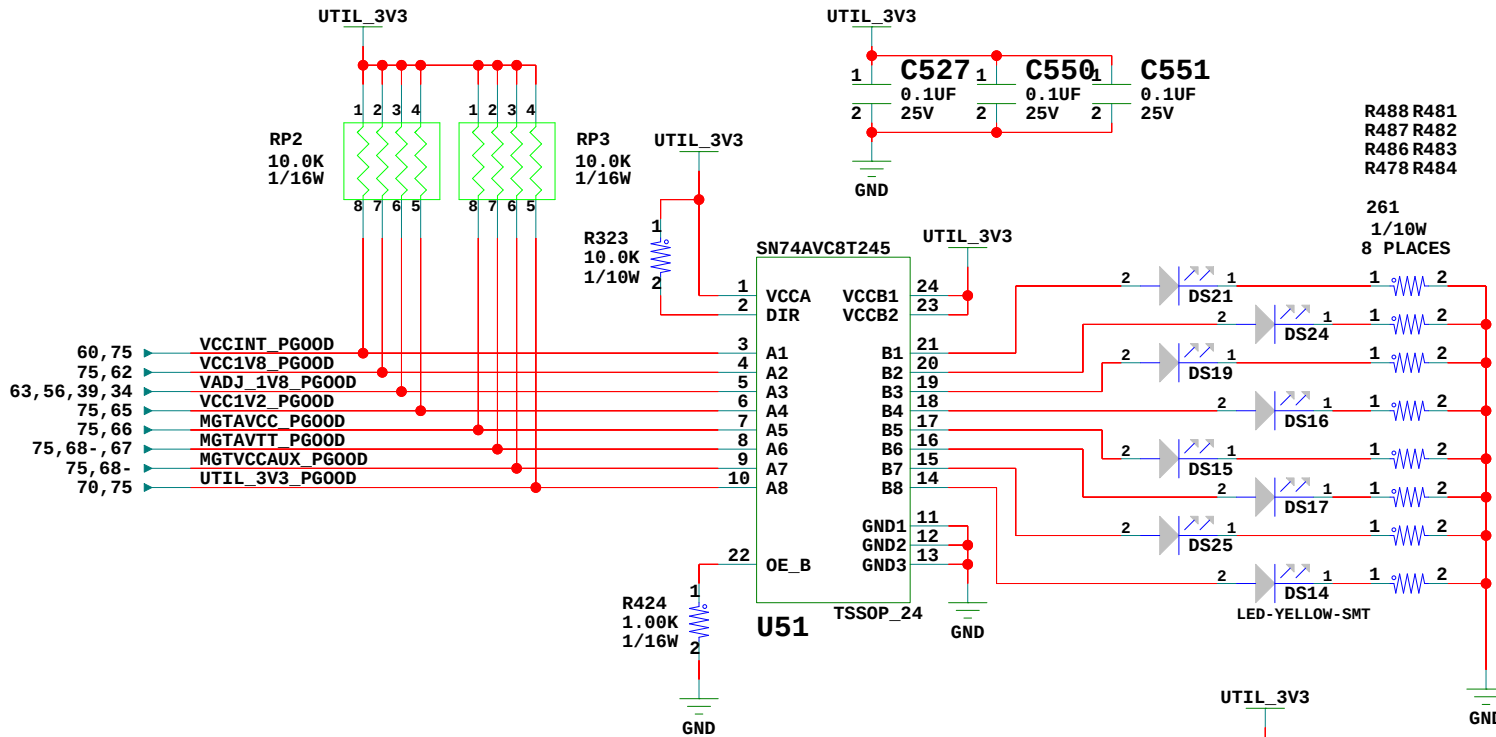


Place sense near RLD3 C3 components

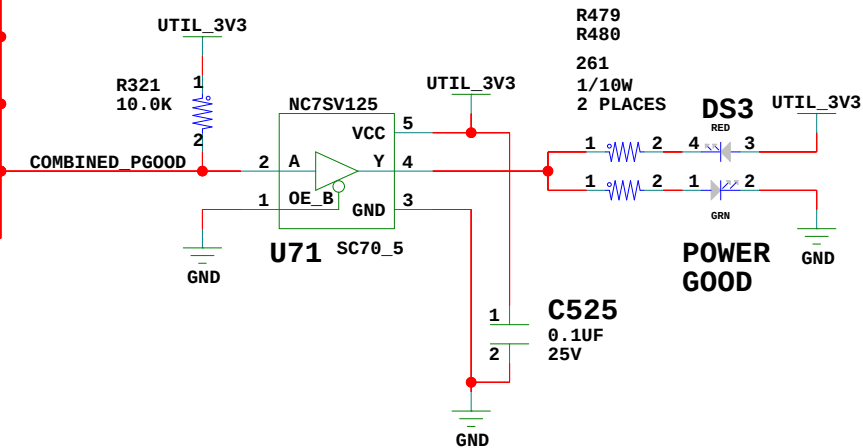
RLD3 Termination Supply

TITLE: RLD3 Termination Supply SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 74 OF 76	DRAWN BY: DN

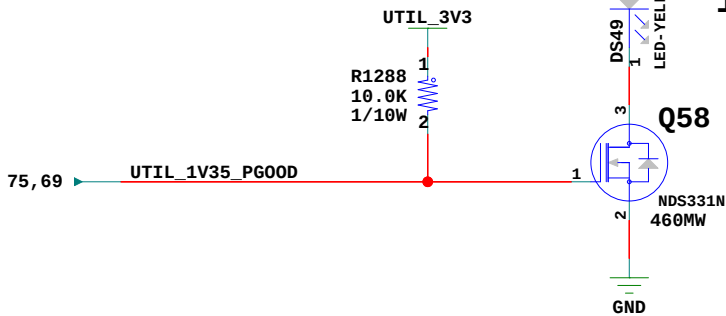
DS19 must be labeled VADJ Power Good Place LEDs near top right corner of board



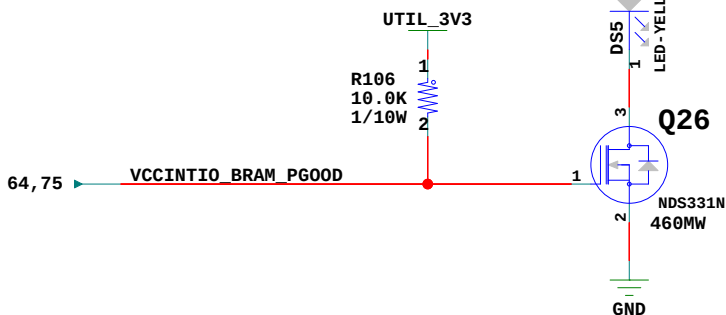
VADJ not included in POWER_GOOD logic



1V35 Power Good

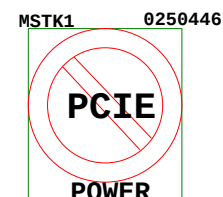
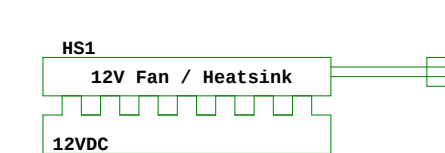


VCCINTIO_BRAM Power Good

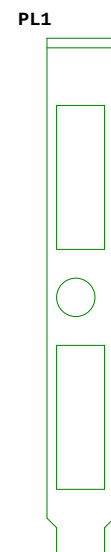


Power Status LEDs

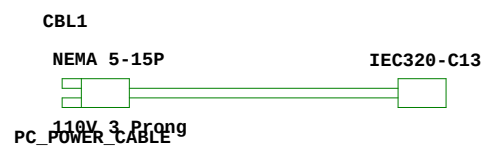
TITLE: Power Status LEDs SCHEM, ROHS COMPLIANT HW-U1-VCU118_REV1_0	
ASSY P/N: 0432001 PCB P/N: 1280906 SCH P/N: 0381739 TEST P/N: TSS0185	
DATE: 11/22/2016:17:59	VER: 1.0
SHEET SIZE: B	REV: 01
SHEET 75 OF 76	DRAWN BY: DN



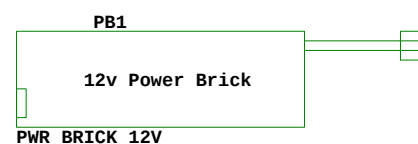
PCIE_POWER_STICKER



VCU118 PCIE PLATE



Power Cord



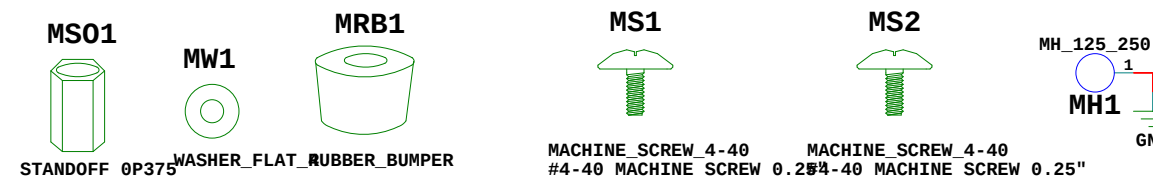
PWR BRICK 12V



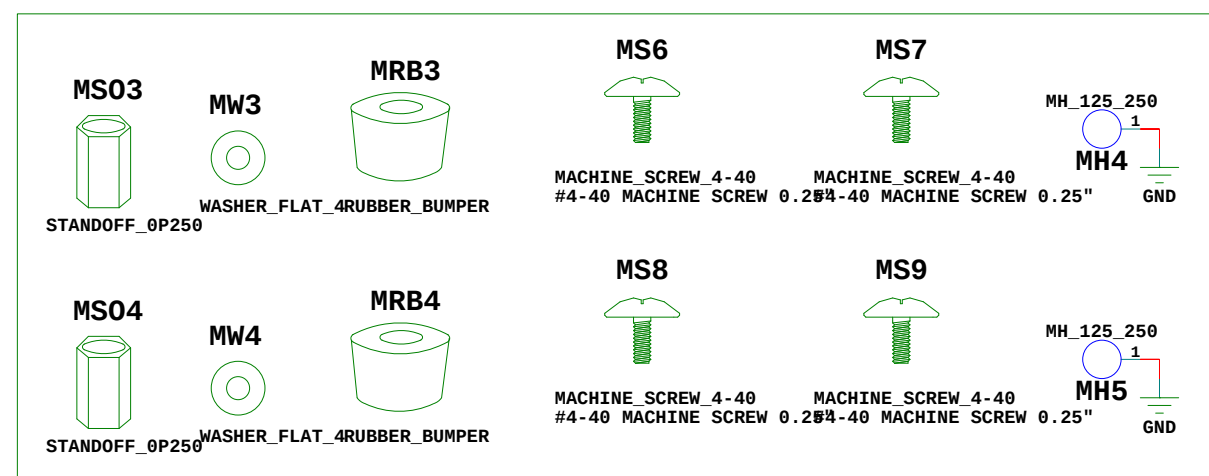
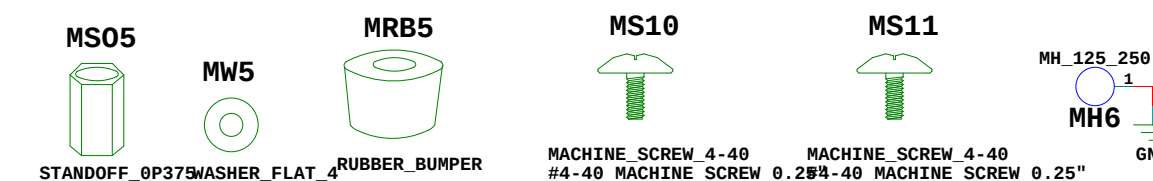
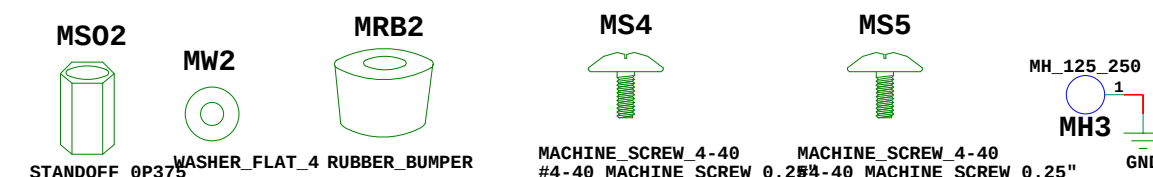
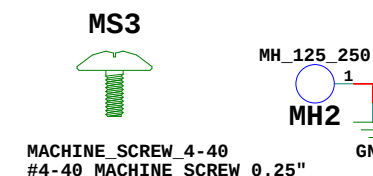
USB Micro-B Cable



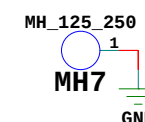
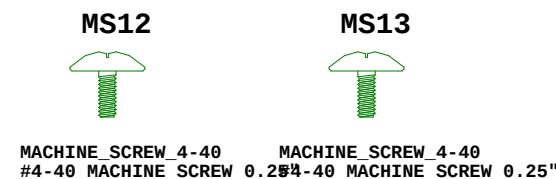
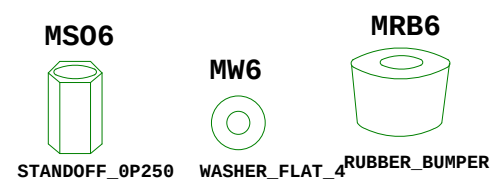
USB Micro-B Cable



NOTE: No standoff foot on MH2



NOTE: The two standoffs above are shorter and must be connected to MH4 and MH5 to level the board properly



Mechanical Components



TITLE: Mechanical Components
SCHEM, ROHS COMPLIANT
HW-U1-VCU118 REV1_0

ASSY P/N: 0432001
PCB P/N: 1280906
SCH P/N: 0381739
TEST P/N: TSS0185

DATE: 11/22/2016:17:59

VER:	1.0
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SHEET SIZE: B

REV:	01
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SHEET **76** OF **76**

DRAWN BY: **DN**